PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System for the Municipality of Oton

In Partial Fulfillment of

the Requirements for the Degree

Bachelor of Science in Information Technology

By:

John Lloyd P. Cabanig

Nelmar B. Buenafe

Justine A. Balmores

John Arvy Z. Lopez

Gian Rovik B. Somes

2027

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System for the Municipality of Oton

John Lloyd Cabanig, Nelmar Buenafe, Justine Balmores, John Arvy Lopez, Gian Rovik Somes

College of Information Technology Education, PHINMA University of Iloilo, Iloilo, Philippines

jopo.cabanig.ui@phinmaed.com|neba.buenafe.ui@phinmaed.com|usap.balmores.ui@phinmaed.com|joza.lopez.ui@phinmaed.com|gibu.somes.ui@phinmaed.com

Abstract

BreedSmart is a centralized web and mobile-based livestock insemination and health management system developed for the Municipality of Oton, Iloilo. The study addresses the operational challenges of manual, paper-based record-keeping, communication delays, and fragmented livestock tracking identified among local farmers and agricultural personnel. Utilizing an iterative System Development Life Cycle (SDLC) and Agile principles, BreedSmart implements role-based access controls (RBAC) tailored for three primary users: Farmers, Livestock Technicians, and Municipal Administrators.

The system's architecture connects a React and React Native frontend with a Node.js, Express, and MongoDB backend. BreedSmart facilitates the complete lifecycle of bovine management—including animal registration, artificial insemination (AI) service requests, technician scheduling, and reproductive milestone tracking (such as estrus monitoring, pregnancy diagnosis, and calving documentation). Furthermore, it provides distinct workflows for animal health requests, transitioning from initial farmer reports to advisory notices, medicine pickups, or on-site clinical treatments recorded as official medical records. To address intermittent field connectivity, the platform incorporates a localized mobile offline queueing and synchronization mechanism.

Evaluated by Information Technology experts using the ISO/IEC 25010 software quality model, BreedSmart achieved a "Very Satisfactory" grand mean rating of 4.02, demonstrating strong functional stability, security, reliability, and technical flexibility. While the platform successfully standardizes municipal livestock services and improves information accessibility, it operates as an organizational and communication tool rather than an independent clinical diagnostic instrument. Ultimately, BreedSmart provides a scalable, role-based digital model for modernizing agricultural information systems in local government units.

Index Terms - Artificial insemination, livestock management, mobile application, reproductive records, veterinary services.

Table of Contents

I. INTRODUCTION....1
A. Background of the Study....1
B. Statement of the Problem....2
C. Objectives of the Study....2
D. Significance of the Study....3
E. Scope and Delimitation....3
F. Definition of Terms....4
II. REVIEW OF RELATED LITERATURE AND STUDIES....7
B. Related Studies....8
C. Synthesis and Research Gap....9
III. METHODOLOGY....12
B. System Development Methodology....12
C. System Architecture....15
D. Development Tools....31
E. Data Gathering Procedure....32
F. Testing and Evaluation Plan....32
IV. RESULTS AND DISCUSSION....34
A. Overview of the Developed System....34
1) Summary of Published Releases....68
3) Workflow Refinement in v1.0.5....69
4) Internal Validation Build in v1.0.7 Preview 1....70
5) Subsequent Refinements in v1.0.8....70
6) Discussion of Release Progression....70
Source References....70
V. CONCLUSION AND RECOMMENDATIONS....72
A. Introduction....72
B. Summary of the Proposed Study Design and Implementation....72
C. Summary of Findings....72
D. Conclusion....73
E. Recommendations....73
F. Impact of the Study....74
REFERENCES....75
APPENDICES....77

List of Figures

Figure 1. IPO Model with Feedback Loop..... 11
Figure 2. System Development Methodology of BreedSmart..... 12
Figure 3. BreedSmart Web-Mobile-Backend Architecture..... 15
Figure 4. BreedSmart Context Diagram (Level 0)..... 15
Figure 5. BreedSmart Data Flow Diagram (DFD) Level 1..... 16
Figure 6. Entity Relationship Diagram..... 17
Figure 7. Use Case Diagram..... 18
Figure 8. BreedSmart Network Architecture..... 19
Figure 9. Request-to-Official-Record Operational Model..... 19
Figure 10. Responsibility and Data Authority Flow..... 20
Figure 11. Complete BreedSmart Reproductive Lifecycle..... 20
Figure 12. Artificial Insemination Request and Service Flow..... 20
Figure 13. Breeding Follow-up and Return-to-Heat Flow..... 21
Figure 14. Pregnancy Diagnosis and Confirmation Flow..... 21
Figure 15. Calving and Newborn Registration Flow..... 21
Figure. 16. Calving-to-Postpartum Eligibility Flow..... 21
Figure 17. Health Request Review and Handling Flow..... 22
Figure 18. Offline Queue and Synchronization Flow..... 22
Figure 23. Calving validation and related transaction updates..... 29
Figure 24. Account-bound queue, dependency resolution, and retry flow..... 30
Figure 25. Farmer App Register / Login Page..... 33
Figure 26. Farmer Home Dashboard..... 34
Figure 27. Farmer / Add Animal..... 35
Figure 28. Farmer / My Animals..... 36
Figure 29. Farmer / Animal Details..... 37
Figure 30. Farmer / Request AI Service..... 38
Figure 31. Farmer / Report Health Concern..... 39
Figure 32. Farmer / Service Requests..... 40
Figure. 33. Farmer / AI Request Details..... 41
Figure 35. Farmer / Notifications..... 43
Figure 36. Farmer / Profile..... 44
Figure 37. Technician Mobile Dashboard..... 45
Figure 38. Technician Availability and Profile Readiness..... 46
Figure 39. Technician Mobile Requests and Visit Scheduling.Technician Mobile Insemination Recording..... 47
Figure 40 . Technician Mobile Insemination Recording..... 48
Figure 41. Technician Mobile Health-Service Recording..... 49
Figure 42. Technician Mobile Pregnancy Diagnosis and Verification..... 50
Figure 43. Technician Mobile Calving and Offspring Recording..... 51
Figure 44. Technician Mobile Official Records..... 52
Figure. 45. Technician Web Dashboard..... 53
Figure 46. Technician Web Work Queue and Requests..... 54
Figure 47. Technician Web Schedule and Visit Context..... 55
Figure. 48. Technician Web Farmer and Animal Dashboard..... 56
Figure. 49. Technician Web Official Records and Detail Dashboard..... 56
Figure. 50. Technician Web Monthly Accomplishment Reports. Screenshot to be supplied..... 57

Figure. 51. Municipal Admin Web Dashboard....58
Figure 52. Admin User Directory and Account Management....59
Figure 53. Municipal Admin Livestock Registry and Animal Profile....59
Figure. 54. Municipal Admin Technician Workload....60
Figure 55. Municipal Admin Barangay Management Interface....61
Figure 56. Municipal Admin Audit History....61
Figure 57. Municipal Admin Archived Records and Restoration....62
Figure. 58. Farmer Official Record Detail. Screenshot to be supplied....64
Figure. 59. Mobile Synchronization Status and Recovery. Screenshot to be supplied....65

List of Tables

Table 1. Research evidence and implications for BreedSmart....10
Table 2. Implementation components and responsibilities....23
Table 3. Core records and their meaning....25
Table 4. Insemination information and workflow use....26
Table 5. Health handling modes and evidence requirements....27
Table 6. Time-related fields and their distinct meanings....28
Table 7. Pregnancy confirmation and continuation responsibilities....28
Table 8. Offline mechanisms and verification scenarios....30
Table 9. Hardware Requirements....31
Table 10. Hardware Requirement....31
Table 11. Technician mobile tasks and resulting records....52
Table 12. Technician web views and verification focus....57
Table 13. Municipal oversight responsibilities and clinical boundaries....63
Table 14. Documented BreedSmart release progression....64

I. INTRODUCTION

A. Background of the Study

Agriculture continues to support the livelihood of many rural communities in the Philippines, with livestock raising contributing to household income, food production, and local agricultural activity. Cattle and carabaos are particularly important to small-scale raisers because they may serve as productive farm assets and sources of income. Effective livestock management therefore depends not only on animal production but also on the proper recording of breeding activities, health interventions, and other events throughout an animal's productive life.

The Municipality of Oton, Iloilo maintains an active livestock sector as part of its local agricultural economy. According to the Municipality of Oton's Comprehensive Development Plan 2023–2028, livestock and poultry raising is among the major livelihood activities in the municipality. The same municipal profile recorded 611 carabaos and 274 cattle in 2021. It also identifies the Municipal Agriculture Office as one of the offices assisting livestock raisers through services such as diagnosis, treatment, deworming, and artificial insemination, with artificial insemination support coordinated with the Philippine Carabao Center. [1]

These services require the Municipal Agriculture Office and its livestock technicians to maintain information about animals, owners, breeding activities, health concerns, field services, and reproductive events. Artificial insemination, for example, does not end when the procedure is performed. Information regarding the service must remain available for subsequent monitoring, including observations after insemination, pregnancy diagnosis, calving, and later reproductive management. Health-related concerns likewise require a record of the original request, the action taken by the technician, and any treatment that was actually performed. Based on the current practices identified during the requirements-gathering activities of the study, a considerable portion of this information is still handled through paper-based records and direct communication between farmers and agricultural personnel. Farmers may communicate livestock concerns through personal visits, phone calls, or other direct means, while technicians maintain and retrieve information from separate records during field and office activities. Although these practices allow livestock services to continue, they make it more difficult to maintain a consolidated history of an animal when information is recorded at different stages or through different channels.

The problem becomes more significant in reproductive management because several activities may occur over an extended period. An artificial insemination request may lead to a scheduled field service, an insemination record, follow-up observation, pregnancy diagnosis, and eventually a calving record. When these events are documented separately, technicians may need to review several sources to determine the animal's previous service history and current reproductive status. Similar difficulties may occur in health services when the original request, technician response, farm visit, and resulting treatment record are not maintained as parts of one organized history.

Record accessibility is also important for farmers and municipal personnel. Farmers need understandable updates regarding submitted service requests and scheduled activities, while technicians need access to previous animal information when reviewing requests or performing field services. At the administrative level, the Municipal Agriculture Office requires organized records for monitoring livestock activities and reviewing the services delivered within the municipality. A fragmented record-keeping process can make retrieval and consolidation more time-consuming, particularly as the number of animals, requests, and completed services increases.

Digital information systems provide an opportunity to organize these activities within a centralized environment. Previous research on digital agriculture has emphasized the value of digital records and mobile information systems for improving access to agricultural information and supporting coordination between field personnel and producers. However, the usefulness of such systems depends on whether they are designed around the actual processes, responsibilities, and technological conditions of their intended users rather than simply transferring paper forms into electronic format.

For this reason, the study proposes BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System for the Municipality of Oton. BreedSmart is intended to provide a centralized platform for managing animal information, livestock service requests, reproductive monitoring, health-service documentation, and official service records used by farmers, livestock technicians, and authorized municipal personnel. The study focuses on organizing these existing livestock-management activities into a more structured digital workflow while maintaining the responsibilities of agricultural personnel in making clinical and service-related decisions.

B. Statement of the Problem

As technology continues to evolve, many government agencies are adopting digital solutions to improve the efficiency and accuracy of their operations. However, the livestock monitoring and artificial insemination services of the Municipal Agriculture Office still rely primarily on manual processes and paper-based records for managing livestock information, breeding schedules, animal health treatment records, pregnancy monitoring, and other related data.

The use of manual documentation makes livestock records susceptible to loss, damage, and misplacement. Retrieving and updating records also requires considerable time and effort, especially when historical information is needed during field operations. In addition, livestock-related reports from farmers are commonly communicated through phone calls or personal visits, since most farmers proceeding technicians are commonly senior citizens making it difficult to maintain a centralized and updated record of livestock activities.

Agricultural technicians also manually monitor breeding schedules and reproductive cycles for different cattle and carabao breeds. As the number of livestock records increases, managing these schedules becomes more challenging and may result in delays or inconsistencies in recording and reporting information. Furthermore, records collected during field activities are consolidated manually before being reflected in municipal reports, making it difficult to maintain updated information for monitoring and decision-making. These existing challenges affect the efficiency of livestock monitoring, record management, communication between farmers and agricultural technicians, and the preparation of accurate municipal livestock reports.

C. Objectives of the Study

1) General Objective

The general objective of this study is to design and develop BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System for the Municipality of Oton, Iloilo, that provides a centralized digital platform for livestock records, artificial insemination and reproductive monitoring, health-service coordination, and municipal livestock management.

2) Specific Objectives

Specifically, the study aims to:

1. Analyze the existing livestock management practices and challenges experienced by farmers and livestock technicians in Oton, particularly in record keeping, artificial insemination, reproductive monitoring, health services, and communication.
2. Design and develop role-based web and mobile interfaces for Farmers, Livestock Technicians, and Administrators for managing livestock information and accessing functions appropriate to their responsibilities.
3. Develop a centralized livestock record system for maintaining animal profiles and official records related to artificial insemination, pregnancy diagnosis, calving, and health treatments.
4. Implement an artificial insemination and reproductive management workflow that supports service requests, technician scheduling and service recording, breeding follow-up, pregnancy diagnosis, reproductive milestones, and calving documentation.
5. Implement a livestock health-service workflow that allows Farmers to report health concerns and enables Technicians to provide advice, arrange medicine pickup, schedule farm visits when necessary, and document treatments performed.
6. Provide notification, synchronization, and administrative monitoring functions for service updates, scheduled activities, reproductive milestones, livestock records, requests, and municipal-level monitoring.
7. Evaluate the developed BreedSmart system through functional and workflow testing and, where supported by collected respondent data, assess selected software quality characteristics and user acceptance.

D. Significance of the Study

The study is significant to the stakeholders involved in livestock management and agricultural service delivery in the Municipality of Oton, Iloilo. Through the development of a centralized web and mobile-based system, the study provides a digital approach for organizing livestock records, service requests, reproductive activities, health-related services, and official livestock records.

Farmers. The study provides farmers with a mobile platform through which they can maintain their registered animals, submit artificial insemination and health-service requests, monitor the status of submitted requests, receive service updates, and view relevant livestock records and reproductive milestones. This provides farmers with a more organized means of accessing information related to their animals and communicating service needs to livestock technicians.

Livestock Technicians. The study provides livestock technicians with tools for reviewing and managing farmer requests, scheduling field visits when required, recording artificial insemination and health services, conducting pregnancy-related follow-up activities, documenting calving events, and accessing animal histories. Centralizing these activities allows related service information to be maintained within a common system instead of being distributed across separate records and communication channels.

Municipal Agriculture Office of Oton. The study provides authorized municipal personnel with centralized access to registered livestock information, service activities, official records, user accounts, and administrative monitoring functions. These records can support the office in reviewing livestock-related activities and maintaining organized information for municipal livestock management.

BSIT Students and Future Researchers. The study may serve as a reference for students and researchers interested in the development of web and mobile information systems for agriculture, particularly systems involving livestock records, field-service coordination, role-based access, offline data handling, and reproductive and health-service workflows. The documented development process, system architecture, and implementation of BreedSmart may also provide a basis for examining similar information-system applications in local government or agricultural settings.

E. Scope and Delimitation

1) Scope

The study covers the design, development, integration, and testing of BreedSmart within the livestock-service operations of the Municipality of Oton, Iloilo. The system is organized around three authorized user roles: Farmer, Livestock Technician, and Administrator. Each role is provided with functions that correspond to its responsibilities in livestock record management, artificial insemination and reproductive monitoring, health-service coordination, and municipal oversight.

Farmer Module. Farmers access BreedSmart through the mobile application. Within their authorized records, they can manage profile information, register and view their animals, submit artificial insemination and health-service requests, monitor request status and scheduled activities, submit breeding observations, view technician advice or medicine-pickup instructions, review relevant animal records and reproductive milestones, and notify the technician of calving events for follow-up and official recording.

Livestock Technician Module. Livestock technicians access BreedSmart through mobile and web interfaces. They can review and claim service requests, schedule farm visits using designated service periods, perform and record artificial insemination services, record walk-in services, review breeding observations, perform and record pregnancy diagnosis, document calving and newborn information, handle health requests through advice, medicine pickup, or farm visits, and create official treatment records when clinical services are performed. The technician interfaces also provide access to assigned work, animal information, and completed service records.

Administrative Module. Authorized municipal administrators access the web application to manage and verify user accounts, review the municipal animal registry, monitor service requests and assignments, reassign requests when necessary, review and correct authorized official records, and view available system summaries and administrative reports.

Reproductive Management. BreedSmart records artificial insemination information and uses recorded reproductive events to

generate follow-up dates and monitoring milestones. These functions support post-insemination observation, pregnancy-diagnosis scheduling, pregnancy monitoring, calving documentation, and postpartum follow-up. System-generated dates and reminders are intended as monitoring support and do not replace clinical examination or the judgment of an authorized livestock technician.

Health-Service Management. Farmers can submit health concerns and medicine-related requests. Livestock technicians review these requests and may respond with advice, arrange office medicine pickup, or schedule a farm visit when on-site assessment or treatment is necessary. An official medical record is created when a clinical treatment is performed and documented.

Scheduling, Notifications, and Synchronization. Farm visits are scheduled using defined morning or afternoon service periods. BreedSmart provides status updates and notifications for relevant service and reproductive events. Supported mobile operations may be stored locally when connectivity is unavailable and synchronized with the server after a network connection is restored.

2) Delimitation

The study is geographically and operationally limited to the livestock-management activities of the Municipality of Oton, Iloilo. Its biological scope focuses on bovine livestock managed within the identified municipal workflows, particularly cattle and carabao. The system is intended for the Farmer, Livestock Technician, and Administrator roles defined in BreedSmart and does not represent a multi-municipality livestock information platform.

BreedSmart is an information-management and service-coordination system and does not independently diagnose disease, determine pregnancy, or determine whether an insemination resulted in conception. Reproductive dates and milestones generated by the system are based on recorded events and configured monitoring rules. Clinical pregnancy diagnosis, treatment decisions, and official clinical records remain under the responsibility of authorized livestock personnel.

Offline capability is limited to supported mobile operations that can be queued locally. An internet connection is still required to synchronize queued information with the central server and to access functions that depend on current server data. The presence of an offline queue should therefore not be interpreted as complete offline availability of every system function.

The study does not depend on physical Internet of Things sensors, wearable livestock devices, automated biometric identification, or other hardware-based monitoring equipment. It also does not directly integrate BreedSmart with external national or provincial livestock information systems; information intended for external reporting remains subject to the procedures of the Municipal Agriculture Office.

The study primarily focuses on the design, development, integration, and technical validation of BreedSmart, together with pilot feedback obtained from its intended users during initial system use. Technical validation is concerned with determining whether the implemented functions and workflows operate according to the identified system requirements, while pilot feedback provides initial information regarding the usability and practical suitability of the system. The study does not attempt to establish long-term effects such as increased technician efficiency, improved farmer productivity, higher conception rates, improved livestock health outcomes, or measurable improvements in overall municipal service performance. Establishing such outcomes would require extended operational use of the system, collection of longitudinal data, and an impact evaluation comparing relevant conditions before and after implementation. These forms of evaluation are therefore considered beyond the scope of the present system-development study.

F. Definition of Terms

The following terms are operationally defined according to their use in the study and in BreedSmart:

Artificial Insemination (AI)
A livestock breeding procedure in which semen is introduced into a female animal by trained personnel. In BreedSmart, the AI service is documented as part of the animal's reproductive history and may be linked to subsequent reproductive monitoring.

Service Request
A request submitted through BreedSmart for livestock assistance, such as artificial insemination or a health-related service. A service request represents the need for assistance and is distinct from the official record created after a service is completed.

Reproductive Monitoring
The recording and tracking of reproductive events and follow-up activities after artificial insemination, including breeding observations, pregnancy diagnosis, pregnancy-related milestones, and calving. Monitoring information supports follow-up but does not itself constitute a clinical diagnosis.

Reproductive Milestone
A system-generated date or monitoring window derived from recorded reproductive events and used to remind users of an expected follow-up activity, such as heat-return observation, pregnancy diagnosis, or expected calving.

Return to Estrus (Heat Return)
The return of observable estrus or heat signs following an insemination cycle. BreedSmart can provide a monitoring window for this observation, but the system does not automatically detect estrus or determine conception failure.

Pregnancy Diagnosis
A clinical assessment performed by authorized livestock personnel to determine the pregnancy status of an animal after breeding. BreedSmart records the result and related reproductive information but does not independently diagnose pregnancy.

Calving
The event in which a pregnant animal gives birth. BreedSmart records the calving event and, when applicable, information about the newborn animal as part of the reproductive history.

Health Request
A farmer-submitted request concerning an animal health issue or medicine-related need. The request may be handled through technician advice, office medicine pickup, or a scheduled farm visit depending on the required response.

Medical Record
An official clinical record created when a livestock technician documents a treatment or other clinical service performed for an animal. It is distinct from the original health request that initiated the service.

Official Record
A verified record representing the outcome of a completed livestock service or clinical activity, such as an insemination, pregnancy diagnosis, medical treatment, or calving event. Requests, schedules, and work tasks are not treated as official service outcomes by themselves.

Livestock Technician
An authorized agricultural personnel member responsible for livestock services such as artificial insemination, reproductive follow-up, pregnancy diagnosis, health-request handling, treatment documentation, and calving recording within the municipal livestock program.

Offline Synchronization
The process in which supported mobile actions are stored locally when network connectivity is unavailable and are submitted to the central server after connectivity is restored.

Role-Based Access Control (RBAC)

A system security approach that limits functions and data according to the responsibilities assigned to a user role, such as Farmer, Livestock Technician, or Administrator.

Sire

The male parent or breeding bull whose semen or breeding information is associated with an animal or artificial insemination record.

Dam

The female parent of an animal. BreedSmart may maintain the dam relationship as part of an animal's recorded lineage.

II. REVIEW OF RELATED LITERATURE AND STUDIES

A. Related Literature

Digitalization of Agricultural and Livestock Records

Digital technologies are increasingly used in agriculture to organize information, improve access to records, and support communication among farmers, service providers, and public institutions. The Food and Agriculture Organization notes that the value of digital agriculture depends not only on the availability of technology but also on infrastructure, affordability, skills, and the conditions in which intended users operate [2]. The World Bank similarly emphasizes that digital technologies can improve access to information and public services, while cautioning that their benefits depend on connectivity, skills, trust, affordability, and complementary investments [3]. These observations are relevant to livestock information systems because digitization should be treated as a means of organizing and supporting existing services rather than as automatic evidence of improved productivity or service performance.

For BreedSmart, the relevance of digitalization is primarily in the organization of livestock information that would otherwise be distributed across paper records and separate communication channels. A digital platform can maintain animal profiles, service requests, completed service records, and reproductive or health-related events in a common environment. The literature therefore supports the use of digital records as an information-management approach, while the actual effect of BreedSmart on users and municipal operations remains subject to system testing and user evaluation.

Livestock Record Integrity and Traceability

Livestock records have value when events can be connected to the correct animal and retrieved as part of a continuing history. FAO guidelines on multipurpose animal recording systems describe animal identification and recording as useful for integrating animal registration, traceability, animal-health information, and performance records within a structured information system [28]. The same guidelines emphasize that recording systems should be designed around clearly defined purposes and institutional conditions rather than around technology alone.The World Organisation for Animal Health also identifies animal identification and traceability as important components of livestock management. Its guidance states that relevant animal events and information should be registered in a timely and accurate manner and that an information system should support the collection, storage, and retrieval of registration data [29]. This principle is relevant to BreedSmart because the system maintains animal information together with service histories. In the system, a submitted request, a scheduled activity, and an official record represent different stages of a workflow and should not be treated as interchangeable records.

Digital Agriculture in the Philippine Context

The Philippine setting presents both opportunities and constraints for agricultural digitalization. Briones, Galang, and Latigar examined the adoption of digital technologies in Philippine agriculture and identified continuing concerns regarding the digital divide and the inclusion of smallholder farmers [11]. Their assessment indicates that digital agriculture should be considered in relation to actual access, capabilities, infrastructure, and the circumstances of intended users.

This local context is important to BreedSmart because the system is intended for municipal livestock services rather than for large commercial farms with specialized infrastructure. The use of mobile and web interfaces, role-specific functions, and supported offline synchronization is therefore better understood as a design response to local operating conditions. These design choices do not by themselves establish user acceptance or effectiveness; instead, they define how the system attempts to make its functions accessible within the conditions identified during requirements gathering.

Mobile Systems for Agricultural and Veterinary Service Coordination

Mobile applications can provide a practical channel for field data entry and communication when users are geographically separated from an office or central information source. FAO documents examples of mobile applications used in agricultural information and animal-health settings, while also emphasizing the challenges of reaching remote communities [2]. The World Bank likewise notes that digital public services and mobile access can reduce some information barriers, although the benefits of such systems remain dependent on surrounding infrastructure and user capabilities [3].

Within BreedSmart, mobile access supports the submission and review of livestock service requests, viewing of service updates, field recording, and access to relevant animal information. The purpose of these functions is to provide a common

channel through which existing livestock-service activities can be recorded and coordinated. The literature supports mobile systems as a possible mechanism for field information exchange, but it does not establish that BreedSmart will necessarily shorten response times or make technicians more efficient without direct evaluation of those outcomes.

Artificial Insemination and Reproductive Record Management

Artificial insemination is not a single isolated event; it is associated with information that may be needed for later reproductive follow-up. In the Philippine context, Bondoc examined artificial insemination of cattle and carabaos and noted the limited availability of performance records for progeny produced through AI programs [30]. This highlights the importance of maintaining breeding and reproductive information beyond the date of insemination itself.For BreedSmart, reproductive management therefore centers on documentation and follow-up rather than on automated biological prediction. The system records artificial insemination information and can organize subsequent activities such as breeding observations, pregnancy diagnosis, reproductive milestones, and calving. These system-generated milestones serve as monitoring support. They do not determine conception or replace the clinical judgment of a livestock technician.

Livestock Health Information and Service Records

Reliable animal-health information depends on consistent data collection and management. WOAH states that the effectiveness of animal-health surveillance is dependent on a reliable process for data collection and management and recognizes that these processes may use either paper or electronic records [26]. The guidance also emphasizes consistency, data quality, communication among participants, and the ability to address missing or inaccurate information.Although BreedSmart is not designed as an epidemiological surveillance or diagnostic system, these principles remain relevant to its health-service functions. Health concerns submitted by farmers are maintained separately from official clinical treatment records. Technician advice or medicine-pickup instructions may resolve a request without creating a treatment record, while an on-site clinical service can result in an official medical record. Maintaining this distinction helps preserve the meaning of each record within the animal's service history.

Rural Connectivity and User-Centered System Design

Connectivity and user capability can directly influence whether a rural digital system is practical. The Philippine assessment by Briones et al. identifies digital inclusion and unequal access as continuing issues in agricultural digitalization [11]. International guidance on animal identification and traceability also recommends considering technology literacy, communication needs, and spoken languages when implementing information systems [29].

BreedSmart addresses connectivity constraints through supported mobile operations that can be queued locally and synchronized when connectivity becomes available. This approach does not make every system function fully available offline. Instead, it is intended to reduce disruption for selected field operations when a network connection is temporarily unavailable. Interface simplicity, clear role separation, and limited workflow choices are likewise design considerations that are evaluated during pilot use rather than assumed to be successful in advance.

B. Related Studies

1) Foreign Studies

Beyene et al. compared smartphone-based cattle disease reporting with a traditional paper-based reporting process in public veterinary clinics in Ethiopia [10]. The smartphone approach produced more timely and complete reporting, but the researchers also documented delays caused by limited connectivity, particularly in more rural locations. The study demonstrates that mobile reporting can improve the movement and completeness of field information while still being constrained by network conditions. For BreedSmart, the study is relevant to the decision to support structured mobile data entry and synchronization, but its results cannot be treated as direct evidence of BreedSmart's performance in Oton because the population, workflow, and application are different.. Njenga et al. evaluated the rollout of the Kenya Animal Biosurveillance System, which combined a mobile application with web-based management and monitoring tools [31]. The system was integrated with existing veterinary reporting processes and was used by government and private animal-health officers. Following deployment, the study reported a substantial increase in the number and geographic coverage of disease reports compared with the previous reporting period. The relevance to BreedSmart lies in the demonstrated feasibility of *combining mobile field data collection with a web-based institutional interface. However, BreedSmart serves a different*

municipal workflow and is not presented as a disease-surveillance platform.

2) Local Studies

Briones et al. provide a Philippine-wide assessment of digital technology adoption in agriculture and the implications for smallholder farmers [11]. Their findings emphasize that digitalization can create opportunities while also creating exclusion if connectivity, affordability, and user capability are not considered. This supports the need for BreedSmart to be evaluated within the actual conditions of intended municipal users rather than assuming that the presence of a digital platform automatically results in successful adoption.Bondoc's study of artificial insemination in Philippine cattle and carabaos identified the lack of performance records associated with AI progeny as an important limitation in breeding programs [30]. Although the study is concerned with the genetics and economics of AI rather than information-system development, it demonstrates the continuing importance of reproductive records in Philippine livestock programs. BreedSmart addresses this information-management aspect by maintaining insemination records and linking them with later reproductive events.More recent Philippine studies further illustrate why pregnancy status should remain a clinical determination rather than a software-generated conclusion. Atabay et al. evaluated controlled breeding protocols in postpartum buffaloes and conducted pregnancy diagnosis 35 to 40 days after timed artificial insemination [32]. Tadeo et al. compared methods for early pregnancy detection in water buffaloes and used transrectal ultrasonography as the reference method for pregnancy diagnosis [33]. These studies demonstrate that pregnancy confirmation involves biological and clinical assessment. Accordingly, BreedSmart is designed to record and organize pregnancy-diagnosis results entered by authorized personnel rather than independently determining whether an animal is pregnant.

C. Synthesis and Research Gap

The reviewed literature and studies show that digital technologies can support the organization, collection, transmission, and retrieval of agricultural and livestock information. International guidance emphasizes reliable record management, traceability, data quality, and the need to design systems around their actual operating environment [2], [26], [28], [29]. Studies of mobile veterinary reporting demonstrate that mobile and web technologies can support field-to-office information flows, while also showing that connectivity and implementation conditions remain important constraints [10], [31]. Philippine literature further highlights both the continuing digital divide in agriculture and the importance of maintaining reproductive records within cattle and carabao programs [11], [30], [32], [33].

The literature does not establish that a software platform alone will improve conception rates, animal-health outcomes, farmer productivity, or technician efficiency. Those effects depend on biological, organizational, and user-related factors and require separate evaluation. The more immediate information-system need identified in this study is the organization of related municipal livestock activities that may otherwise be handled through separate records and communication channels.

BreedSmart is developed in response to this information-management gap. It integrates animal profiles, artificial insemination requests and records, reproductive follow-up, pregnancy diagnosis records, calving information, health-service requests, clinical treatment records, scheduling, notifications, and administrative oversight within a role-based web and mobile environment for the Municipality of Oton. Its contribution is therefore evaluated primarily through the correctness of its implemented functions and workflows, together with pilot feedback from intended users, rather than through unverified claims of long-term agricultural or organizational impact.

Additional Evidence on Recording, Adoption, and Field Connectivity

Resti et al. reviewed digital and manual recording tools for smallholder dairy farming in developing countries. Their review included 21 publications describing 24 tools, covering animal identification, traceability, health, and production information. They identified connectivity, continuity of data entry, and feedback to Farmers as considerations for further development [36]. For BreedSmart, this evidence supports linking animal identities to service histories and returning relevant records to the Farmer. The review provides a design rationale; its findings do not establish that the Oton implementation has already improved record quality.

Muromba, Keeni, and Fuyuki synthesized 14 empirical studies applying the Technology Acceptance Model to mobile agricultural services in sub-Saharan Africa. Perceived usefulness was the more influential adoption factor in their synthesis, while ease of use reflected interface simplicity, user support, and infrastructure. Education, income, device type, and network availability also appeared as contextual influences [37]. This study informs the decision to evaluate whether BreedSmart helps users complete actual livestock-service tasks. Its regional findings should be treated as comparison evidence rather than as measurements of the preferences of Oton Farmers.

Gaundare et al. described an online and offline field-data collection system for a dairy genetic-improvement project in

India. Their implementation combined field recording with supervisory web access and synchronization to a central repository [38]. The relevant comparison is the separation between information captured in the field and information available for centralized review. BreedSmart adopts a different implementation and municipal purpose. A locally queued action must still be accepted by the backend before it is treated as a synchronized official record; the results of the Indian project are not transferable performance results for BreedSmart.

Sivakumar et al. evaluated 25 smartphone applications for Indian agricultural stakeholders using dimensions including usability, accessibility, updates, ratings, and downloads. Their study highlights the need to compare available features with stakeholder needs and to evaluate applications systematically [39]. This supports examining the complete BreedSmart task sequence, including error handling and record retrieval, rather than relying solely on the availability of an application build. The study does not supply a validated questionnaire or acceptance threshold for the present municipal pilot.

Table 1. *Research evidence and implications for BreedSmart*

Study	***Relevant evidence***	***Application and limitation***
Resti et al. [36]	Review of smallholder dairy recording tools.	Link animal identity and service history; connectivity remains a design consideration.
Muromba et al. [37]	Review of mobile agricultural-service adoption.	Evaluate usefulness and ease of use locally; avoid transferring regional ratings.
Gaundare et al. [38]	Field capture and centralized review through online/offline software.	Document queue and synchronization states; their project is not a BreedSmart benchmark.
Sivakumar et al. [39]	Systematic evaluation of agricultural smartphone applications.	Evaluate task suitability and accessibility; application publication is not user acceptance.

Implications for the Municipal Study

The research gap addressed by BreedSmart is operational: animal information, requests, schedules, Technician activities, reproductive follow-up, and official service records must be connected within the same municipal workflow. A request list alone cannot provide this connection because it does not establish whether a service occurred, which animal received it, who recorded it, or what follow-up remains due. The study therefore examines linked workflows and clearly assigned responsibilities. This is the researchers' interpretation of the reviewed evidence in relation to the implemented municipal system.

Interface evaluation should consequently cover both entry and retrieval. A participant may successfully submit an insemination request but still be unable to recognize its scheduled visit or locate the completed service record. Likewise, a Technician may record a procedure but find its follow-up difficult to identify in assigned work. The evaluation should observe these task boundaries, record the point at which assistance is required, and distinguish navigation problems from backend rejection of an invalid operation.

The study also separates system quality from biological and organizational impact. Correctly storing an insemination date demonstrates a recording function. Confirming pregnancy requires an authorized clinical process. Establishing that the system improves conception rates, reduces municipal workload, or changes livestock-health outcomes requires a separate evaluation design and sustained operational evidence. The present manuscript therefore treats these as possible subjects of future study rather than outcomes inferred from the existence of software.

D. Conceptual Framework

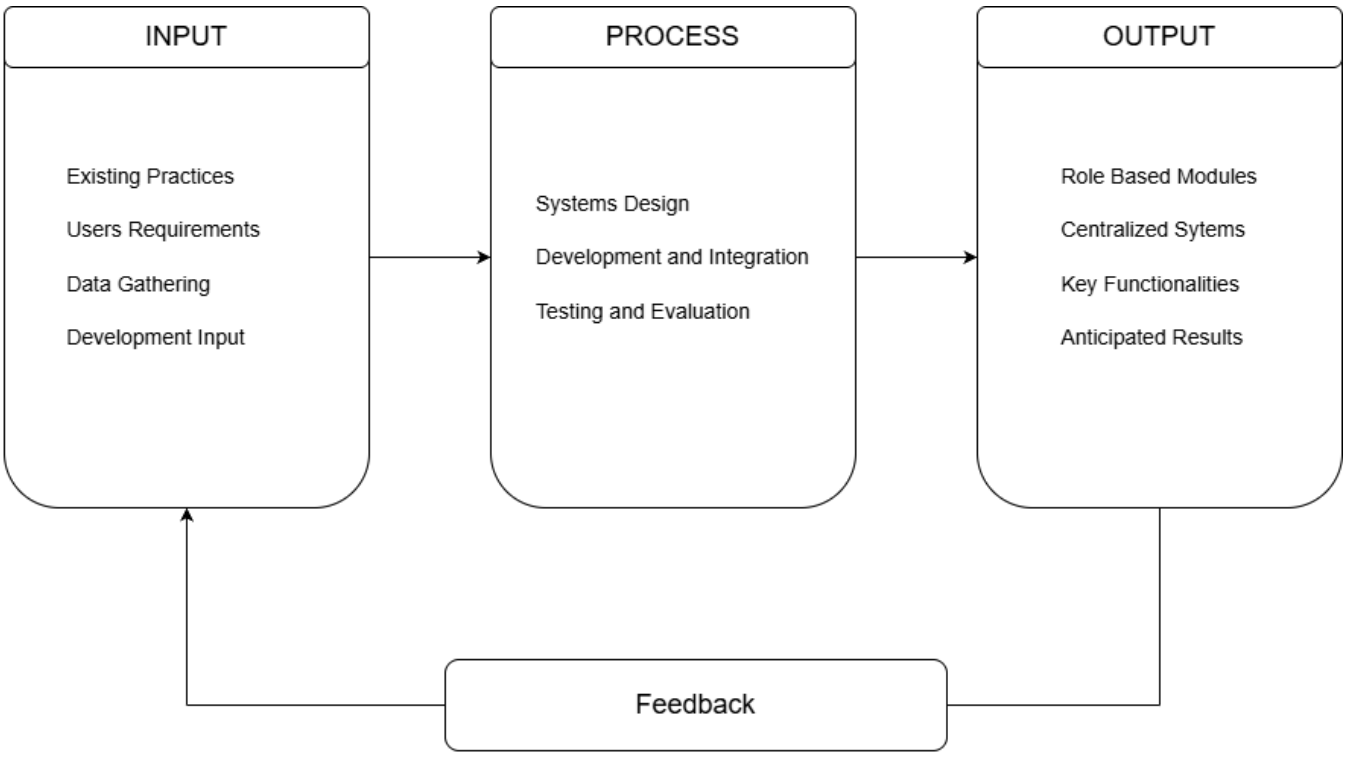


Figure 1. IPO Model with Feedback Loop

The conceptual framework of the study follows the Input–Process–Output (IPO) model. It presents how information about existing livestock-management practices and identified system requirements is used in the design, development, testing, and evaluation of BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System for the Municipality of Oton.

The input consists of information regarding the existing livestock-management practices of the Municipal Agriculture Office of Oton, the operational challenges encountered in record keeping and service coordination, and the requirements identified for the proposed system. Requirements gathering is supported by consultation, observation, review of available records, and a researcher-developed questionnaire for the livestock technician focused on current artificial insemination procedures, record-keeping practices, operational difficulties, connectivity conditions, and desired system functions.

The process covers requirements analysis, system design, development and integration of the web, mobile, backend, and database components, and functional and workflow testing of the implemented features. The study also includes pilot user evaluation with intended users after they have sufficient exposure to the relevant system functions. Technical testing is used to verify whether the implemented workflows operate according to the identified requirements, while user evaluation is treated separately as feedback on usability and practical suitability.

The output of the study is the developed BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System for the Municipality of Oton. The system provides role-based functions for animal information, livestock service requests, artificial insemination and reproductive records, health-service activities, scheduled services, official records, and administrative monitoring.

III. METHODOLOGY

A. Research Design

The study uses a Developmental Research Design because it focuses on identifying existing livestock-management problems and user requirements and using these as the basis for designing, developing, testing, and refining BreedSmart. This approach is appropriate because the study aims to develop a web and mobile-based livestock management system that supports livestock insemination, reproductive monitoring, health services, scheduling, and record management in the Municipality of Oton, Iloilo.

B. System Development Methodology

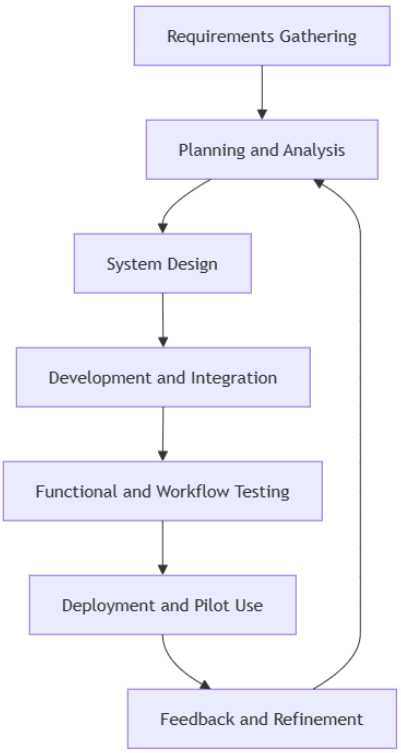


Figure 2. System Development Methodology of BreedSmart.

Figure 2 presents the iterative system development methodology used in developing BreedSmart. It shows the progression from requirements gathering and planning through system design, development and integration, functional and workflow testing, deployment and pilot use, and feedback and refinement. The feedback loop indicates that findings from testing and pilot use may lead to revisions in earlier phases before the system is finalized.

BreedSmart was developed using an iterative System Development Life Cycle (SDLC) approach guided by Agile development principles. The SDLC provided a structured progression through requirements gathering, planning and analysis, system design, development and integration, testing, deployment and pilot use, and feedback and refinement. Agile principles supported the iterative nature of the development process by allowing the researchers to revisit earlier phases when issues, changing requirements, or user feedback identified the need for modification.

Rather than treating each development phase as a one-time activity, features and workflows were progressively implemented, tested, reviewed, and refined throughout the development of BreedSmart. This approach allowed corrections and adjustments to be incorporated while the system was being developed and during its initial pilot use.

Requirements Gathering Phase

The requirements gathering phase focused on understanding the existing livestock-management practices, operational concerns, and information needs of the intended users in the Municipality of Oton, Iloilo. Information was gathered through consultation, observation, review of available records, and the researcher-developed questionnaire for the livestock technician. The gathered information was used to identify requirements related to animal record management, artificial insemination services, reproductive monitoring, health-service requests, scheduling, notifications, official service records, and administrative monitoring. This phase established the initial functional requirements that guided the development of BreedSmart.

Planning and Analysis Phase

During the planning and analysis phase, the gathered requirements were reviewed and organized according to the responsibilities of the three primary user roles: Farmer, Livestock Technician, and Administrator. The researchers defined the scope of the system, identified the functions available to each role, and analyzed the relationships among service requests, scheduled activities, completed services, and official records. The major livestock-service workflows were also examined during this phase. These included artificial insemination requests, technician handling and scheduling, reproductive follow-up, pregnancy diagnosis, calving documentation, health-service requests, and administrative oversight. The resulting requirements served as the basis for the system design and development priorities.

System Design Phase

The system design phase involved creating the structural and visual design of BreedSmart before and during implementation. This included the design of the mobile and web interfaces, system navigation, database structure, backend organization, and major workflow interactions.Role-specific interfaces were designed for Farmers, Livestock Technicians, and Administrators. The mobile application was designed primarily to support Farmer and Technician activities, while the web application supports Technician and Administrator functions. The database and backend structures were designed to maintain relationships among users, animals, service requests, artificial insemination records, pregnancy records, calving records, medical records, schedules, tasks, and notifications. The design also considered the operating conditions of intended users, including the use of mobile devices during field activities and intermittent network connectivity.

Development and Integration Phase

During the development and integration phase, the planned system components were implemented and connected into a working BreedSmart platform. The web application was developed using React, while the mobile application was developed using React Native with Expo. The backend services were implemented using Node.js and Express, with MongoDB used for persistent data storage. Authentication and user identity are handled through Clerk, while additional services are integrated where required for media storage, notifications, and other supporting functions.

The development process included the implementation and integration of major functions such as animal management, artificial insemination service requests and recording, reproductive follow-up, pregnancy diagnosis recording, calving documentation, health-service handling, scheduling, notifications, official records, and administrative monitoring.

Development was performed iteratively. Features and workflows were implemented, reviewed, tested, and revised as issues or additional requirements were identified.

Functional and Workflow Testing Phase

The testing phase focused on verifying whether the implemented functions and workflows behaved according to the identified system requirements. Functional testing was used to examine individual features, while workflow testing was used to verify sequences of actions involving multiple users and system components.

For example, the artificial insemination workflow was tested from the Farmer's submission of a service request, through Technician review, scheduling, and service recording, until the corresponding official insemination record was created. Similar workflow testing was conducted for reproductive follow-up, pregnancy diagnosis, calving, and health-service activities.

Testing was also used to identify functional defects, incorrect workflow transitions, missing information, synchronization

issues, and inconsistencies between the web and mobile applications. Identified issues were corrected and retested as part of the iterative development process.

Deployment and Pilot Use Phase

After the core functions and major workflows underwent technical testing, BreedSmart was prepared for controlled pilot use with its intended users. The mobile and web applications were made available according to the access requirements of Farmers, Livestock Technicians, and Administrators.

Pilot use allows the researchers to observe the system under conditions closer to its intended operational environment and provides users with sufficient exposure to the implemented functions before formal feedback is interpreted.

This phase is treated as pilot use rather than full municipal deployment. The study therefore does not assume that BreedSmart has already replaced existing municipal procedures or established long-term improvements in livestock-service performance.

Feedback and Refinement Phase

Feedback and issues identified during technical testing and pilot use are reviewed and used to refine BreedSmart. Refinement may include correcting software defects, simplifying interfaces, revising displayed information, adjusting workflow behavior, and modifying implemented functions when they do not adequately reflect the identified requirements.

Because the development process is iterative, findings from this phase may require the researchers to return to the planning, design, development, or testing stages. The cycle continues within the approved scope of the study until identified issues have been addressed and the system is considered suitable for continued pilot use and evaluation.

C. System Architecture

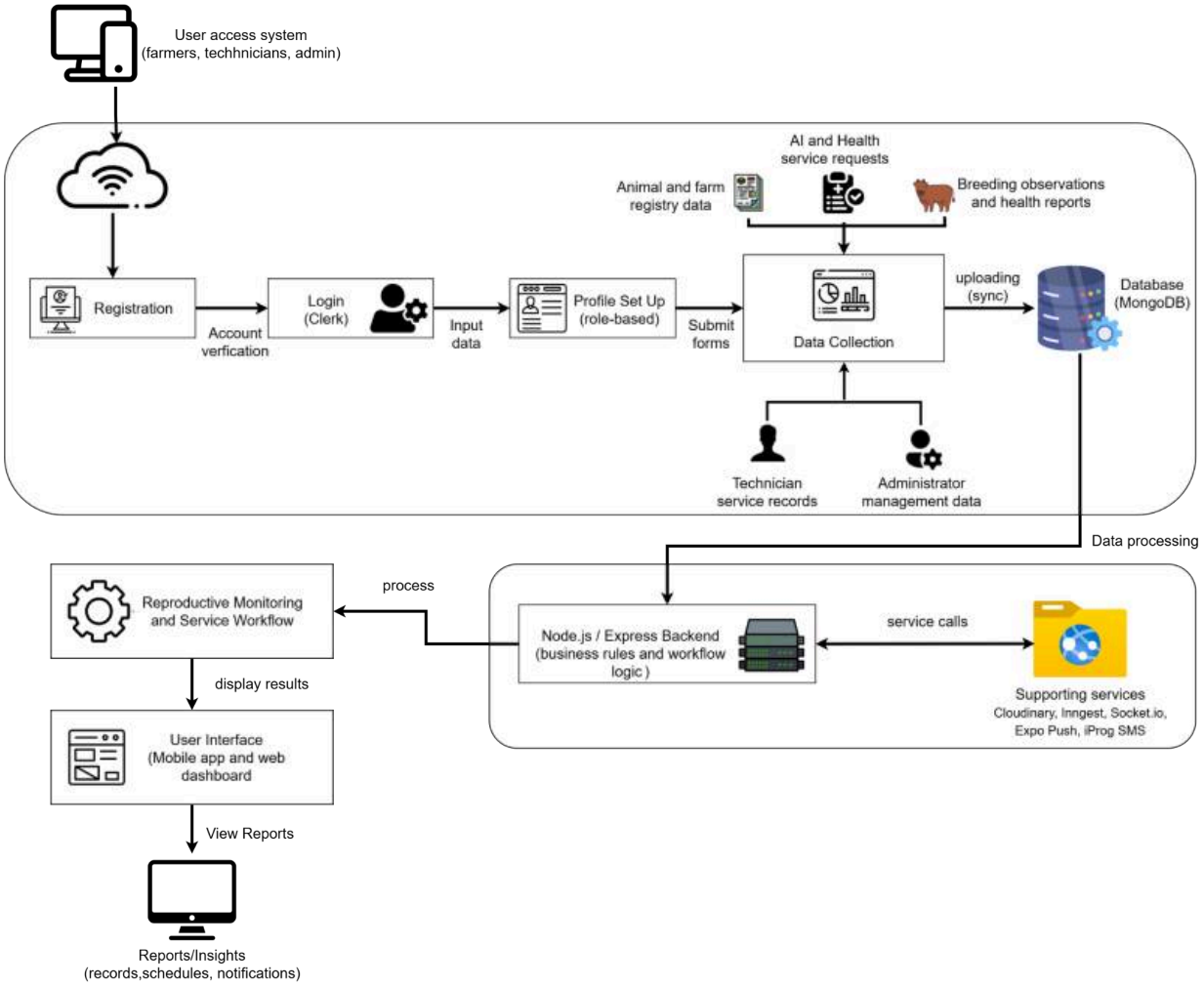


Figure 3. BreedSmart Web-Mobile-Backend Architecture.

Figure 3 presents the web-mobile-backend architecture of BreedSmart. Farmers access the system through the mobile application, while Livestock Technicians and Administrators use role-specific mobile and web interfaces connected to a common Node.js and Express backend. The backend applies identity and authorization controls, processes service workflows, reproductive monitoring, records, and notifications, and stores persistent data in MongoDB through Mongoose. Supporting services provide media storage, background processing, connected-client updates, mobile push notifications, and SMS functions, while selected mobile actions can be queued locally and synchronized after connectivity is restored.

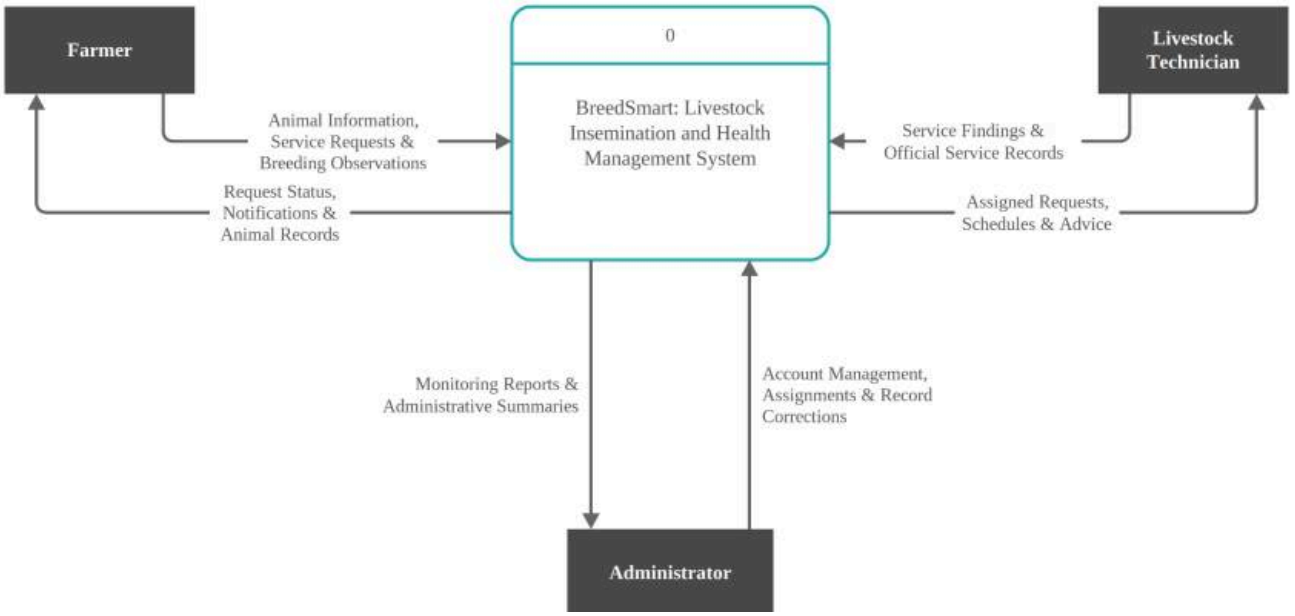


Figure 4. BreedSmart Context Diagram (Level 0).

Figure 4 shows the data exchanged between BreedSmart and its three users: Farmers submit animal information and service requests; Technicians manage assigned work and enter service findings; and Administrators review operations and manage accounts and assignments.

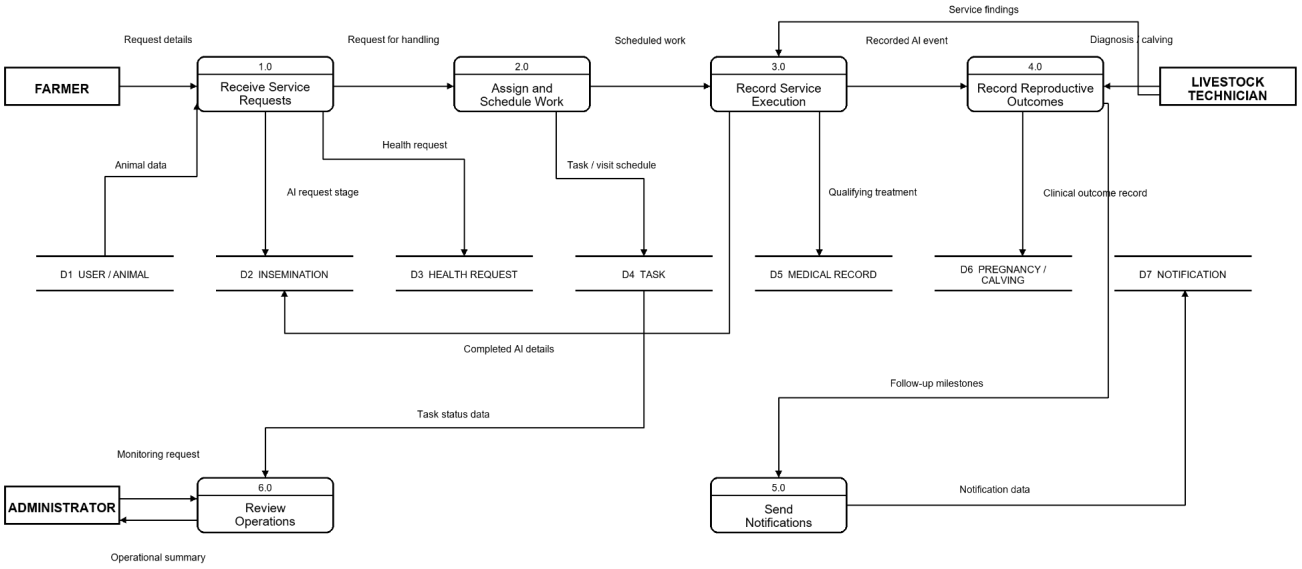


Figure 5. BreedSmart Data Flow Diagram (DFD) Level 1.

Figure 5 presents the Level 1 Data Flow Diagram of BreedSmart, showing the major system processes and the movement of data among users, processes, and data stores. It provides a detailed view of how information is processed within the system.

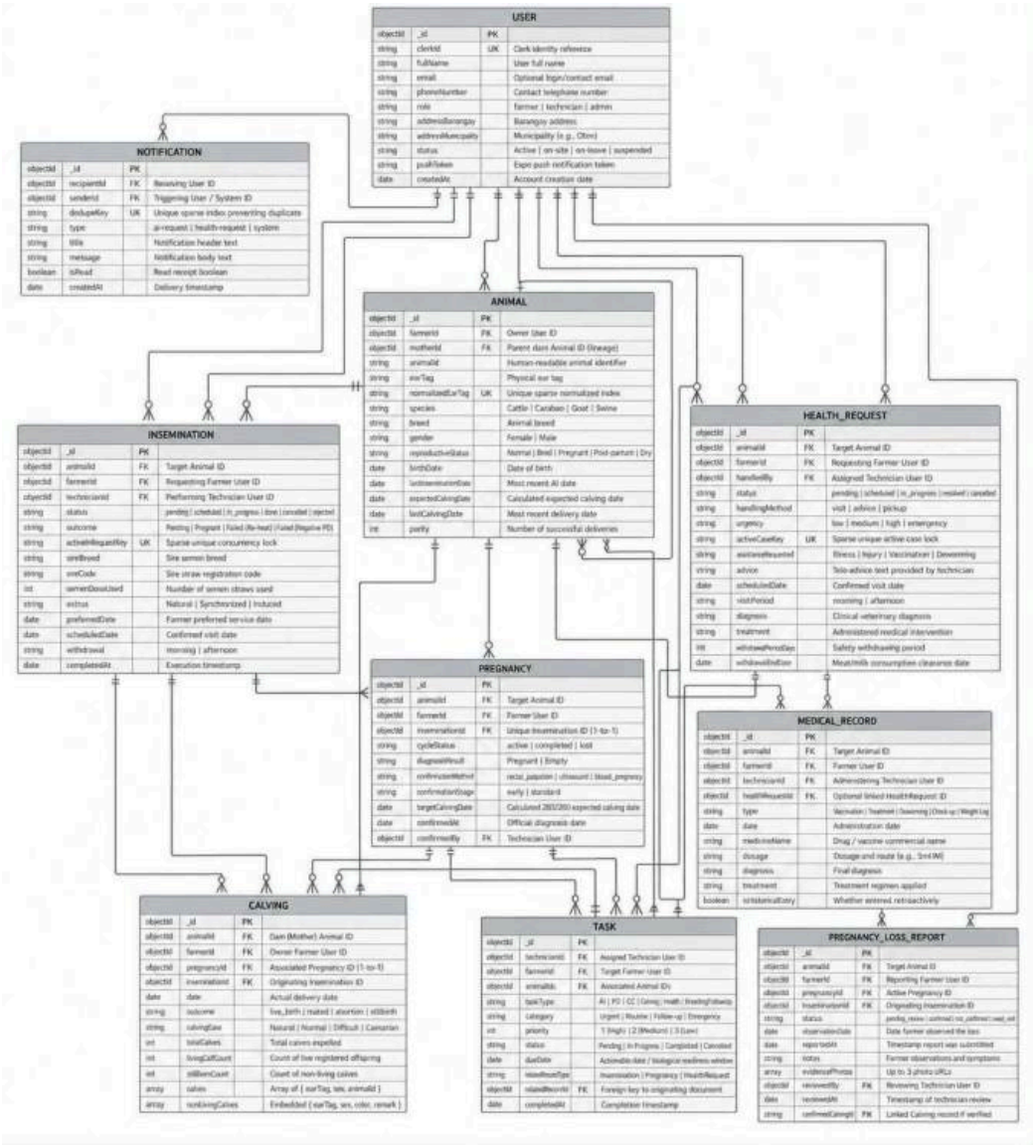


Figure 6. Entity Relationship Diagram.

Figure 6 shows the relationships among BreedSmart's main data entities, including animal records, service requests, clinical records, tasks, and notifications.

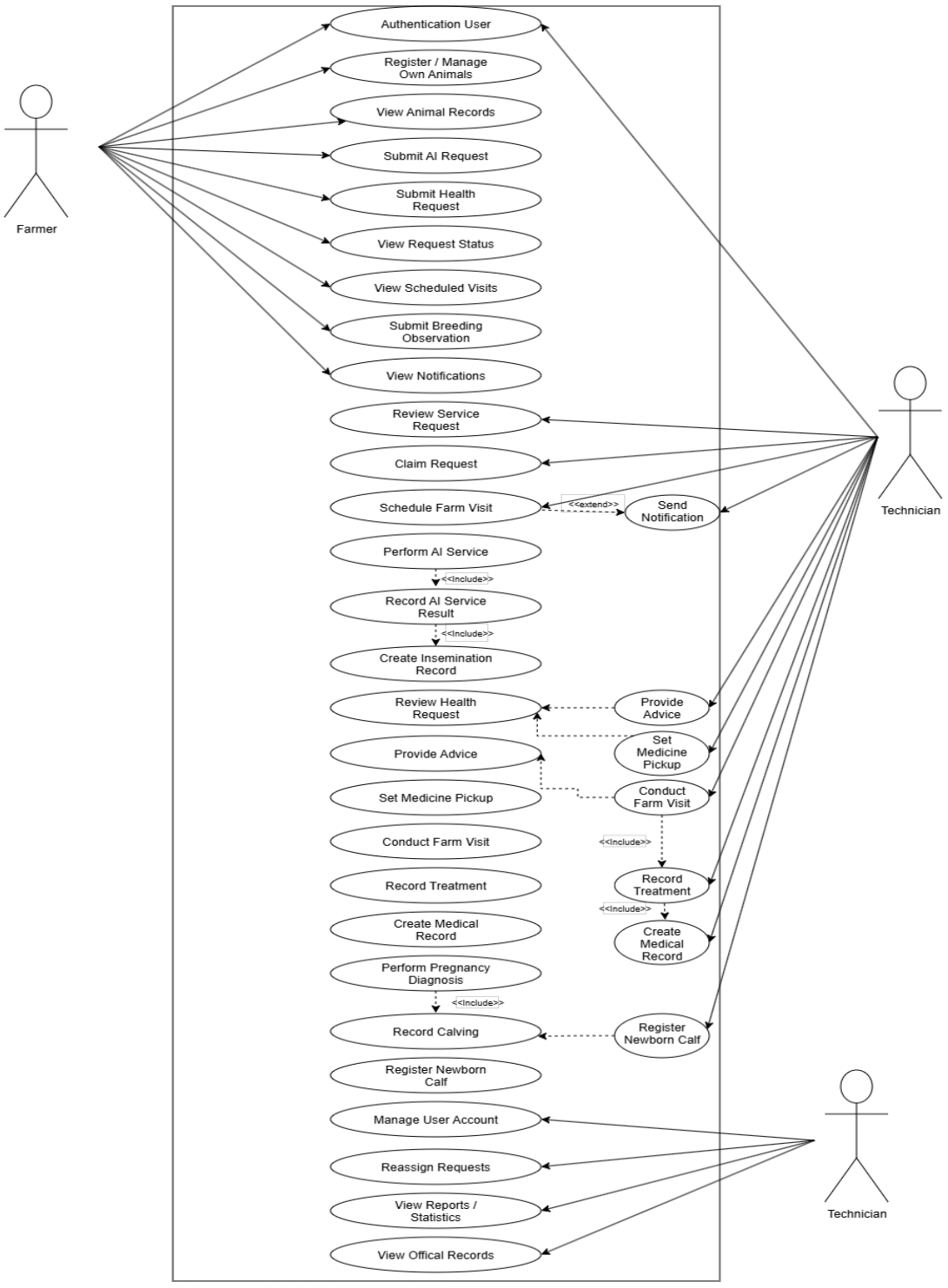


Figure 7. Use Case Diagram.

Figure 7 presents the BreedSmart Use Case Diagram, illustrating how Farmers, Livestock Technicians, and Administrators interact with the system's role-based functions for animal management, livestock services, reproductive monitoring, records, notifications, and administrative activities.

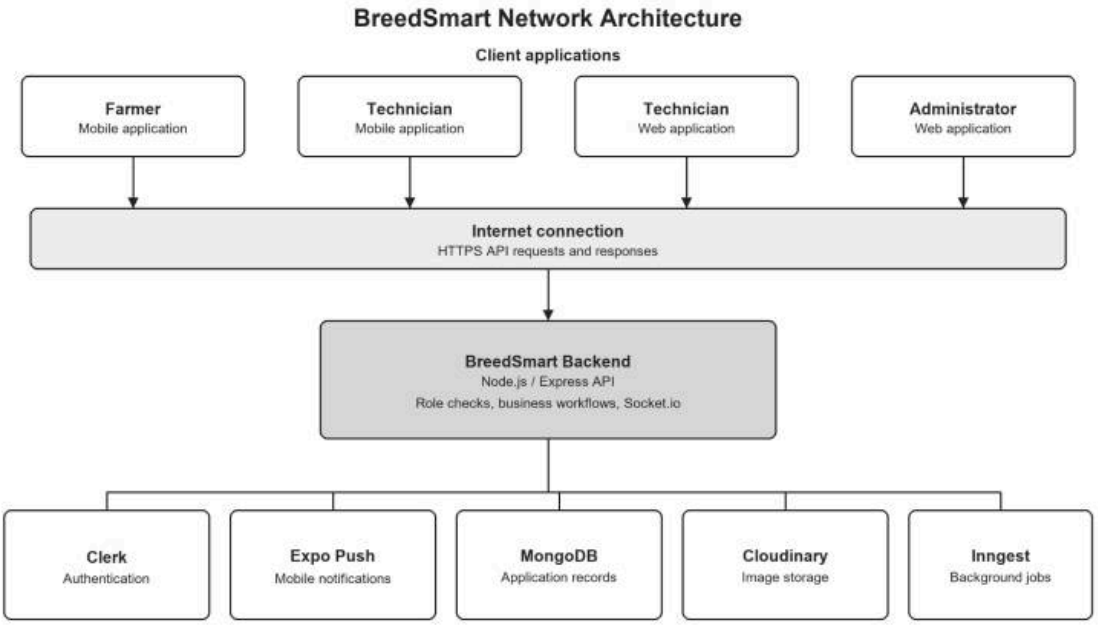


Figure 8. BreedSmart Network Architecture.

Figure 8 shows how BreedSmart's mobile and web applications connect through the internet to the backend, which manages application records and communicates with supporting services.

Connectivity and Synchronization

BreedSmart uses a centralized backend to process and store information submitted through the mobile and web applications. Client requests are validated by the backend before relevant records are stored in the database. Updates to livestock services, records, schedules, and notifications are then made available to authorized users according to their roles.

To support field activities where connectivity may be intermittent, selected mobile operations can be stored locally on the device and submitted to the server when a network connection is restored. This synchronization capability applies only to supported actions and does not provide complete offline access to all BreedSmart functions.

Workflow Design

This section describes the operational and reproductive workflows implemented in BreedSmart. A request or scheduled task does not itself constitute an official service record.



Figure 9. Request-to-Official-Record Operational Model.

Figure 9 shows how a Farmer request moves through Technician review, scheduling, and service execution. BreedSmart creates an official record only when the authorized service result is recorded; the request, assignment, and visit remain distinct workflow items.

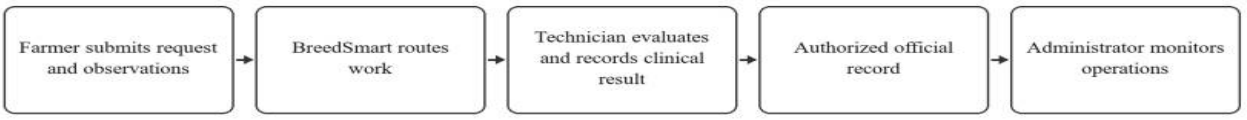


Figure 10. Responsibility and Data Authority Flow.

Figure 10 distinguishes Farmer requests and observations from Technician clinical decisions and official service records. BreedSmart routes work and retains linked data, while the Administrator monitors operations without replacing Technician authority.

Reproductive Workflow and Date Calculations

BreedSmart uses the actual date recorded for a completed artificial insemination service as the reference for subsequent monitoring. Its standard workflow begins Farmer observation on Day 18, marks Day 21 as an expected estrous-cycle monitoring point, schedules Technician follow-up on Day 25, and marks pregnancy diagnosis due on Day 60 and overdue on Day 75 in the default pathway. These are operational milestones, not exact predictions of biological events; cattle estrous cycles average about 21 days but vary [34]. Lack of observed heat does not confirm pregnancy. An authorized Livestock Technician performs and records the diagnosis; BreedSmart manages readiness, reminders, and the resulting workflow. An active method-based policy may permit earlier examination and require continuation or recheck, while absent or invalid active policy falls back to the default Day-60 pathway. Veterinary guidance recognizes method-dependent examination timing [35]. A positive diagnosis updates the linked pregnancy workflow; a negative diagnosis or Technician-verified return to heat resolves that attempt without erasing its history. BreedSmart calculates an estimated expected-calving date by adding the applicable gestation period from the animal's implemented reproductive profile to the recorded insemination date. Actual calving is recorded separately. It calculates postpartum eligibility from the actual calving date plus the applicable profile waiting period, then applies other eligibility checks; reaching that date is not a clinical declaration of full recovery.

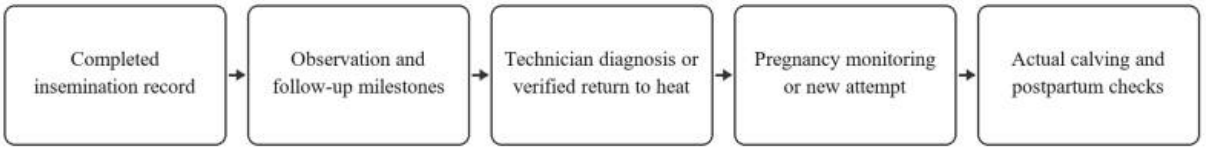


Figure 11. Complete BreedSmart Reproductive Lifecycle.

Figure 11 begins with a completed artificial insemination record and follows observation, Technician assessment, and pregnancy or return-to-heat outcomes. BreedSmart retains distinct insemination attempts and, after an authorized calving entry, moves to postpartum eligibility checks rather than treating calculated dates as clinical findings.

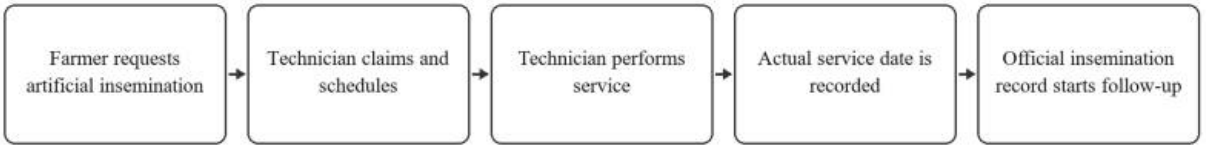


Figure 12. Artificial Insemination Request and Service Flow.

Figure 12 begins when a farmer requests artificial insemination. The Technician claims and schedules the work, performs the service, and records its actual date and result; only that completed service creates the official insemination record and follow-up milestones.

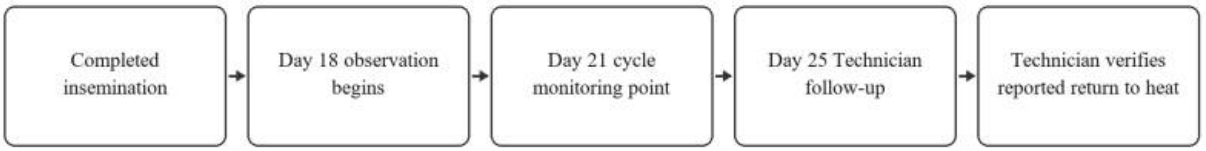


Figure 13. Breeding Follow-up and Return-to-Heat Flow.

Figure 13 shows BreedSmart's Day 18 observation, Day 21 cycle-monitoring, and Day 25 Technician follow-up milestones after a completed insemination. A Farmer may report return-to-heat signs, but a Technician verifies the outcome before the current attempt is resolved and another request can proceed [34].

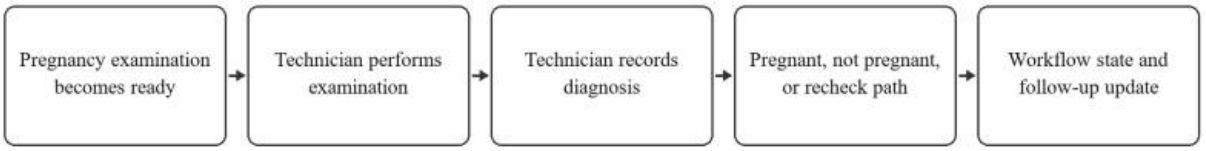


Figure 14. Pregnancy Diagnosis and Confirmation Flow.

Figure 14 starts when the applicable pregnancy policy makes an examination ready. The Technician performs and records the clinical diagnosis; BreedSmart then updates the pregnancy, not-pregnant, or recheck path and its linked reminders. The software does not diagnose pregnancy independently [35]

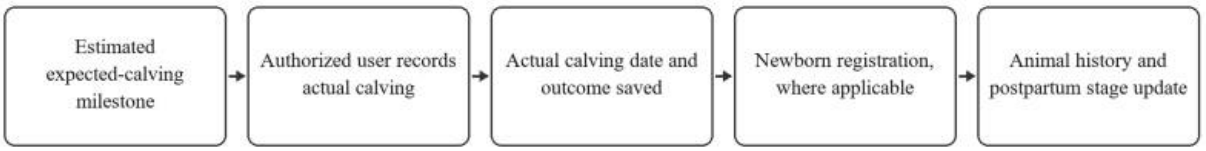


Figure 15. Calving and Newborn Registration Flow.

Figure 15 distinguishes the system-calculated expected-calving milestone from actual calving. When calving occurs, an authorized Farmer or Technician records the actual date and outcome, and newborn registration may be linked to the animal history where applicable.

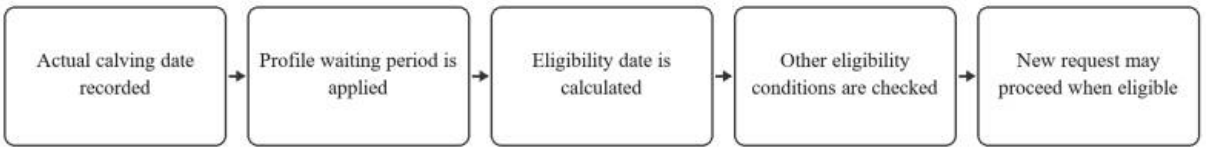


Figure. 16. Calving-to-Postpartum Eligibility Flow.

Figure 16 starts from the actual recorded calving date, not the estimated due date. BreedSmart applies the animal's reproductive-profile waiting period and other eligibility rules before a subsequent artificial insemination request can proceed; the calculated date does not certify clinical recovery.

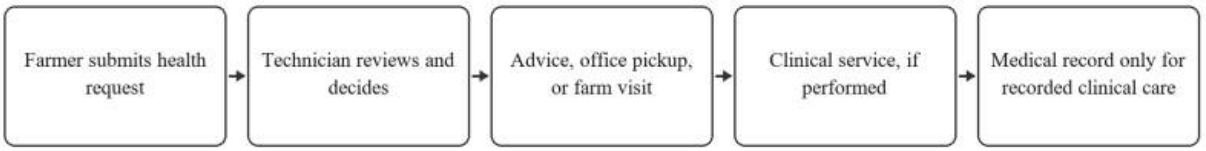


Figure 17. Health Request Review and Handling Flow.

Figure 17 begins with a Farmer health request and a Technician review. The Technician may provide advice, arrange office medicine pickup, or schedule a farm visit; an official MedicalRecord is created only when clinical care is actually performed and recorded, not merely when the request is submitted.

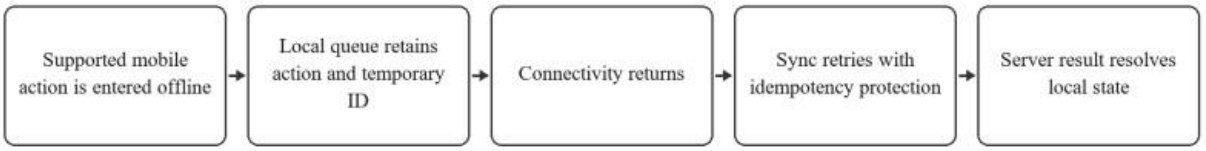
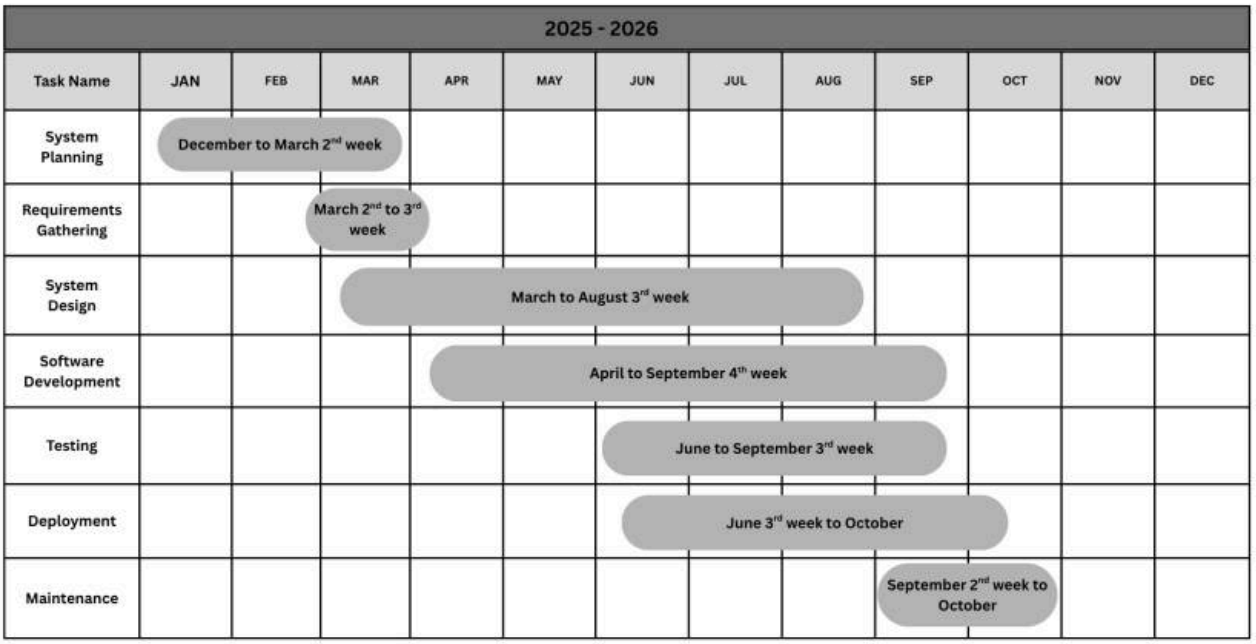


Figure 18. Offline Queue and Synchronization Flow.

Figure 18 shows a supported mobile action retained locally when connectivity is unavailable and retried when a connection returns. BreedSmart resolves temporary local identifiers and uses idempotency protection during synchronization; this limited queue does not make every mobile function available offline.

Project Gantt Chart (Weekly Estimation)

The timeline spans 21 weeks, from December 10, 2025, to May 2, 2026.



Implementation Detail and Record Integrity

The implementation separates interface presentation, API authorization, workflow validation, and persistence. Mobile and web clients submit operations to the same Node.js and Express backend. The backend resolves the authenticated account, checks its role and record authority, and applies workflow rules before updating MongoDB documents. This arrangement places operational rules in a shared service layer, while allowing the mobile and web interfaces to present tasks in forms suited to their users.

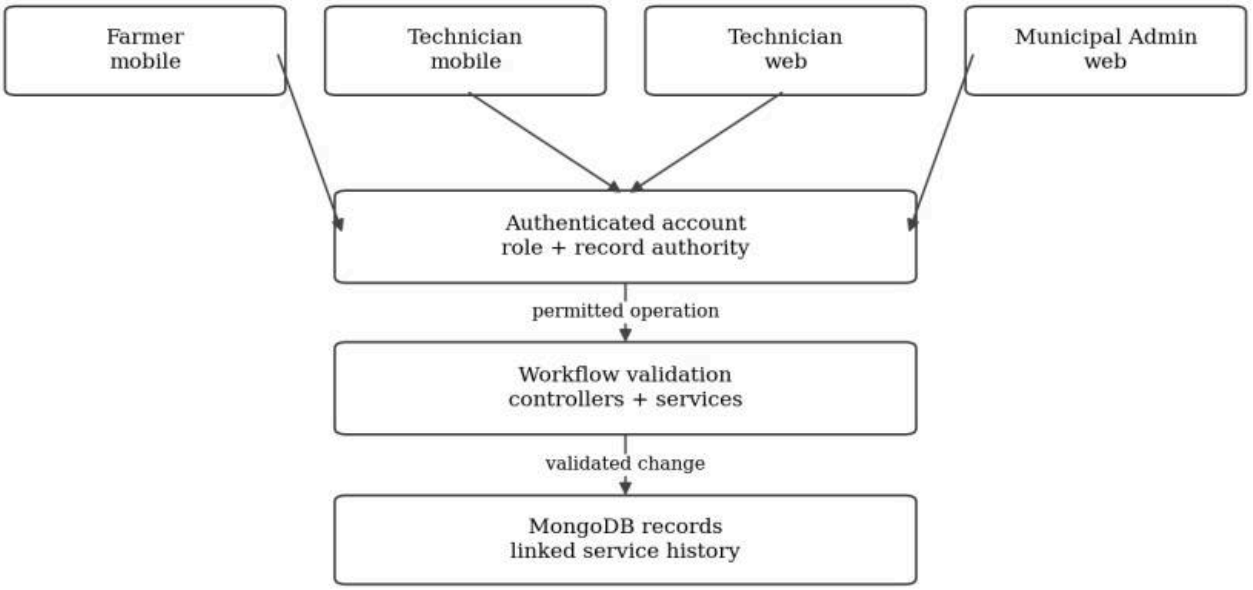


Figure 19. Client, authorization, workflow, and persistence boundaries.

Figure 19 expands the component view by identifying the boundary at which an interface action becomes an authorized server operation. Farmer mobile, Technician mobile, Technician web, and municipal Admin web interfaces share the backend but have different permitted actions. Authentication identifies the account; subsequent checks determine whether the account may act on the requested record. The diagram represents the reviewed source implementation and does not certify the security of an untested deployment.

Table 2. Implementation components and responsibilities

Component	Responsibility	Verified source area
Mobile application	Role-specific navigation, input, cached information, and supported queued mutations.	mobile/features; mobile/lib/offlineQueue.ts
Web application	Technician workspace and municipal oversight interfaces.	web/src/pages/technician; web/src/pages/admin
Authentication and authorization	Bind the authenticated account and enforce role and record authority.	backend/src/middleware/auth.middleware.js; backend/src/policies
Workflow controllers and services	Validate requests and coordinate related record changes.	backend/src/controllers; backend/src/services

Component	Responsibility	Verified source area
MongoDB models	Represent linked users, animals, services, tasks, and records.	backend/src/models
Notification delivery	Create eligible notifications and dispatch supported delivery channels.	backend/src/services/notification-delivery.service.js

Record Relationships and Canonical Identity

An Animal document provides the shared subject for reproductive and health information. User references identify the Farmer and, where applicable, the responsible Technician. Insemination records may begin as requests and later contain completed service information. Pregnancy records link the clinical confirmation to the associated insemination. Calving records retain the dam and available reproductive links, while living offspring are represented through linked Animal records. HealthRequest and MedicalRecord documents separate the reported concern from the officially recorded clinical service.

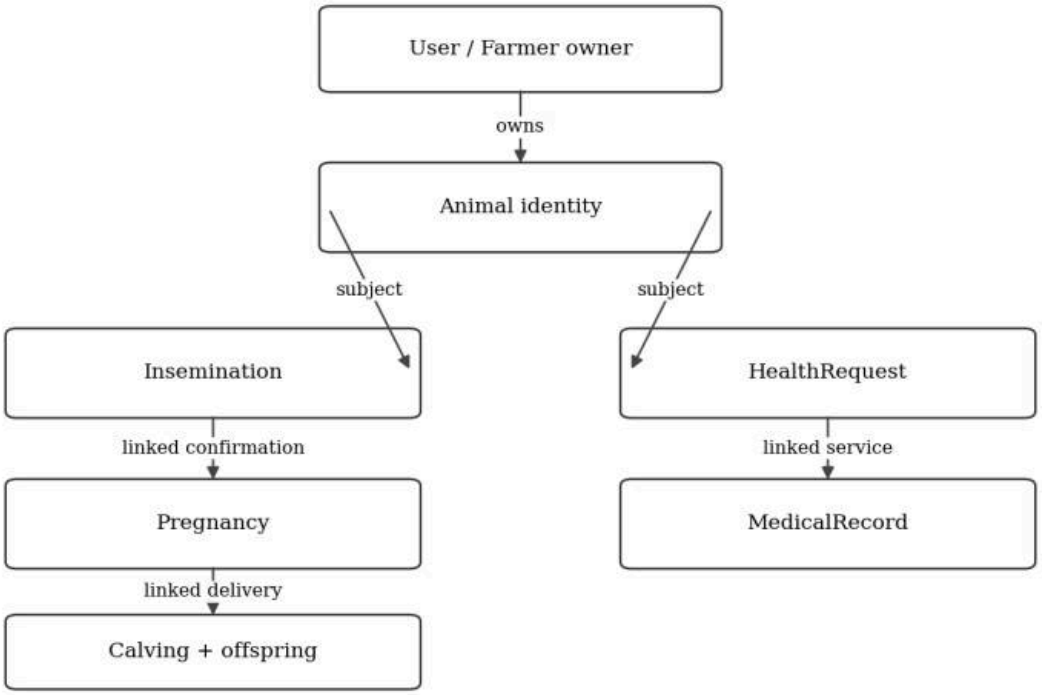


Figure 20. Principal record relationships in reproductive and health workflows.

Figure 20 presents the principal relationship paths rather than every schema field. These relationships allow a user to open the same animal from a request, service record, or historical view. A temporary mobile identifier has a different purpose: it identifies an object before successful synchronization. It must be resolved to the server identifier before a dependent operation can become a valid central record. The presence of several interface cards for one animal therefore does not mean that several Animal documents should be created.

Table 3**.** Core records and their meaning

Record	Meaning in the workflow	Key relationship
User	Authenticated Farmer, Technician, or Administrator account.	Clerk identity and local role; Farmer and Technician references.
Animal	Registered livestock identity and current stored attributes.	Farmer owner; links from service records; maternal lineage where recorded.
Insemination	AI request or completed insemination service according to its lifecycle.	Animal, Farmer, and responsible Technician.
Pregnancy	Clinical diagnosis and pregnancy-cycle information.	Animal and associated insemination; confirming Technician.
Calving	Actual delivery and recorded birth outcomes.	Dam, available pregnancy/insemination links, and live offspring.
HealthRequest	Farmer concern or medicine-related request and handling progress.	Animal, Farmer, and assigned Technician.
MedicalRecord	Officially recorded health service or treatment.	Animal, Technician, and optional originating HealthRequest.
Task	Assigned or discoverable work linked to a workflow.	Responsible Technician and related source record.
Notification / AuditLog	User-facing event information / administrative activity history.	Recipient or actor and the relevant event or entity.

Insemination Requests and Actual Service Recording

The insemination lifecycle distinguishes pending, scheduled, in-progress, done, cancelled, and rejected records. A Farmer request contains information for review; it is not proof that an insemination was performed. Technician claiming and scheduling establish responsibility and an intended visit period. Completion uses the actual service fields, including the insemination date and time, estrus type, sire information, semen doses, and Technician note when provided. The backend validates the relevant transition and required service information.

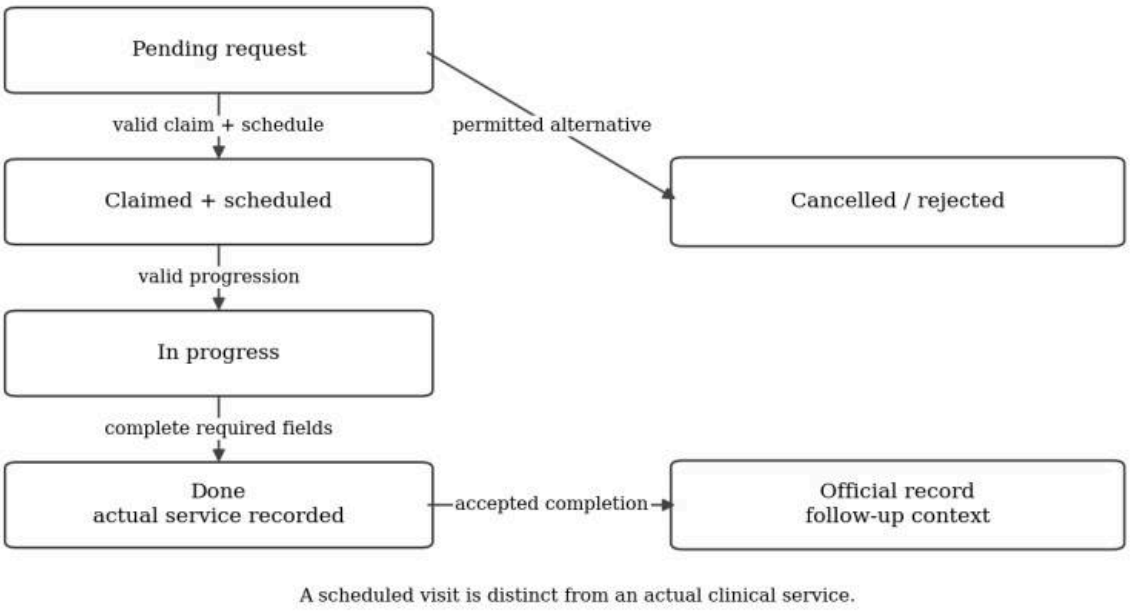


Figure 21. Insemination request lifecycle and service-record boundary.

Figure 21 places the official service boundary at validated completion. Cancellation and rejection are alternative request outcomes rather than successful services. A walk-in service enters through a dedicated recording path because it may not have begun as a Farmer request. A previous insemination entry likewise has its own context for historical recording and continued tracking. These routes should not be merged in the manuscript into a single unrestricted form because their purpose and validation differ.

Table 4. Insemination information and workflow use

Information	Workflow use	Interpretation
Animal and Farmer references	Identify the service subject and owner.	Must refer to the intended registered animal.
Scheduled date and visit period	Represent an intended visit.	Different from the actual time of insemination.
Actual insemination date and time	Record when the service occurred.	Basis for subsequent reproductive monitoring.
Estrus type	Record Natural, Synchronized, or Induced context.	A stored service descriptor, not an automatically diagnosed condition.
Sire breed and sire code	Identify recorded semen/sire information.	Supports traceability of the completed service entry.
Semen doses used	Record the quantity used for the service.	Requires valid input according to the recording workflow.
Technician note	Retain additional service context when supplied.	Does not replace required official service fields.

Health Handling and Official Treatment Records

Health handling is based on the reported concern and the Technician's chosen response. The current vocabulary groups health concerns and medicine requests and distinguishes advice, office pickup, and visit handling. The Farmer's description remains separate from the Technician's findings. A response that gives advice or pickup instructions should not be described as an on-site clinical treatment. An official MedicalRecord is created through the corresponding service-recording path when clinical information is submitted.

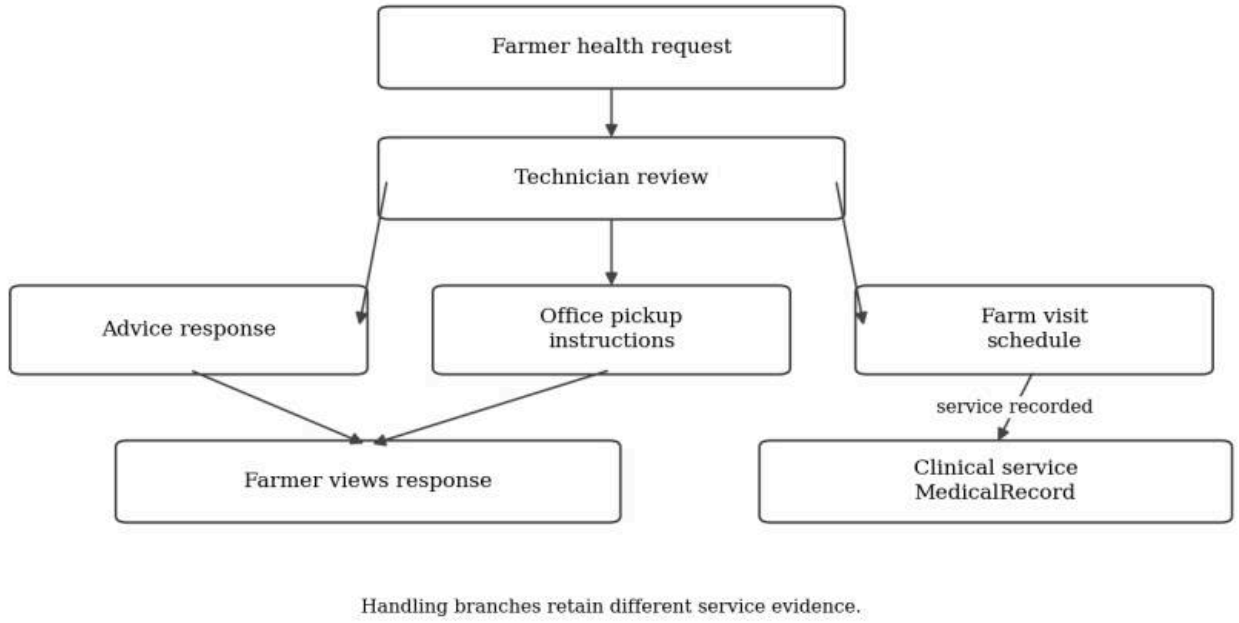


Figure 22. Health request handling and clinical-record branches.

Figure 22 shows the health diagram shows a common review point followed by distinct handling branches. A visit branch introduces scheduling and service recording; advice and office pickup communicate the chosen response without inventing a farm visit. Follow-up work remains linked to the relevant workflow. The implementation should be evaluated using each branch, including invalid transitions and missing information, because a successful visit test does not verify the advice and pickup paths.

Table 5. Health handling modes and evidence requirements

Handling mode	Recorded interaction	Evidence needed for evaluation
Advice	Technician response communicated to the Farmer.	Request before response; saved response; corresponding Farmer view.
Office pickup	Instructions for medicine-related office handling.	Pickup instructions and resulting request status; confirmation fields where used.
Farm visit	Visit scheduling and, after service, clinical recording.	Schedule, actual service submission, and linked MedicalRecord.
Direct health recording	Technician records a service outside an existing Farmer request.	Selected Farmer/animal and the completed official record.

Handling mode	Recorded interaction	Evidence needed for evaluation
Follow-up	Additional work associated with the relevant service.	Source record, follow-up work item, and subsequent completion evidence.

Calendar Dates, Visit Periods, and Clinical Timing

A visit consists of a Philippine calendar date and a morning or afternoon period. The scheduling domain rejects invalid dates and past calendar days. For date-only storage it anchors the selected day at noon in Asia/Manila; this is a storage convention rather than an appointment at noon. Same-day scheduling also checks whether the selected period is still available and whether explicit confirmation is required late in that period. Actual clinical service time is recorded separately.

Pregnancy readiness is governed by a configurable policy. The policy identifies enabled examination methods, applicable earliest intervals after insemination, species overrides, and continuation-recheck requirements. A fallback policy is available when a valid enabled configuration is absent. The manuscript should therefore describe method-dependent readiness rather than claim that every diagnosis always occurs on one universal day. Configuration and policy support do not establish that a method was performed during the municipal pilot.

Table 6**.** Time-related fields and their distinct meanings

Field or rule	Meaning	Reporting boundary
Preferred date	Farmer’s requested timing, where supplied.	Does not confirm a Technician assignment.
Scheduled date + visit period	Accepted visit day and morning/afternoon period.	The date-only noon anchor is not an appointment time.
Actual service date/time	Time recorded for the completed clinical service.	Use this for service-history interpretation.
Days after insemination	Interval used for reproductive readiness and monitoring.	Depends on actual AI date and applicable policy.
Expected calving date	Calculated monitoring estimate from reproductive data.	Must remain distinct from actual calving.
Actual calving date	Recorded delivery event date.	Basis for the applicable postpartum eligibility calculation.
Task due date	Date assigned to follow-up work.	A due task does not demonstrate that its clinical activity occurred.

Table 7**.** Pregnancy confirmation and continuation responsibilities

Stage	Required distinction	Responsible authority
Farmer observation/report	Evidence or concern for review; not a clinical diagnosis.	Farmer submits within the authorized animal context.

Stage	Required distinction	Responsible authority
Task or record selection	Select the linked animal and reproductive event.	Technician subject to assignment and ownership checks.
Readiness assessment	Apply the enabled policy and method-specific interval.	Backend policy; clinical method chosen by authorized personnel.
Diagnosis recording	Record the supported clinical result and confirmation details.	Technician clinical workflow.
Pregnancy update	Link confirmation, animal state, and reproductive cycle.	Validated backend service operation.
Continuation or recheck	Record further confirmation where the policy requires it.	Technician continuation workflow; linked source pregnancy.

Calving Transactions and Offspring Registration

Calving recording joins several related changes: the actual delivery, birth outcome, reproductive cycle, dam state, and living offspring references. The calving service coordinates these changes through the livestock transaction mechanism. Living offspring and non-living birth outcomes are handled differently, avoiding the assumption that every recorded calf becomes an active animal. The calving outcome may represent live birth, stillbirth, abortion, or mixed delivery as supported by the current model and recording context.

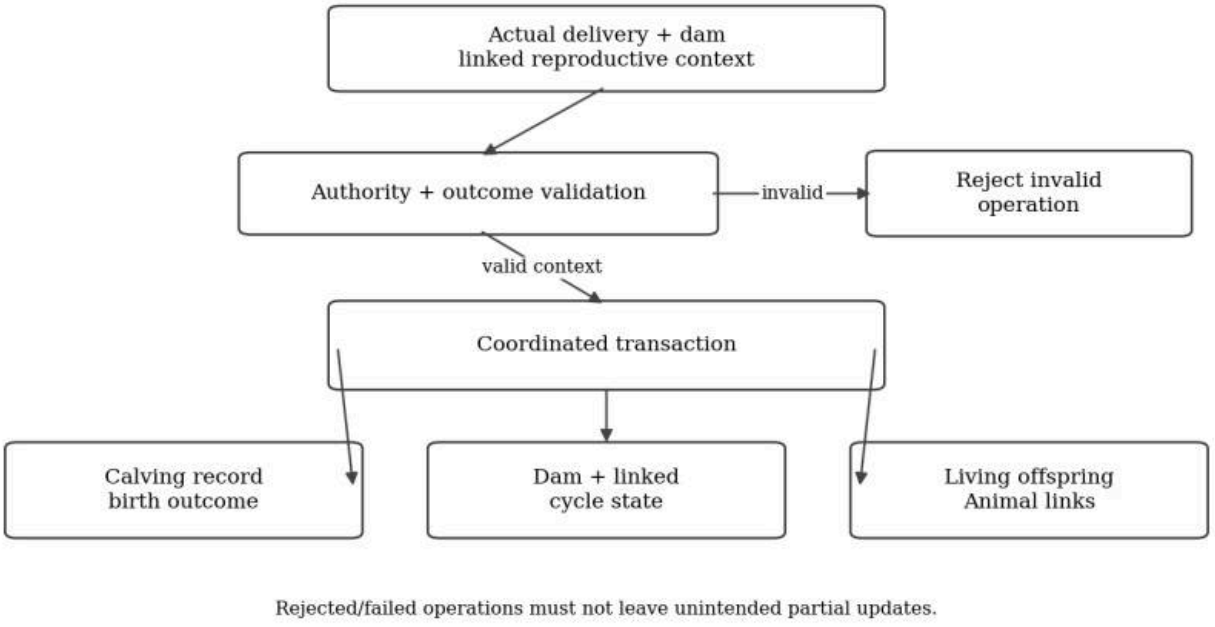


Figure 23. Calving validation and related transaction updates.

Figure 23 identifies the changes that must remain consistent when a calving operation succeeds. A rejected or failed transaction should not leave only part of the delivery stored. The figure is an implementation explanation, not a result of a fault-injection test. Verifying transaction rollback requires an executed test with before-and-after database evidence. Likewise, a calculated postpartum waiting interval is an information-system eligibility rule and does not certify that an

animal has clinically recovered.

Supported Offline Operations and Synchronization

The mobile offline queue retains supported mutations for an identified account. Each queued item includes an operation identity, status, retry information, and the authoritative owner. The implementation rejects queuing without an owner identifier. It resolves temporary references for dependent records, caches supported attachments locally, and submits operations through the normal authenticated API path. An operation saved locally remains pending until the server accepts it.

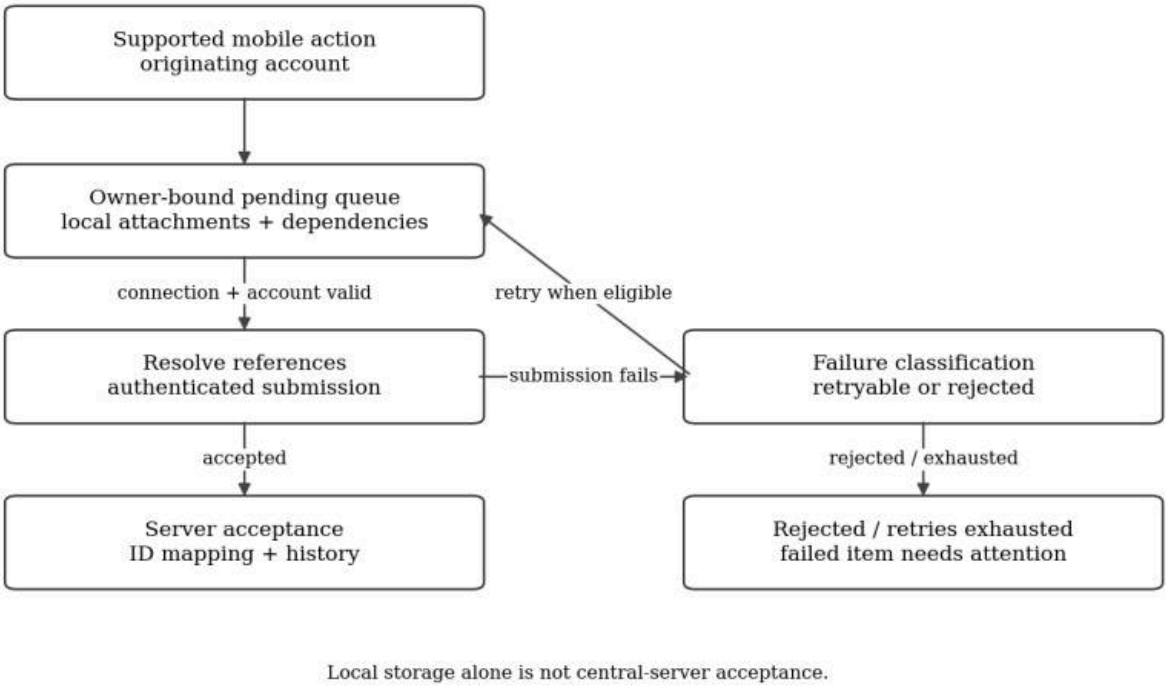


Figure 24. Account-bound queue, dependency resolution, and retry flow.

Transient failures may remain pending for retry, while non-retryable errors or exhausted retry attempts produce a failed item requiring attention. Dependencies matter: an insemination that references a locally created animal cannot become a valid central record until the parent animal has synchronized and its temporary identifier has been resolved. The queue also checks account continuity while processing. These mechanisms support selected field operations and do not mean that every screen, authentication step, or workflow is available without connectivity.

Table 8. Offline mechanisms and verification scenarios

Mechanism	Implemented purpose	Scenario to verify
Owner-bound queue	Keep queued operations associated with the originating account.	Switch accounts while one account has unsynchronized work.
Idempotency key	Allow the server to recognize repeated submission of one operation.	Retry after a response is lost; check that no duplicate record is created.
Temporary references	Replace local identifiers after successful parent creation.	Create animal locally, then submit dependent service information.

Mechanism	Implemented purpose	Scenario to verify
Attachment cache	Retain supported image payloads for later submission.	Queue with attachment, restart app, and synchronize.
Dependency ordering	Wait for required earlier changes and propagate unresolved failure.	Fail a parent operation and inspect the dependent item.
Retry classification	Distinguish retryable connection/server failures from rejected input.	Compare a transient network failure with a validation rejection.
Sync history	Retain information about processed operations.	Compare a successful server record with the corresponding history item.

Verification of Shared Rules Across Clients

Web and mobile may differ in navigation and presentation while still relying on the same backend rule. Verification should compare equivalent operations using the same test animal and workflow context. The comparison should cover permitted actions, mandatory inputs, resulting status, and official record retrieval. Screenshots alone verify presentation; API responses and resulting records establish whether the operation was accepted. Neither a web-only test nor a mobile-only test should be generalized to both clients.

The implementation review is pinned to the repository commit e178bf819ea8164348f726b84f4d0eb63d8554e2. A test performed on another commit or installed APK must identify its own version. This is particularly important because the current branch includes changes after the latest published release. Development-branch behavior should not be attributed automatically to the published APK. The configuration appendix provides the fields needed to identify the build, backend environment, and test device without exposing credentials.

D. Development Tools

1) Hardware Requirements

Table 9. Hardware Requirements

Components	Minimum Specification	Recommended Specification
Laptop/ Desktop	Intel Core i3/ AMD Ryzen 3	Intel Core i5 / AMD Ryzen 5
Memory/ RAM	4 GB	8 GB or Higher
Storage	1366 x 768 Resolution	Full HD Display (1920 x 1080
Monitor	10 Mbps	50 Mbps
Internet Connection	4G	5G
Mobile	Android 9.0	Android 13 or Higher

2) Software Requirements

Table 10. Hardware Requirement

Software	**Purpose**
Windows 11 / Ubuntu Linux	Operating System
Visual Studio Code	Source Code Editor
MongoDB Atlas	Database Management
Node JS / Express JS	Backend Server Development
React JS / React Native (Expo)	Frontend Development
Chrome / Microsoft Edge / Firefox	Web Browser
Git / Github	Version Control
Clerk / Socket.io	Authentication & Real Time Services

E. Data Gathering Procedure

The data gathering procedures used to collect relevant information needed for the development of BreedSmart: A Web and Mobile-Based Livestock Insemination and Health Management System include:

1. Surveys/Questionnaires: Structured questionnaires were distributed to selected farmers and livestock personnel to assess current challenges related to manual record-keeping, communication delays, livestock monitoring, and accessibility of agricultural services within different barangays in Oton, Iloilo.

2. Interviews: Direct interviews and consultations were conducted with the livestock technician and selected farmers to gather detailed information about breeding cycles, insemination procedures, livestock health monitoring, and existing reporting practices.

3. Observations: The researchers conducted physical observations and reviews of existing hardbound logbooks, folders, and manual livestock records to identify issues such as data degradation, record inconsistencies, and inefficiencies in the current documentation process.

F. Testing and Evaluation Plan

Testing and evaluation are conducted in two parts: technical validation and user evaluation. Technical validation is used to determine whether implemented functions and workflows operate according to the identified system requirements. Functional testing is used for individual features, while integration and workflow testing verify interactions among the mobile application, web application, backend, database, and supporting services. Test cases compare expected and actual behavior and record pass/fail results, identified defects, and retest status. Major workflows include authentication and role-based access, animal records, artificial insemination requests and service recording, reproductive follow-up, pregnancy diagnosis recording, calving, health-service handling, notifications, and selected mobile synchronization.

User evaluation is planned during controlled pilot use after intended users have sufficient exposure to the relevant system functions. The evaluation will focus on selected ISO/IEC 25010 software quality characteristics, particularly functional suitability, usability, reliability, performance efficiency, and security. A structured evaluation questionnaire based on these characteristics will be used to gather respondent ratings and feedback. The collected responses will be summarized using

appropriate descriptive measures, while qualitative feedback will be reviewed to identify usability concerns and areas for refinement.

Technical test results and user-evaluation results will be analyzed separately. The evaluation is intended to assess the implemented system and its initial suitability for use and does not establish long-term improvements in technician efficiency, conception rates, livestock health outcomes, or overall municipal service performance. Detailed testing and evaluation results will be presented in Chapter IV.

IV. RESULTS AND DISCUSSION

A. Overview of the Developed System

BreedSmart was implemented as a web and mobile-based livestock insemination and health management system for the Municipality of Oton, Iloilo. The developed system provides role-based functions for Farmers, Livestock Technicians, and Administrators and uses a centralized backend and database to maintain livestock information, service requests, schedules, reproductive records, health records, notifications, and administrative data.

The following subsections present the implemented BreedSmart interfaces according to the user role and platform through which the major system functions are accessed.

1) Farmer Mobile Application

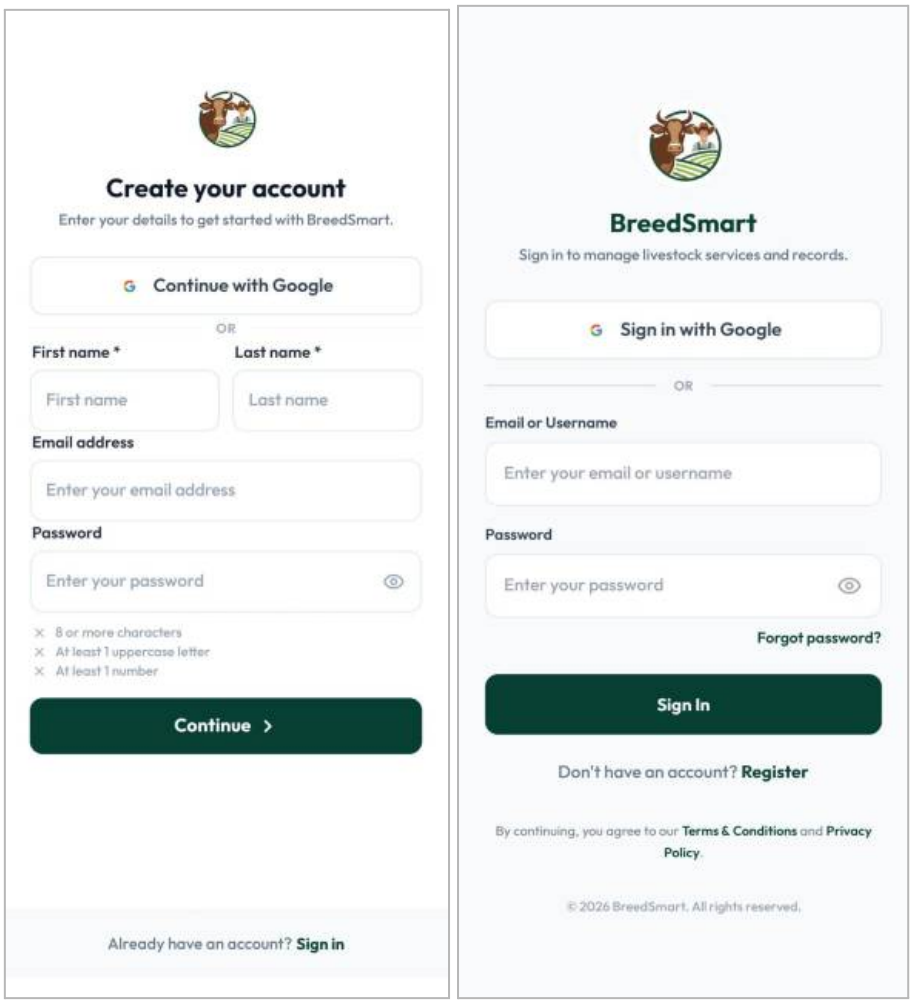


Figure 25. Farmer App Register / Login Page.

Figure 25 represents the first page to login and create a user account. It serves as the main gate that authenticates user email and password before entering the home page. The BreedSmart provides a Role-based Access Control (RBAC) that acts as the security by assigning permission to specific job roles rather than individual users.

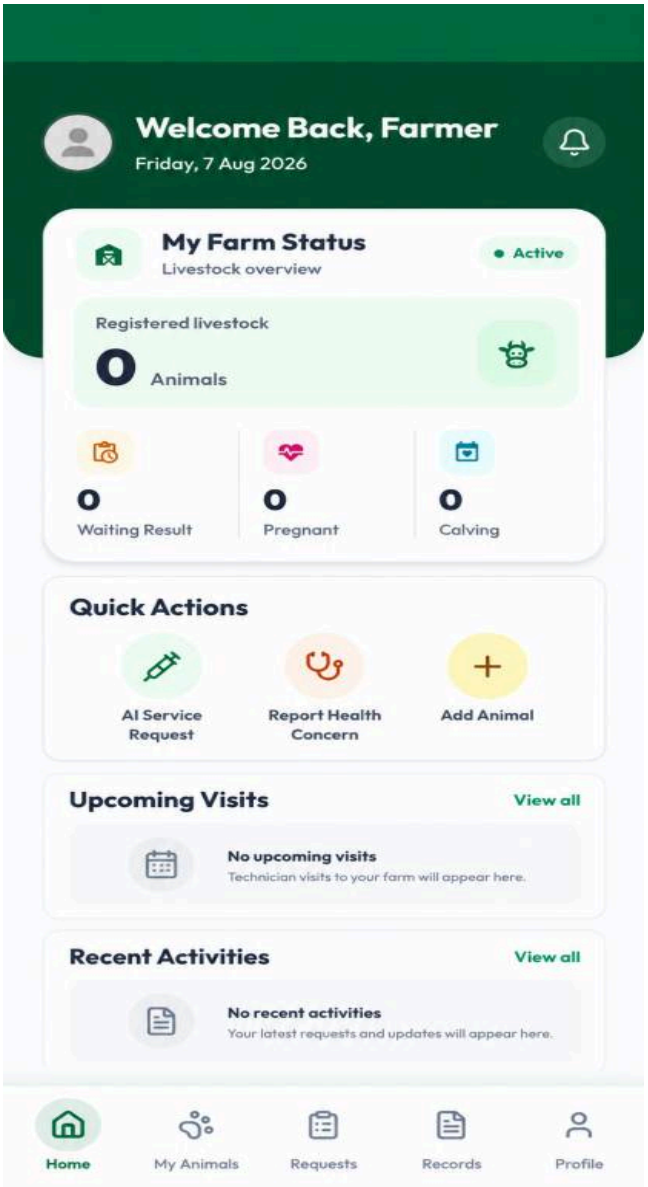


Figure 26. Farmer Home Dashboard.

Figure 26 shows the main dashboard of farmers and their exact needs when taking quick actions such as AI service Request, Report Health Concern, and Add animal. Upcoming visits and recent activities allow farmer users to notify its details and confirmation that come from technician's approval.

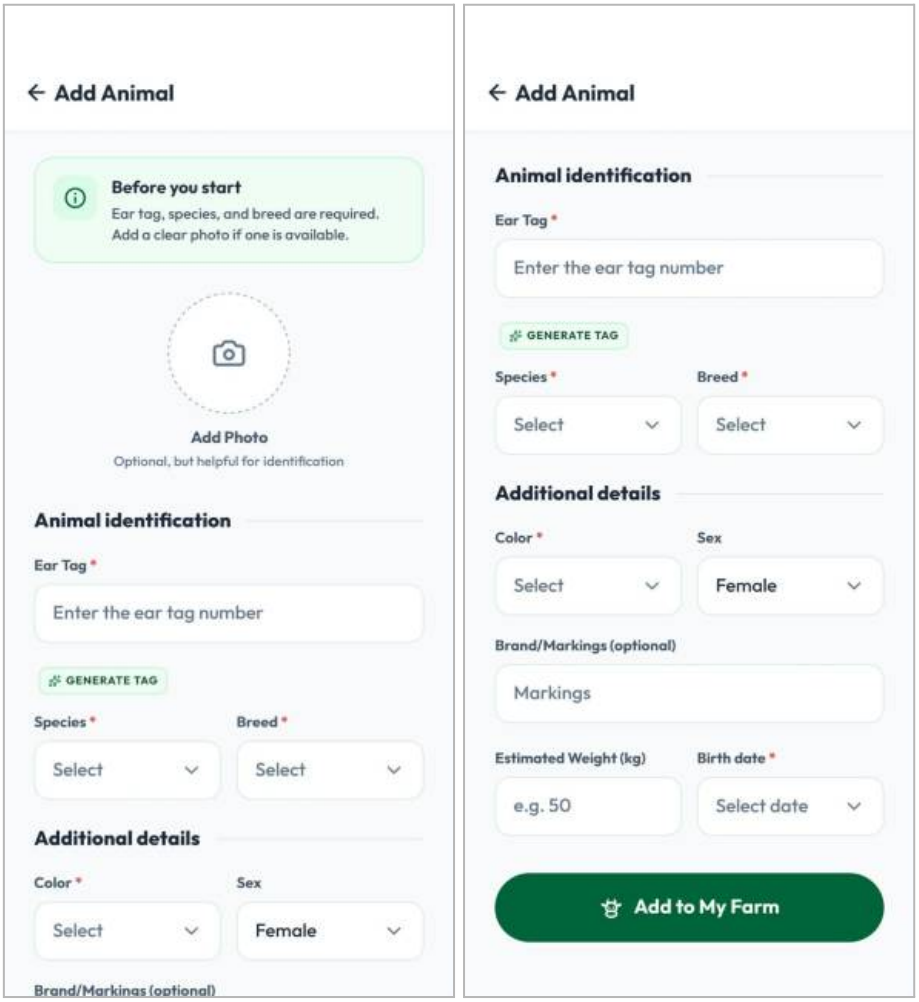


Figure 27. Farmer / Add Animal

Figure 27 presents the add animal page to register the farmer's cattle or carabao where animal identification and other details are required to be input before submitted.

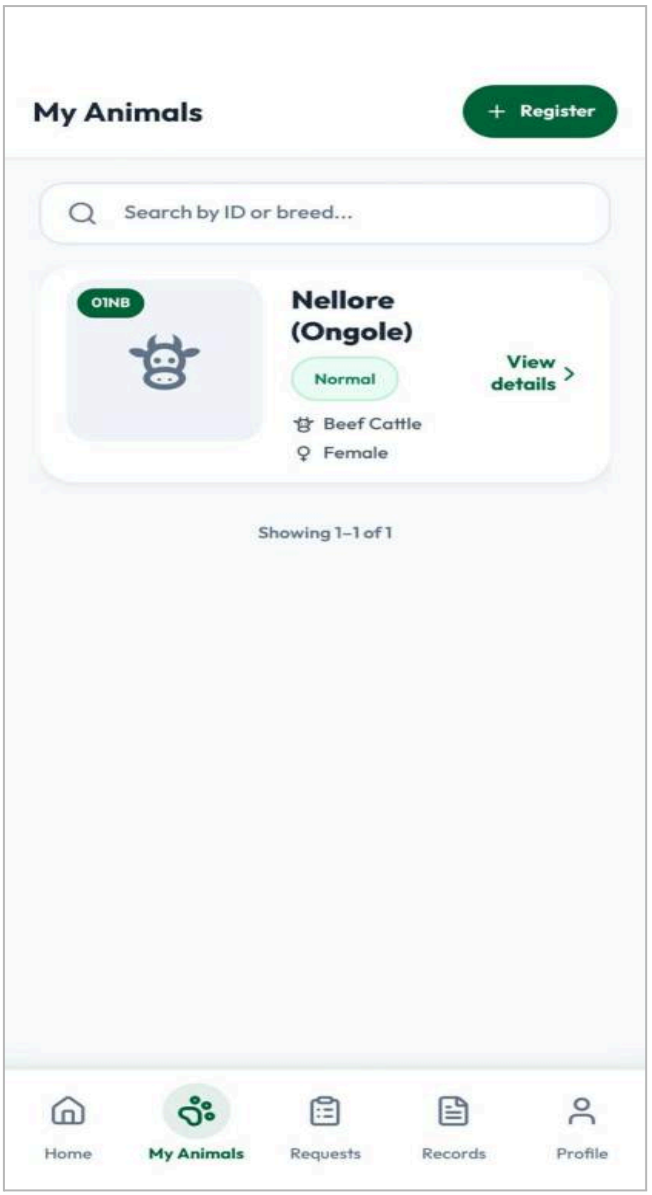


Figure 28. Farmer / My Animals

Figure 28 shows the list of the animals that the farmer registered. Each cattle and carabao have their own unique ID that acts as an identifier to different animals.

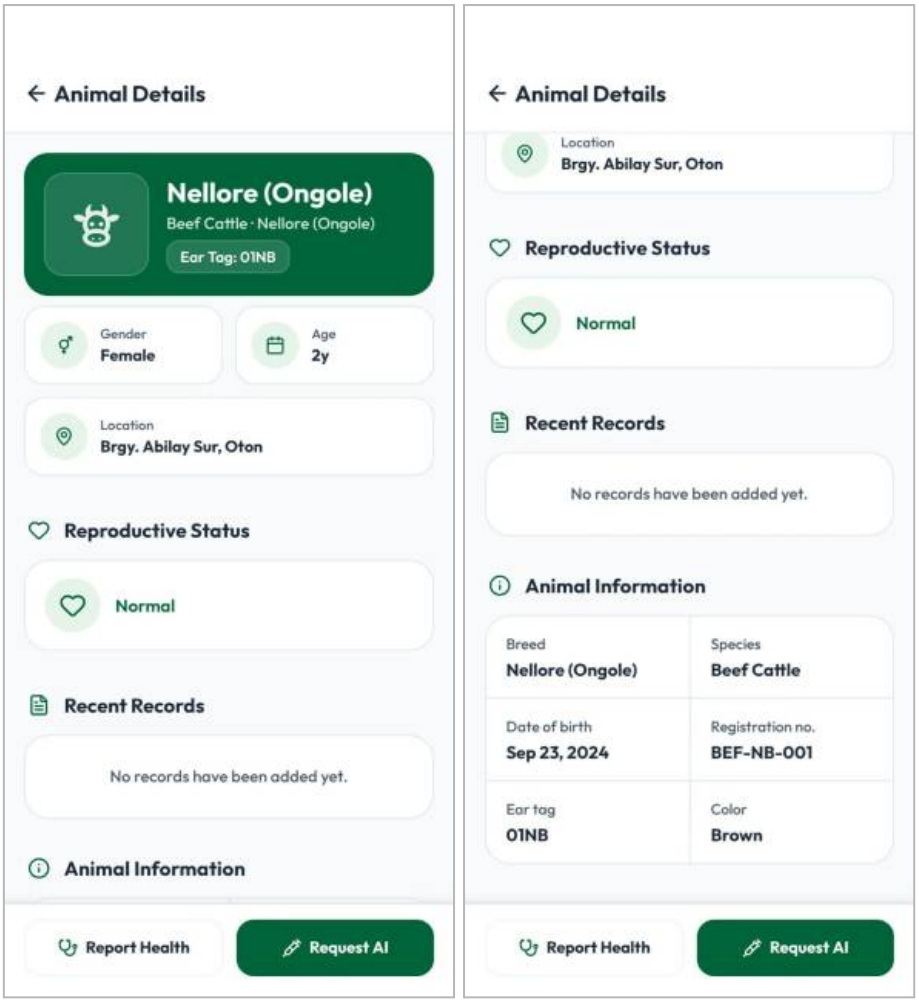


Figure 29. Farmer / Animal Details

Figure 29 provides the showcasing Animal Details. It demonstrates the application's capability to display animal's biological data, location, and operational features like health reporting and AI requests for individual profile management.

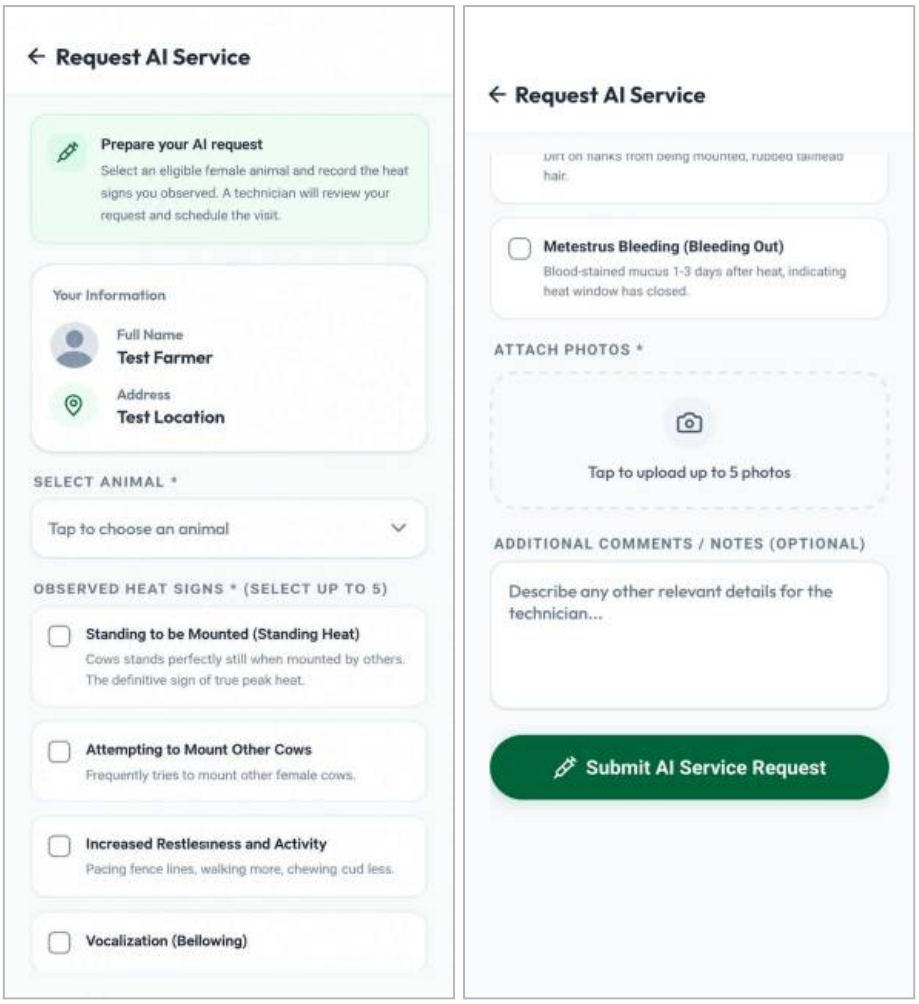


Figure 30. Farmer / Request AI Service

Figure 30 illustrates the Request AI Service interface enabling farmers to submit requests for artificial insemination services. These requests will be submitted via technician's notification to alert them of the specific animal's specific task.

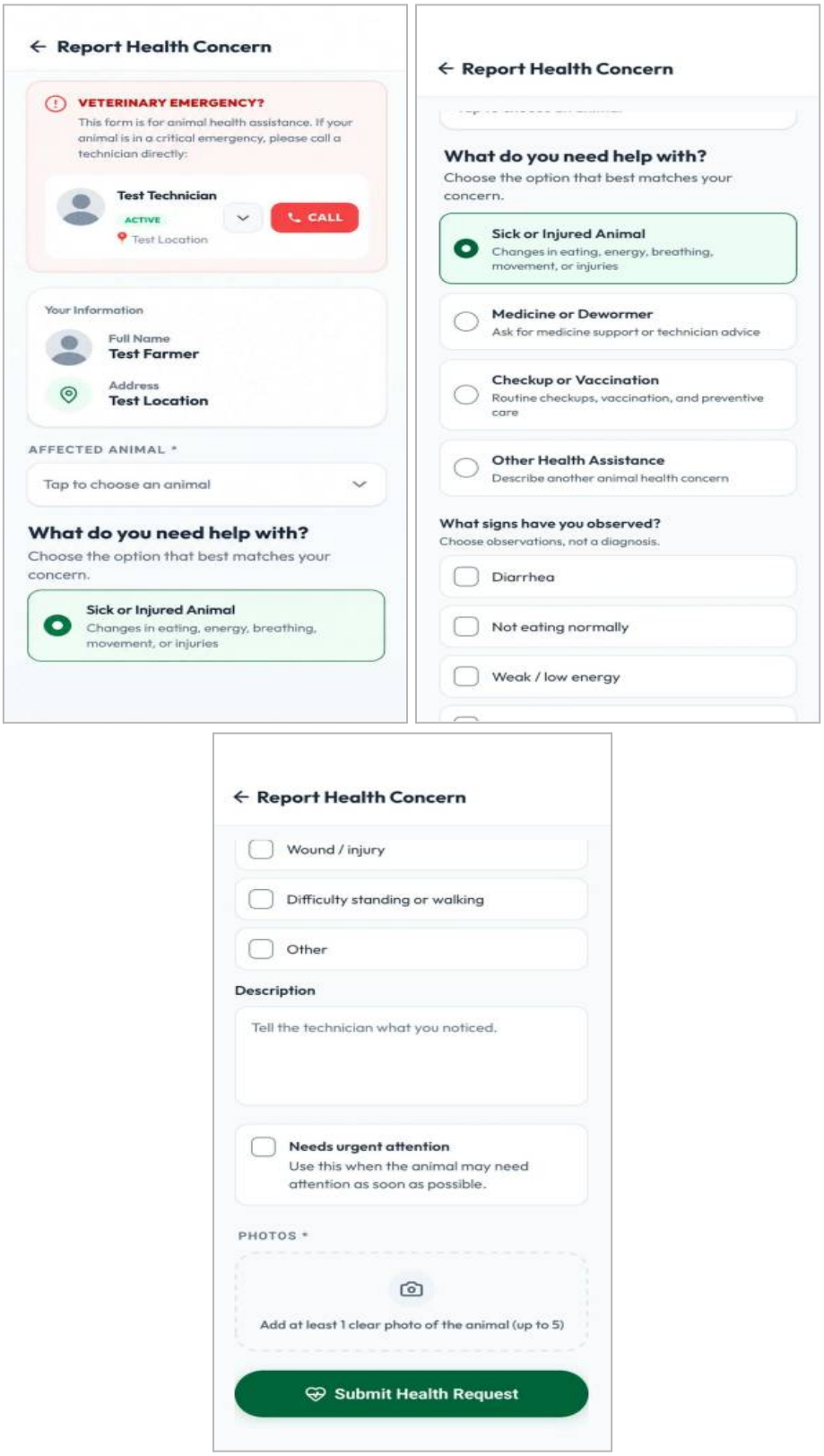


Figure 31. Farmer / Report Health Concern

Figure 31 illustrates the input field for animals with health issues and medicine-related requests to livestock technicians as a response for reminding them of animal status whether it could be scheduled or needs urgent attention as possible.

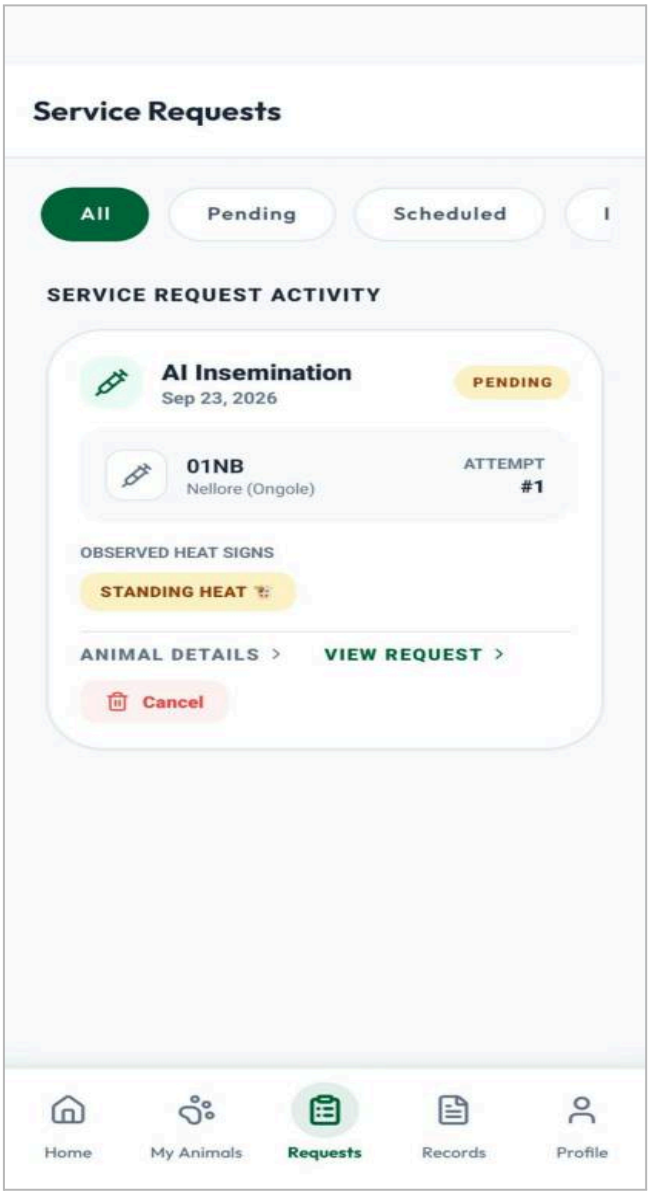


Figure 32. Farmer / Service Requests

Figure 32 tracks the real-time progress, status updates, and handling stages of submitting the AI requests. It has status that tells the request is still pending.

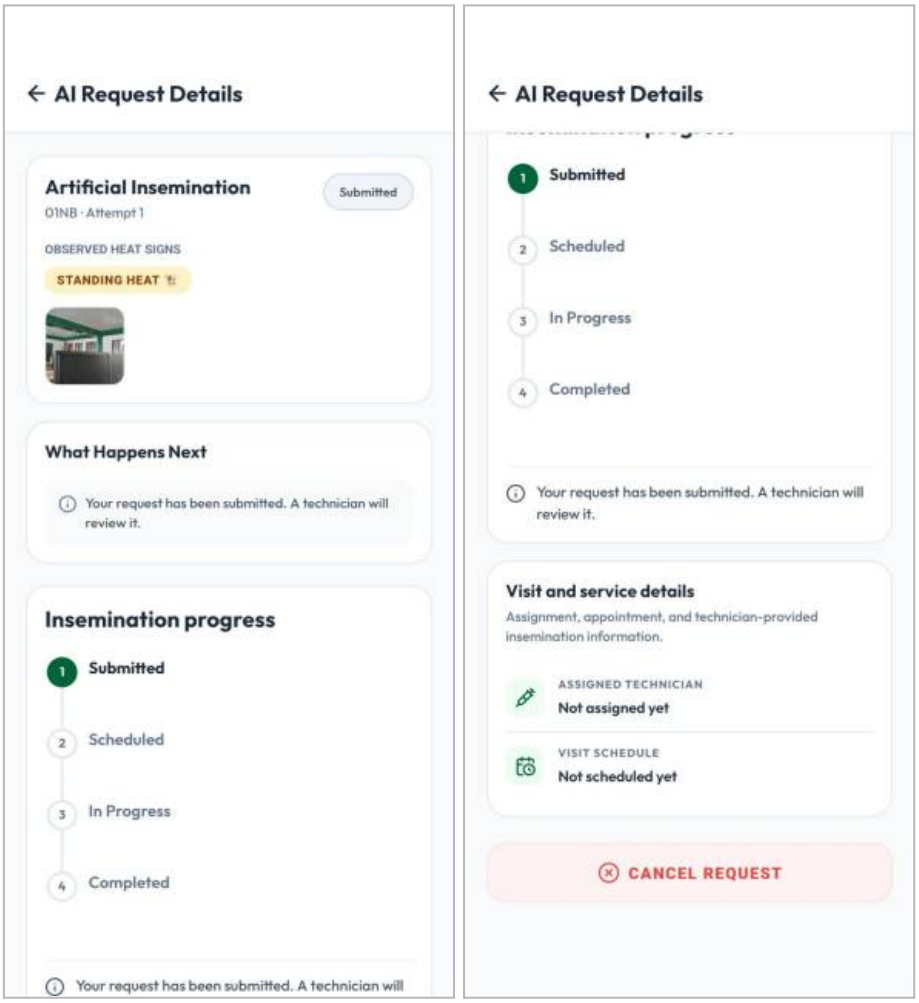


Figure. 33. Farmer / AI Request Details

Figure 33 monitors the specific lifecycle status and technician assignments for an individual artificial insemination request.

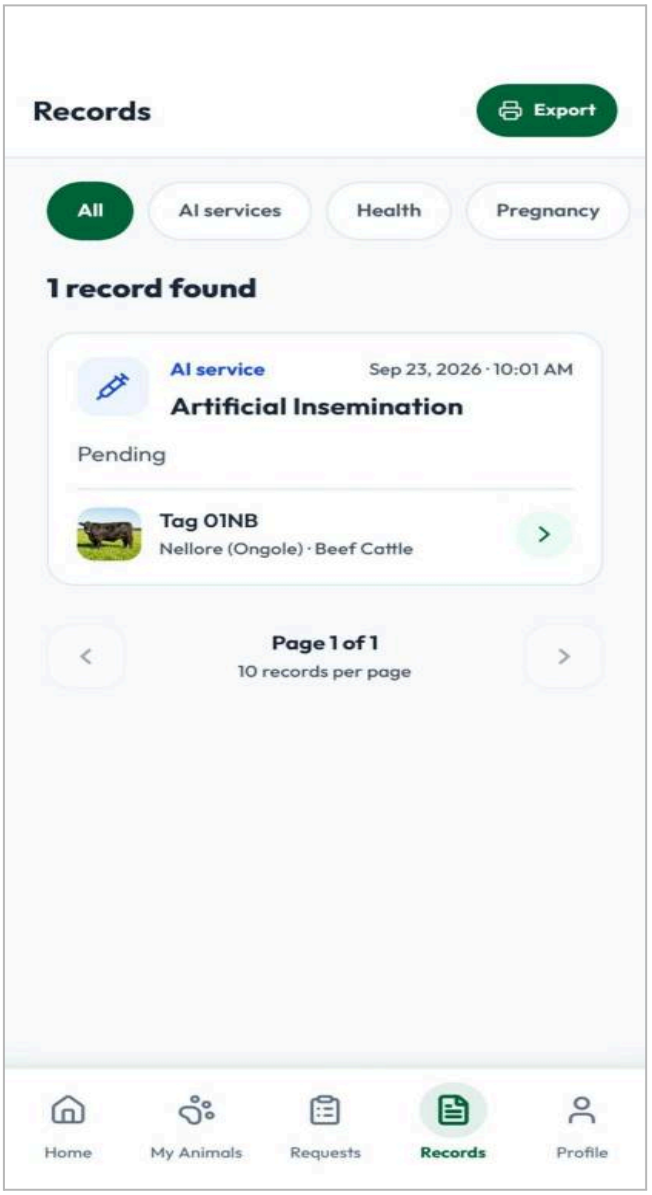


Figure 34. Farmer / Records

Figure 34 retrieves the record where all recent actions are displayed and activities consolidates which the user has the time to review of their specific pending requests.

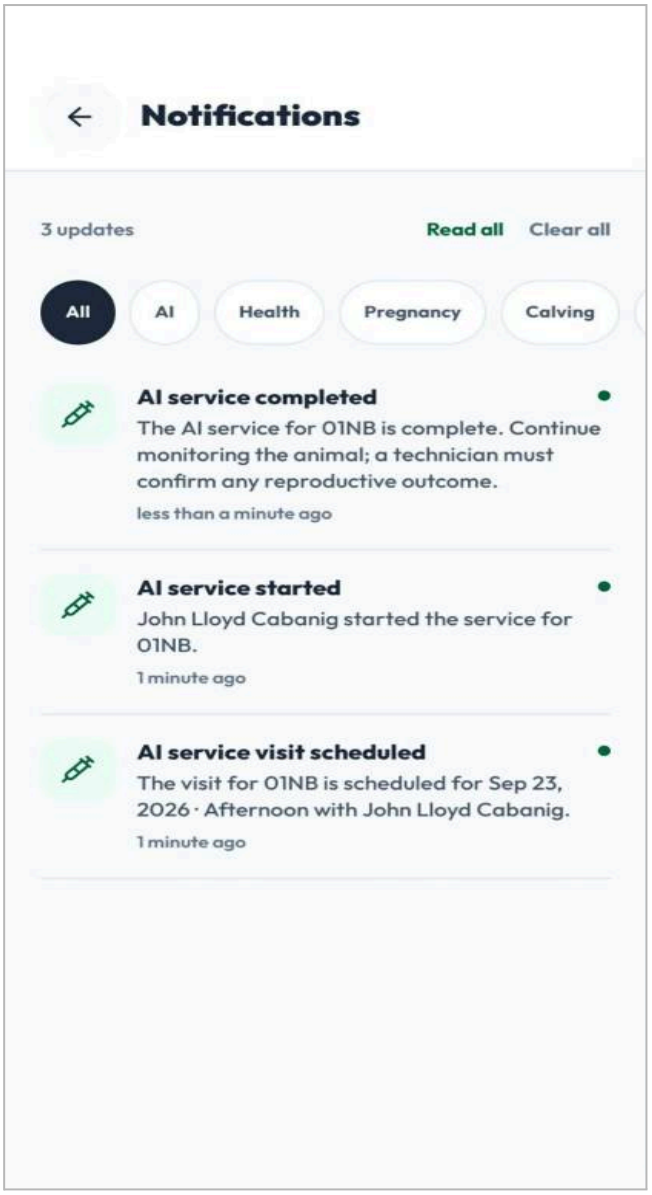


Figure 35. Farmer / Notifications

Figure 35 Delivers automated system alerts for service request updates, request confirmations, and rescheduling information are easy to review and can be also cleared to unnecessary things.

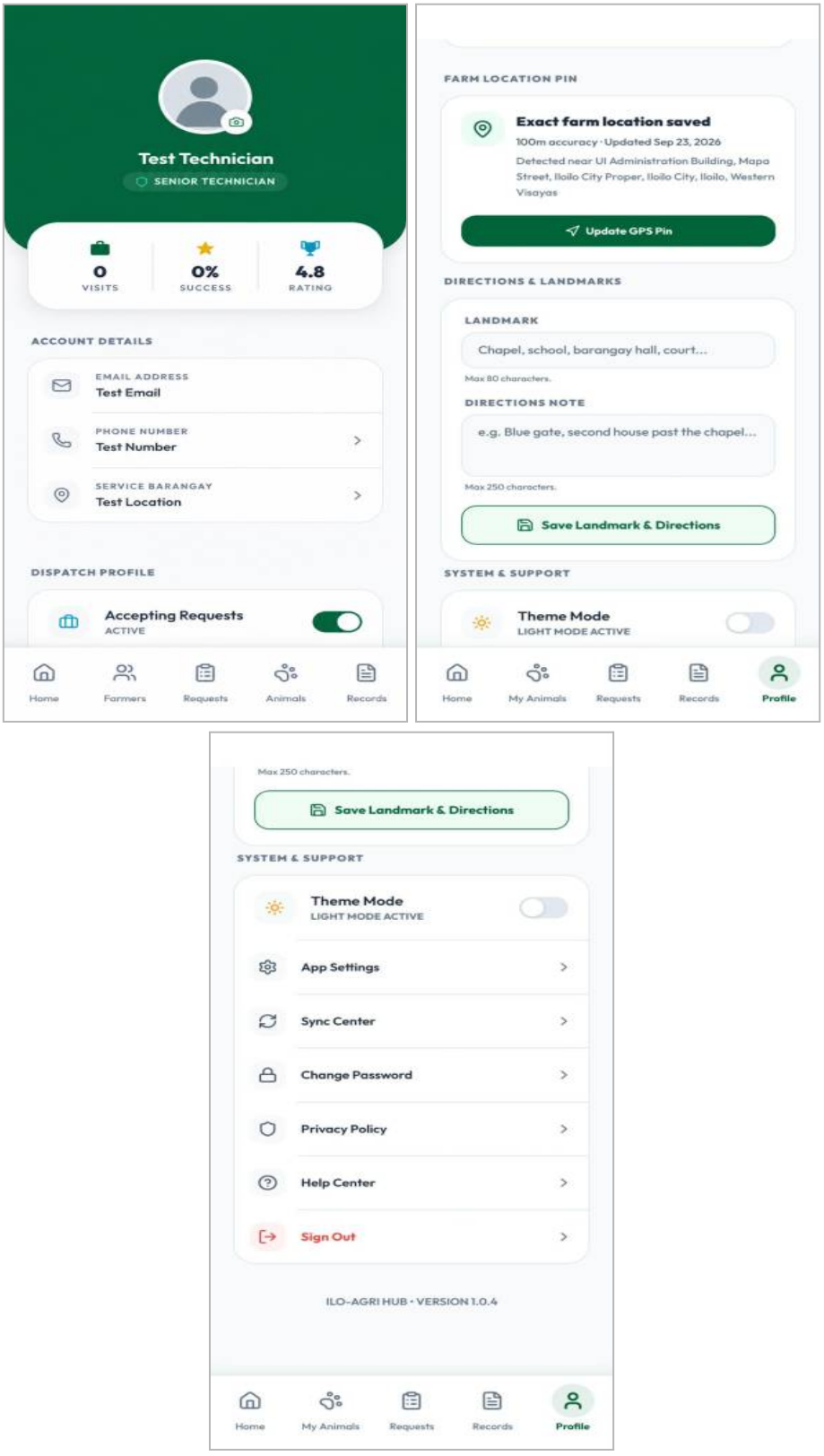


Figure 36. Farmer / Profile

Figure 36 Manages and updates personal account credentials and profile configurations within applications that can be changed such as theme color mode, language, and other functions.

2) Technician Mobile Application

The Technician mobile application supports field access to requests, assigned work, animal and Farmer information, clinical recording, and completed records. The following descriptions are grounded in the reviewed implementation. Figure positions are reserved for actual captures from the corresponding build. Demonstration data and evaluation data must be identified separately; a populated screen is evidence of presentation, not by itself evidence that a workflow passed testing.

Technician Mobile Dashboard

The dashboard brings together a work summary and the entry points used during field activity. The statistics card displays inseminations completed today, monthly inseminations, and the interface metric labeled Success (90d). Six quick actions: open insemination recording, health recording, pregnancy checking, calving recording, Farmer registration, and animal registration. This arrangement lets the Technician begin a service entry or inspect assigned work from a common starting screen.

Today's work presents a preview of relevant work items with their timing, service type, and linked context. The View all action opens the assigned-work section when additional items are available. Selecting a work item navigates according to its workflow type; pregnancy, calving, and other task-based entries require the appropriate task identifier. Farmer request previews provide entry to requests that the Technician may review. Pull-to-refresh updates dashboard data, unread notifications, and the current user profile.

The displayed Success (90d) metric is computed from recent pregnancy confirmations and insemination records in the backend dashboard calculation. Because the numerator and denominator are selected through their respective recent date fields, it is a dashboard summary rather than a matched-cohort conception-rate estimate. Its value must not be reported as the study's effectiveness result. Any screenshot used in the manuscript should show the displayed value faithfully and identify it as demonstration or operational data.

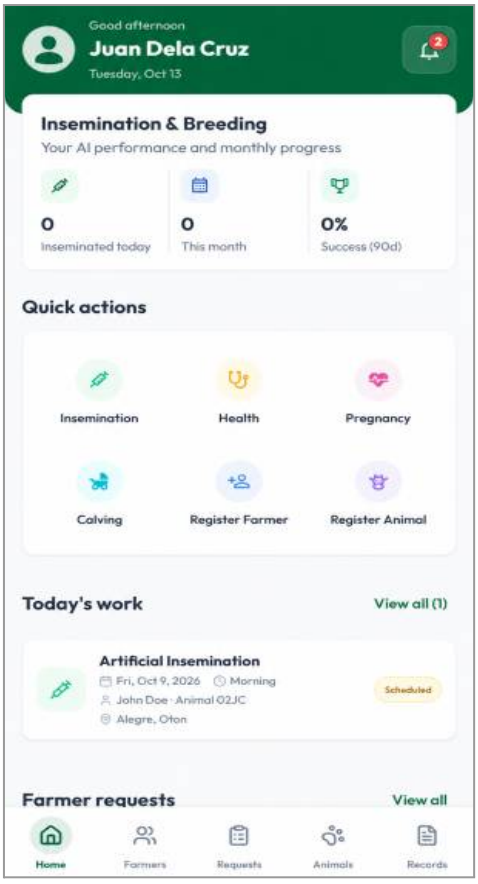


Figure 37. Technician Mobile Dashboard.

Technician Availability and Profile Readiness

The mobile dashboard can present an availability dialog for receiving service requests. Its state is loaded from the current Technician profile, and the update is submitted to the backend dispatch-status operation. Successful submission refreshes the user profile so the interface reflects the stored state. Availability is an operational preference used within request handling; changing it does not automatically complete existing work or grant authority over another Technician's records.

The dashboard also checks profile readiness, including contact information and service barangay details. Where required information is missing, the interface guides the Technician to complete the profile. The availability dialog and profile warning are coordinated so that two blocking messages are not presented simultaneously. The paper should describe this as input and dispatch preparation rather than evidence that the Technician has been clinically qualified or formally appointed.

Verification should compare the selected availability state with the server response and the profile retrieved afterward. A dialog screenshot establishes what the Technician was asked to choose, while the saved-profile readback establishes whether the setting persisted. Account switching and a rejected update are useful additional scenarios because they distinguish a remembered local dialog state from the authoritative dispatch profile.

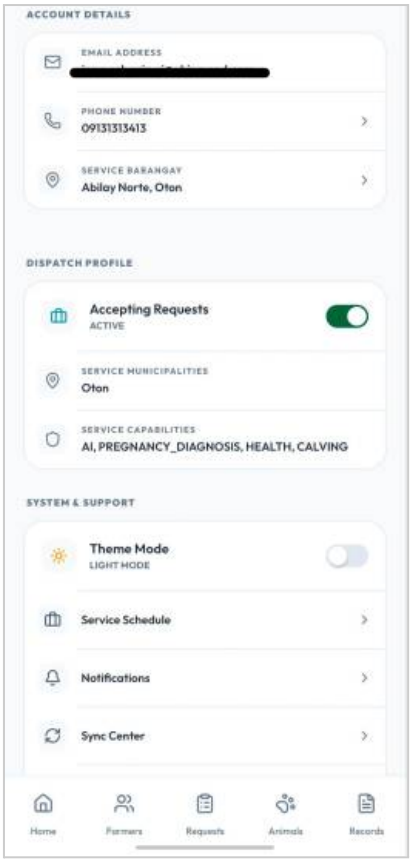


Figure 38. Technician Availability and Profile Readiness.

Technician Mobile Requests and Visit Scheduling

The requests workspace separates available requests from the Technician's own work. Opening a request exposes the permitted review context and the relevant service action. An insemination request detail may show animal information, Farmer notes, location guidance, and the current lifecycle state. Candidate discovery and assigned-request detail are distinct contexts; the paper should not assume that all Farmer contact and location details are exposed before an eligible request is claimed.

Claiming and scheduling associate the request with a responsible Technician and a visit. The scheduling sheet uses a calendar day and morning or afternoon period, while the backend validates date and period availability. Rescheduling changes intended visit timing within the allowed workflow. Skipping an available request returns the Technician to the request list after a successful action; it does not create an official service record.

A complete demonstration should therefore follow the selected request into My Work and show its saved status and timing. Testing should also exercise a request already claimed by another Technician, a past date, and missing schedule information. Those cases help distinguish interface input validation from server-side authority and transition checks. The dates shown in the screen are intended visits until an actual service has been recorded.

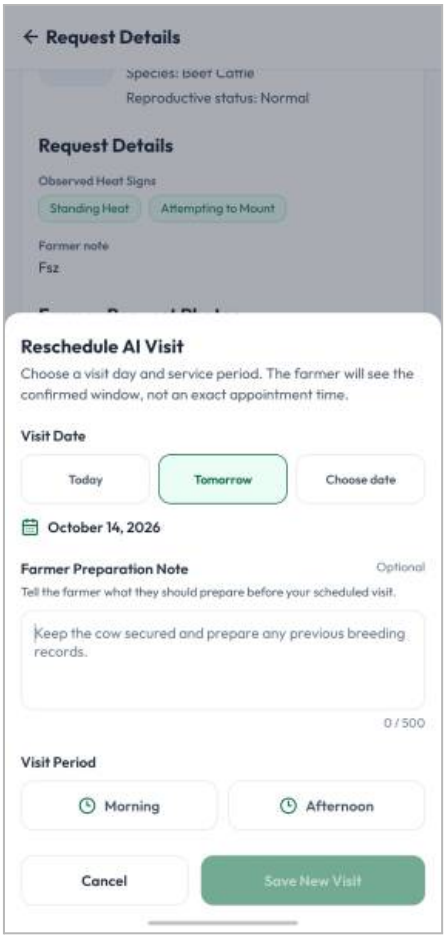


Figure 39. Technician Mobile Requests and Visit Scheduling.Technician Mobile Insemination Recording

The insemination form records clinical service information rather than only by a request status. The current fields include the actual insemination date and time, estrus type, sire breed, sire code, semen doses used, and Technician note. The date and time represent when the service took place and are distinct from the scheduled visit. The review modal gives the Technician an opportunity to inspect the collected information before submission.

The recording feature distinguishes request-linked work from direct recording. Request-linked submission completes the appropriate AI workflow, while a direct or walk-in service identifies the Farmer and animal through its own recording context. Historical entries also have a separate purpose. These distinctions preserve the origin of the record and prevent the manuscript from implying that a generic form can bypass request ownership, lifecycle progression, or reproductive eligibility.

After an accepted submission, the resulting service should be retrieved through the official-record view and checked against the entered fields. A pending local queue item is not the same as a completed central record. The evidence package should include the submitted values, response, and saved record identifier. Validation results must come from actual execution; the presence of a review screen does not establish that missing or invalid fields were rejected.

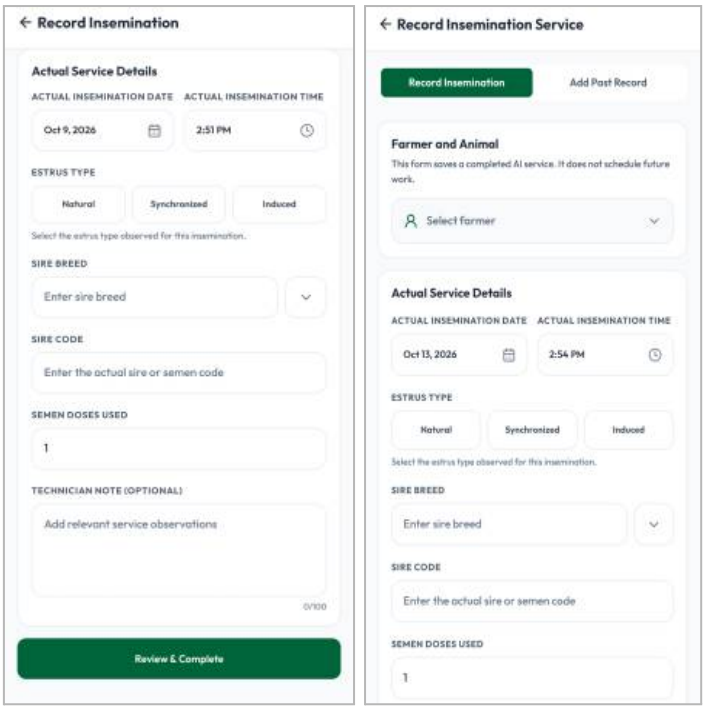


Figure 40 . Technician Mobile Insemination Recording.

Technician Mobile Health-Service Recording

The health-recording form separates the Farmer's concern from the Technician's clinical information. The current recording fields include findings and diagnosis, treatment provided, medication given, dosage, an optional withdrawal period, and optional care advice. Request-linked recording retains the originating concern, while direct recording allows the Technician to identify the Farmer and animal for a service outside an existing Farmer request.

Medication and withdrawal information are entered by the responsible personnel. The interface is a documentation mechanism; it should not be described as automatically prescribing a drug, determining a safe dosage, or validating clinical suitability. Likewise, the recorded withdrawal period should be interpreted as submitted service information rather than a software guarantee of food safety. The surrounding manuscript should explain the record's role without giving clinical treatment instructions.

A saved official medical record should be compared with the input and the associated animal history. For request-linked work, the evidence should also show the resulting request state. Advice-only handling and office medicine pickup belong to their own request-response paths and should be captured separately if used in the pilot. This prevents a successful clinical-record submission from being generalized to every health handling branch.

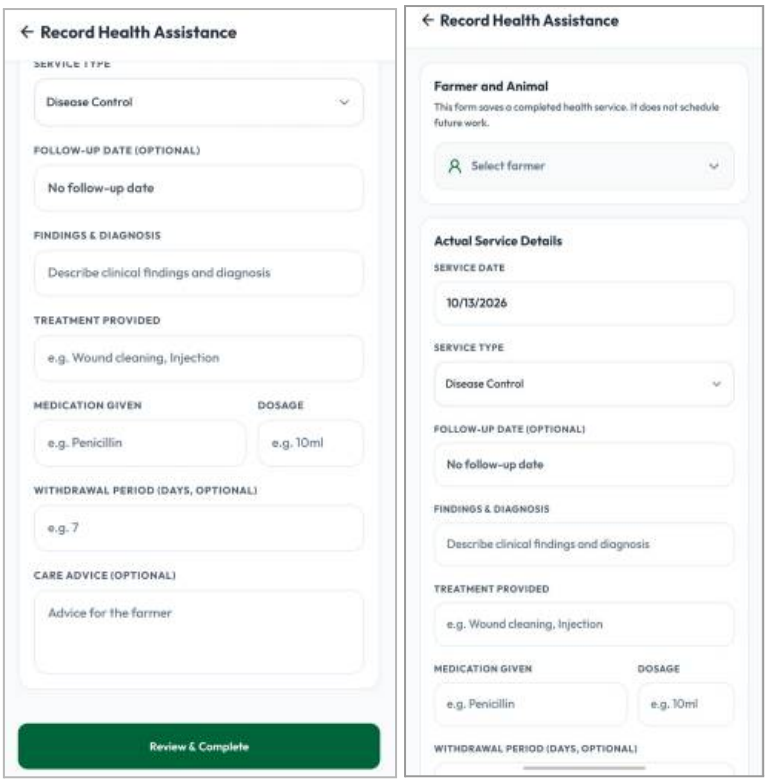


Figure 41. Technician Mobile Health-Service Recording.

Technician Mobile Pregnancy Diagnosis and Verification

The pregnancy screens connect the examination to an animal and its relevant reproductive event. Clinical diagnosis records the supported result and confirmation information, while verification screens address the related Farmer report or assigned follow-up context. A Farmer's observation remains evidence for review until the authorized Technician performs the applicable clinical action. The backend checks the Technician's authority and the selected record relationship.

Readiness depends on the enabled confirmation policy and examination method. The interface may therefore expose readiness information, a required method, or a recheck requirement rather than accept every submitted confirmation immediately. The manuscript should avoid using a single fixed diagnosis day to describe every permitted method. An enabled option also does not prove that the method was actually performed during testing.

Accepted confirmation links the official pregnancy information to the animal and originating insemination. Continuation or recheck work remains associated with the existing pregnancy instead of being presented as an unrelated new cycle. The evidence should show both the submitted confirmation and the resulting linked record. A method-policy rejection, an unassigned task, and a conflicting owner require their own executed test results before the authority controls can be reported as verified.

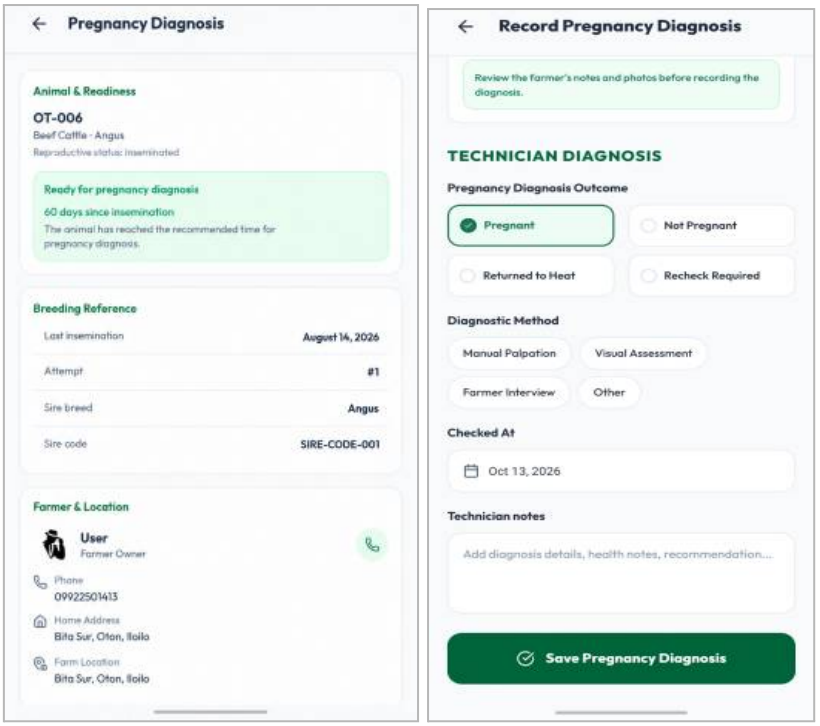


Figure 42. Technician Mobile Pregnancy Diagnosis and Verification

Technician Mobile Calving and Offspring Recording

Calving recording captures an actual delivery event. The Technician selects the dam and relevant reproductive context, records delivery information, and supplies offspring details appropriate to the outcome. Live births and non-living outcomes are distinguished; mixed delivery can retain both categories. The expected calving milestone assists monitoring, but it does not establish that the delivery occurred on the estimated date.

Living offspring are linked to registered Animal records with maternal references, while non-living details remain in the delivery record according to the supported outcome handling. The dam and reproductive cycle are updated through the coordinated backend operation. This is why the interface should be explained as more than a count of newborn animals: one submission affects the actual delivery history, the maternal record, and linked cycle information.

Verification should compare the recorded outcome, calf counts, dam state, and linked offspring after submission. A stillbirth or abortion should not be described as generating active living-animal records. A duplicate delivery, invalid reproductive link, or interrupted transaction needs separate test evidence. The postpartum eligibility information is calculated from the actual recorded event and applicable profile; it is not a certification of the dam’s clinical recovery.

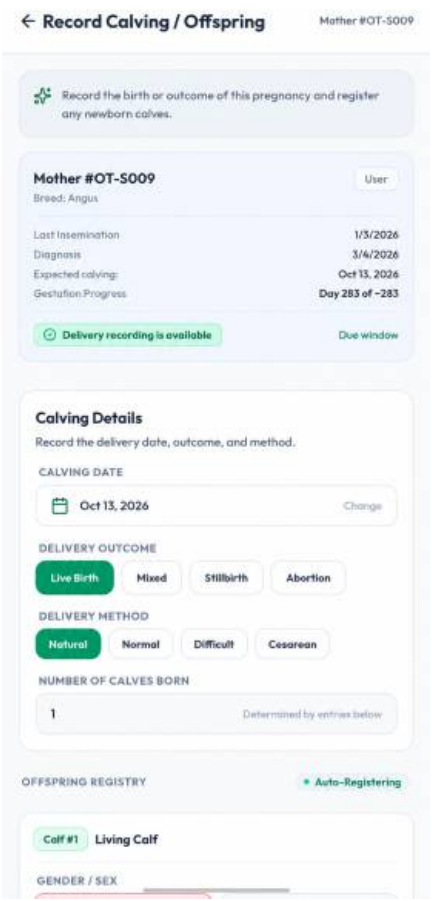


Figure 43. Technician Mobile Calving and Offspring Recording.

Technician Mobile Official Records

The mobile records workspace presents completed official entries by record category and date context. Opening a record shows its own service fields and linked animal information. Insemination, health, pregnancy, and calving entries have different detailed content because they represent different official events. The list should not be described as a collection of every pending request or informal note.

A date-range selection supports review of a particular period, while the summary and list reflect the selected records. The record detail is intended for inspection, and its meaning should remain consistent with the official-record view used elsewhere in the system. Informal field notes may provide useful context but remain distinct from a completed clinical ledger entry.

For cross-platform verification, retrieve one accepted record from both mobile and web and compare its identifier, animal, service date, Technician, and category-specific fields. If one view uses a different date label or count, report that observation as a presentation issue until its source is confirmed. A successful list load alone does not verify every record category or its authority boundary.

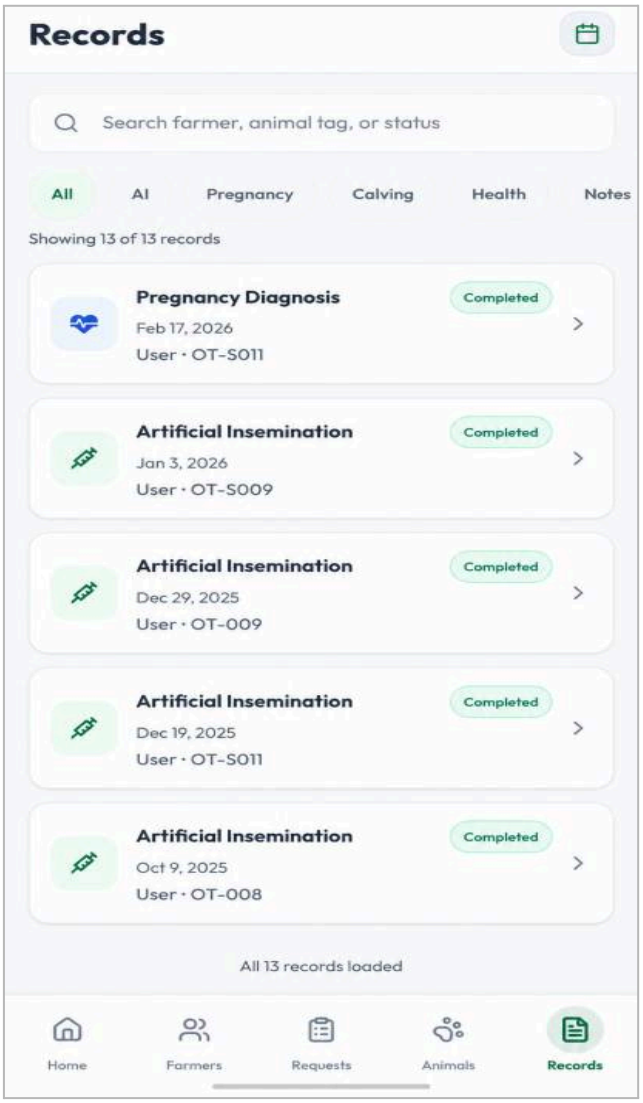


Figure 44. Technician Mobile Official Records

Table 11**.** Technician mobile tasks and resulting records

Task	Main interface action	Result to verify
Review and claim work	Open eligible requests and choose the permitted action.	Assigned source workflow and correct Technician.
Schedule a visit	Choose the date and morning/afternoon period.	Stored intended visit and updated work item.

Task	Main interface action	Result to verify
Record insemination	Submit actual service fields through the appropriate context.	Completed insemination and linked animal history.
Record health service	Submit clinical findings and treatment information.	Official MedicalRecord and request update where linked.
Confirm pregnancy	Submit the authorized method/result and context.	Linked pregnancy confirmation and cycle state.
Record calving	Submit actual delivery and offspring information.	Delivery, dam update, and living offspring references.
Review records	Filter and open a completed official entry.	Consistent record identity and category-specific detail.

3) Technician Web Application

The Technician web workspace provides a browser-based view of the same municipal service processes. Its layout supports reviewing lists, schedules, records, and reports on a larger display. Equivalent web and mobile actions remain subject to backend authorization and workflow rules. The descriptions below explain the reviewed interface behavior; successful use must be supported by the actual test evidence.

Technician Web Dashboard

The web dashboard summarizes the Technician's recent service activity and provides entry to recording and assigned work. Its metrics include inseminations recorded today, monthly inseminations, and the recent-period success summary. Quick actions provide routes to common service and registration tasks. A schedule overview and today's work preview bring intended visits and due activities into the same workspace.

The dashboard is a navigation and monitoring screen. A statistic does not replace an individual service record, and a work preview is not a complete history of every municipal case. The recent success metric has the same reporting limitation described for mobile: it should be explained as an implementation summary and not as a validated conception-rate estimate or user-evaluation score.

The screenshot should demonstrate which summary and work items are visible for the selected Technician. Cross-platform checking should compare the underlying sample records and date boundaries, not assume that identical card arrangements are required. Missing, loading, and empty states should also be recorded when evaluated, since these determine whether users can distinguish unavailable information from a genuine zero count.

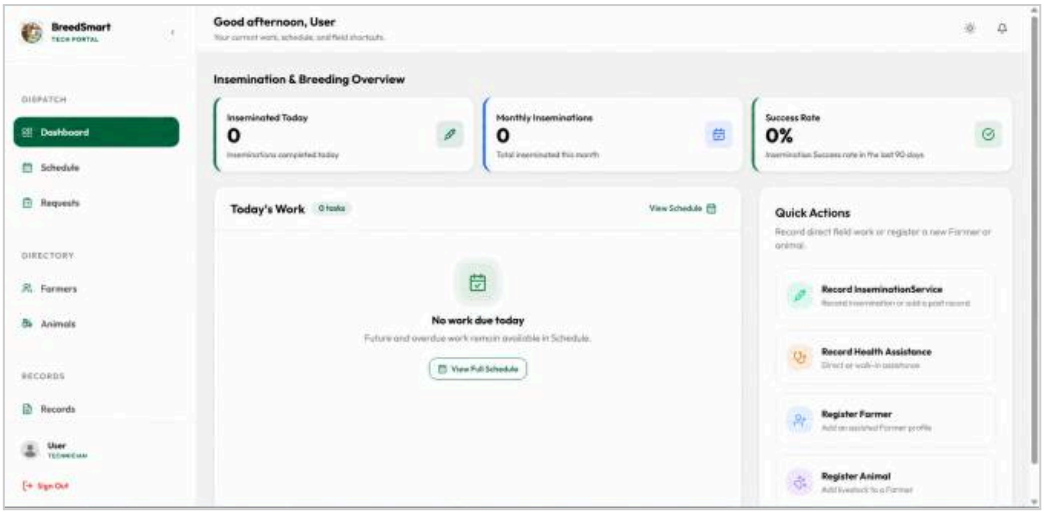


Figure. 45. Technician Web Dashboard

Technician Web Work Queue and Requests

The web workspace distinguishes available requests, assigned active work, and completed work. A linked task carries the identifiers required to open its appropriate workflow. Pregnancy checks, calving, breeding follow-up, insemination, and health work cannot all be completed through one unrestricted generic action. The type of work determines the destination and the service information required.

Completed work is meaningful only when the corresponding workflow or official service has reached its completed state. Opening a completed entry allows the Technician to inspect the official record rather than re-enter the service. The Farmer and animal links provide access to the relevant profiles, allowing the context to be checked before further action.

Testing should compare the same work item before and after completion and confirm that it appears in the appropriate section. A non-completed item should not be counted as completed merely because its due date has passed. The evidence should preserve the source request or task identifier so the transition can be traced through the API response and saved record.

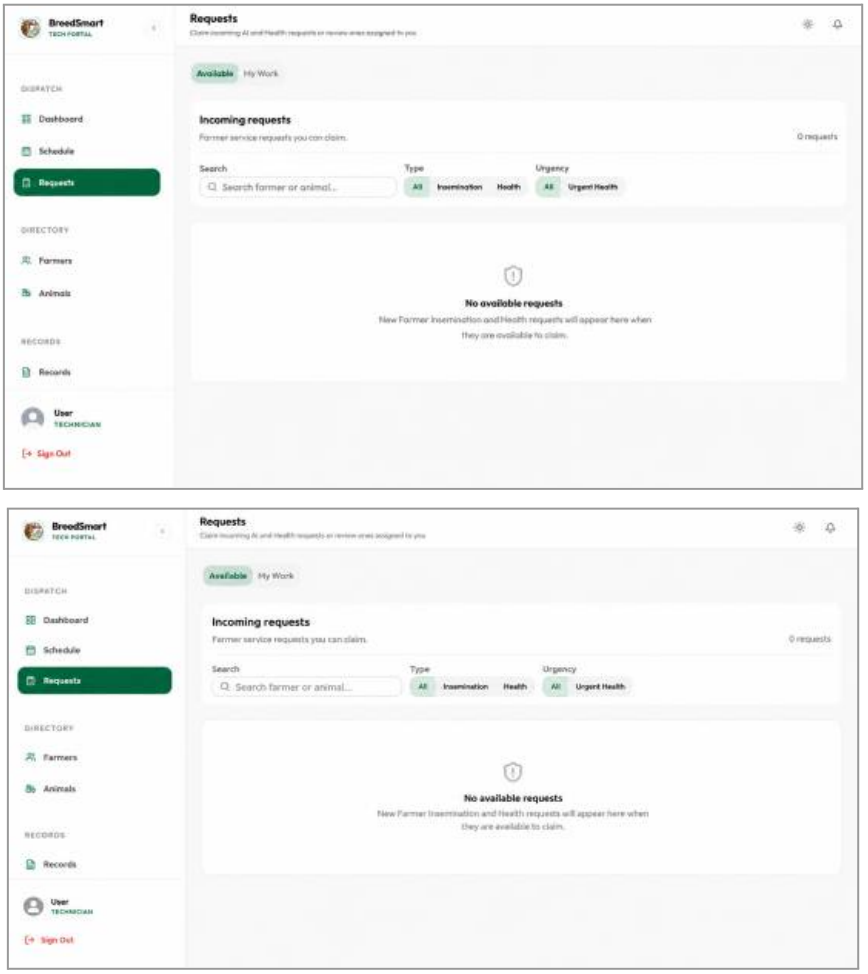


Figure 46. Technician Web Work Queue and Requests

Technician Web Schedule and Visit Context

The schedule view organizes service activity by intended timing. The relevant item retains its service type and links to the request or task that governs the available action. Morning and afternoon are visit periods, while actual procedure time is recorded when the service is performed. The separation helps prevent a schedule from being mistaken for proof of completion.

Changing an intended visit should follow the supported scheduling action and its backend checks. Selecting a past calendar day is not a substitute for recording a service that already happened. The system's dedicated recording context should be used for an actual completed service. The interface and domain rules should be tested together because a calendar can display a date that the server subsequently rejects.

The evidence package should show the stored schedule and the resulting work item after any accepted change. A date shown under a different timezone or year format should be checked against the Philippine calendar date in the source record. The manuscript does not claim an office-hours or Sunday policy beyond the verified implementation.

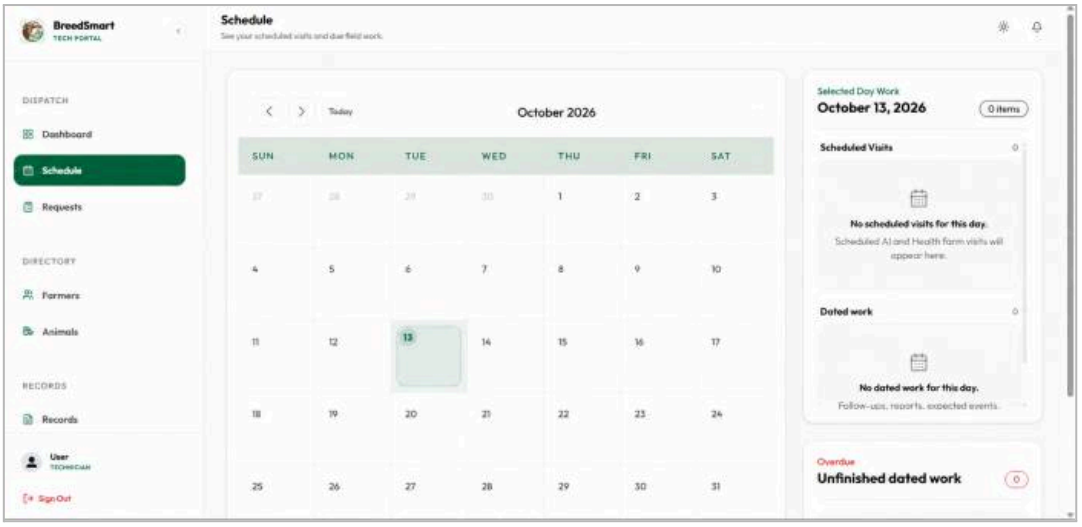


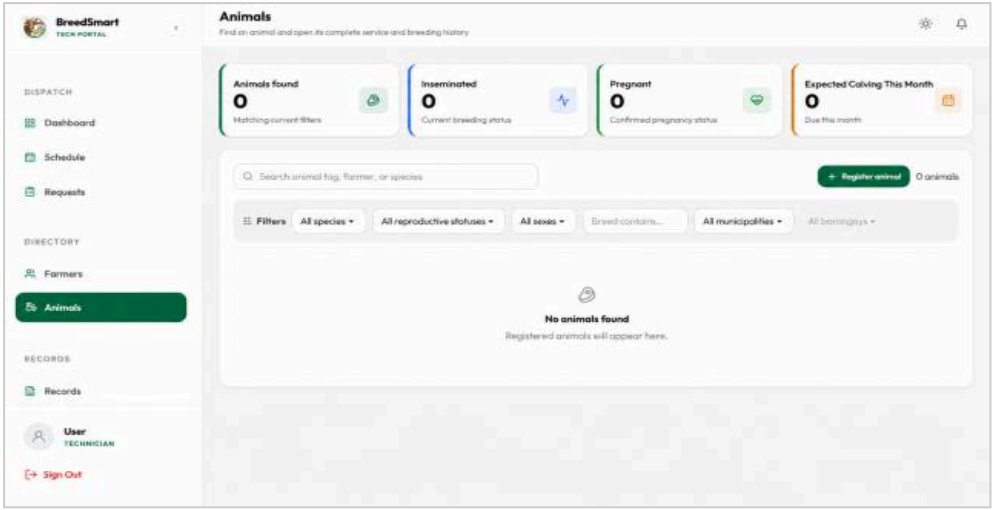
Figure 47. Technician Web Schedule and Visit Context

Technician Web Farmer and Animal Context

Farmer and animal views provide the context needed to identify the service subject. The Farmer directory leads to an individual profile and associated animal information, while animal views retain identification and history. These links allow the Technician to check ownership and prior activity before recording an additional service.

An animal's displayed reproductive state is current stored information supported by its workflow history. It should not be interpreted as a new clinical diagnosis performed by opening the page. Where a record is missing or a link cannot be resolved, the observation should be documented rather than replaced with assumed details. The same principle applies to Farmer contact and farm-location data.

For demonstration, use one Farmer and animal throughout the request, service, and record captures. This makes the sequence readable and supports checking that each operation affected the intended subject. Evaluation of search, registration, and profile updates should record the participant's task and the actual resulting record rather than infer usability from a populated profile.



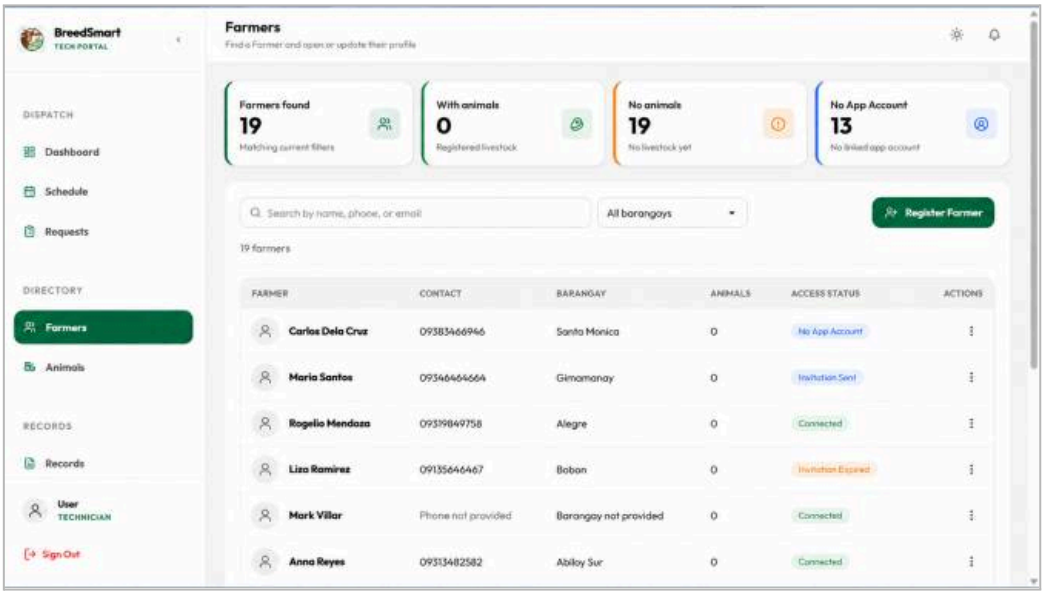


Figure. 48. Technician Web Farmer and Animal Dashboard

Technician Web Official Records and Detail

The web official-record ledger presents completed service entries assigned to the Technician's authorized scope. The record categories include insemination, medical/health service, pregnancy, and calving. Type-specific columns and the shared detail view retain the information relevant to each event. Requests that have not produced an official service record should remain distinguishable from this ledger.

Opening a detail modal retrieves the official record through its animal, category, and record identity. Attachments, where available, provide supporting material for review. The view should be described as record retrieval rather than as evidence that every uploaded attachment has been independently authenticated. Completed work can open the same official detail, maintaining one interpretation of the saved service.

A useful test compares the ledger row with the modal and then with the API record. The comparison should include the service date, animal, Technician, and category-specific outcome. Ownership checks require a different-account test as well as a successful retrieval by the intended Technician. The paper should report the outcomes of those tests only when the logs are supplied.

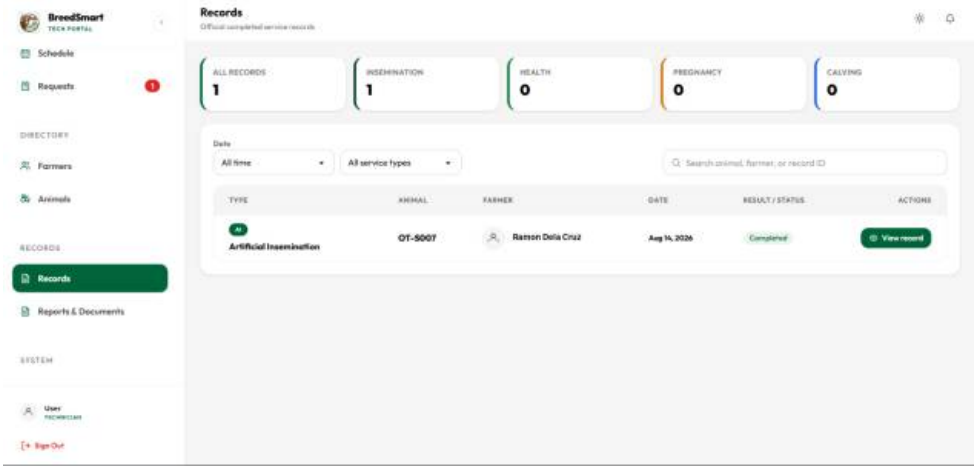


Figure. 49. Technician Web Official Records and Detail Dashboard

Technician Web Monthly Accomplishment Reports

The Reports and Documents screen prepares monthly accomplishment information from recorded activity. The Technician selects a month and service category and reviews the available count and report content. The interface includes print preparation and an Excel export action for the supported monthly accomplishment output.

A generated report remains a representation of the records selected by the reporting query. It should be checked against known test entries, including month boundaries and the chosen service category. Exporting a file establishes that an output was produced; it does not establish that the municipality accepted it as an official submission or that it was transmitted to an external national system.

Verification should compare the source records, screen summary, print layout, and workbook cells. An empty month and an export failure are separate scenarios. The manuscript should describe the current export format supported in the code and avoid claiming an Admin reports screen that is not present in the reviewed web route structure.

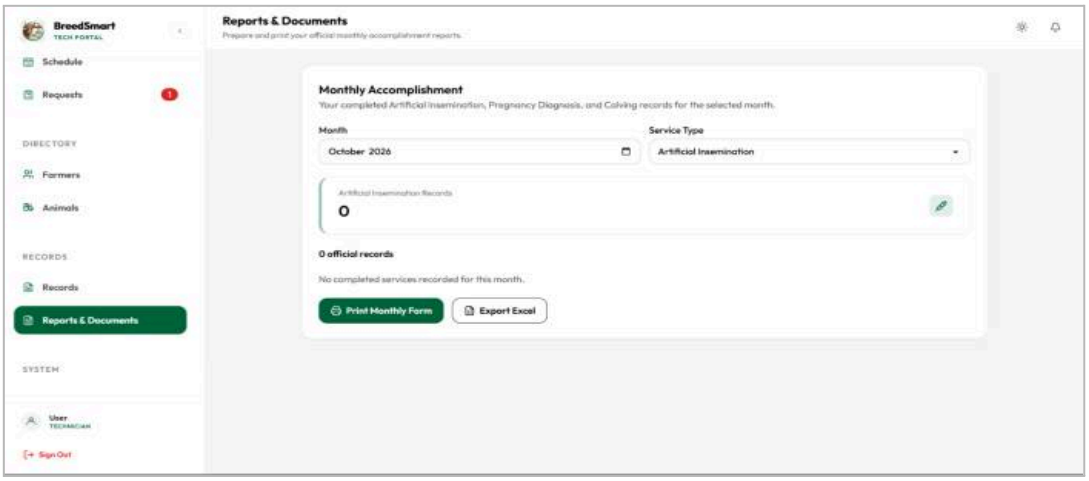


Figure. 50. Technician Web Monthly Accomplishment Reports. Screenshot to be supplied.

Table 12. Technician web views and verification focus

View	Purpose	Verification focus
Dashboard	Review summaries and reach common tasks.	Correct account and date scope; summary linked to known records.
Requests / Work Queue	Review, claim, and follow assigned work.	Authority, lifecycle state, correct workflow destination.
Schedule	Inspect intended visits and date context.	Philippine date and period; schedule separate from actual service.
Farmer / Animal profiles	Identify the owner and service subject.	Linked identities and history for the same test animal.
Official Records	Retrieve completed service entries.	Row, modal, and API show the same record.
Reports & Documents	Prepare monthly print and Excel output.	Selected month/category and matching output values.

4) Administrator Web Application

The municipal Administrator uses the web portal for coordination and oversight. The role includes account management, registry review, workload visibility, request reassignment, administrative audit history, archive restoration, and authorized record correction. It does not replace the Technician's ordinary field-service role. The following municipal oversight views are present in the reviewed web application source.

Admin Web Dashboard

The Admin dashboard summarizes the municipal registry and operational workload. Its statistics include total Farmers, Technicians, and animals. Workload and recent audit panels provide entry to coordination and administrative activity. These summaries support oversight without requiring the Administrator to use a clinical service form.

The current dashboard fetches several sources independently. If a source fails, its unavailable state and partial-data warning are distinct from a valid numeric zero. This matters when interpreting screenshots: a panel marked unavailable should not be reported as an absence of work or animals. The displayed totals also need a defined account and record status scope before they can be treated as municipal census figures.

Verification should compare the dashboard with controlled registry entries and known assignments. The paper can explain how the panels work from the code, but measured loading performance and correctness under failed requests require actual execution evidence. A populated dashboard does not establish complete municipal deployment or complete registry coverage.

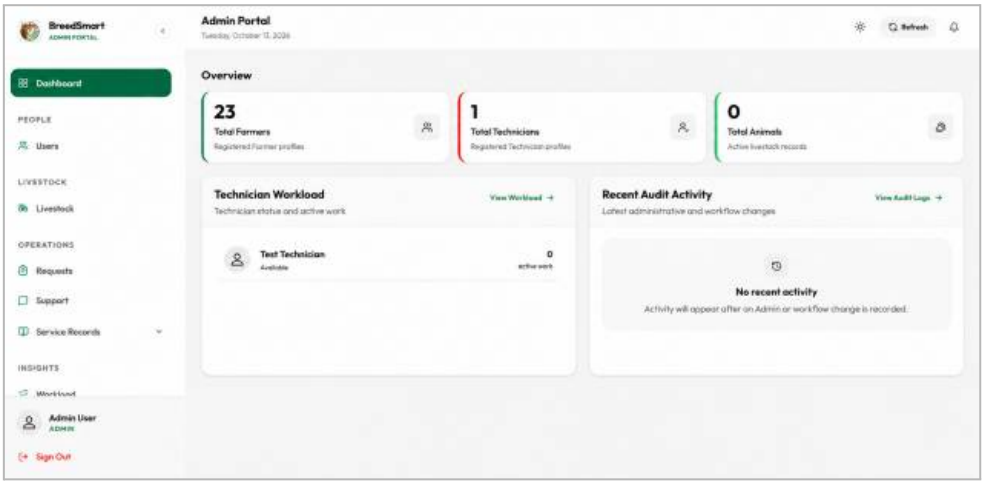


Figure. 51. Municipal Admin Web Dashboard.

Municipal Admin User Directory and Account Management

The Users view separates Farmer and Technician accounts and supports directory search and filtering. Available controls include role-specific views, status, municipality, barangay, and presentation choices. Opening a user detail provides the relevant profile and administrative actions. Account creation and status management are municipal administration functions rather than clinical service activities.

Suspension, reactivation, and archival have different meanings and should be described separately. The backend authentication path checks whether an account is suspended or deactivated before permitting protected operations. A directory action therefore affects subsequent authorized access, not just the color or label of a row.

The evidence should show the account before the action, the administrative response, and its saved status afterward. For suspension testing, a subsequent protected request from the affected account is stronger evidence than a directory screenshot alone. Demonstration records should be used when possible so the manuscript does not disclose personal data or imply that actual municipal staff accounts were altered during a test.

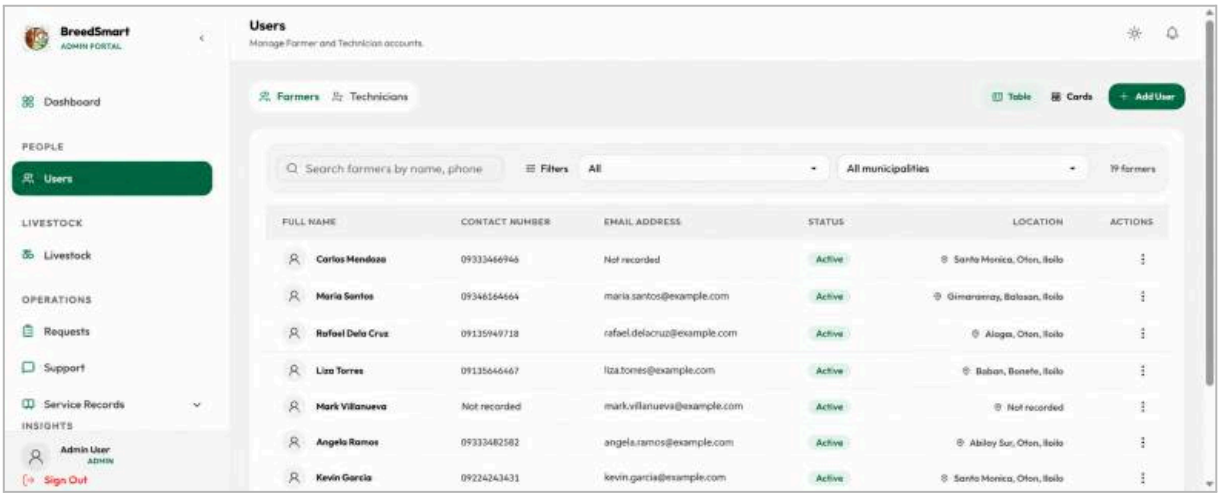


Figure 52. Admin User Directory and Account Management.

Municipal Admin Livestock Registry and Animal Profile

The livestock registry provides a municipal view of registered animals. Search and profile navigation help locate the intended animal and its owner. The individual profile combines stored animal attributes with reproductive and health context and official-record access. This supports identifying related entries without treating each service as a separate animal.

Registry completeness depends on actual registration and maintenance. The presence of a municipal list does not prove that all livestock in Oton have been enrolled. Likewise, profile data reflects recorded events and should not be described as continuous sensor-based monitoring. The study focuses on cattle and carabao workflows even where the broader code model contains additional categories.

Verification should use known animals with distinct identifiers and ownership, then confirm that filters and profile links preserve identity. Changes made through authorized update or correction actions should be checked in the resulting history and relevant record. The paper should distinguish a Farmer-maintained attribute from an officially recorded clinical event.

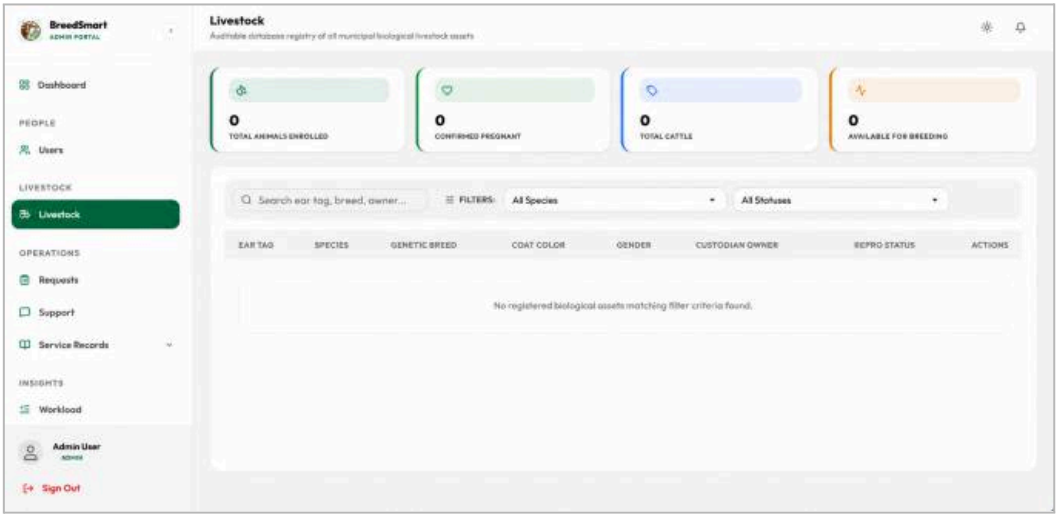


Figure 53. Municipal Admin Livestock Registry and Animal Profile.

Screenshot requirement: Capture registry search/filter controls and one animal profile with identity, owner, reproductive context, and linked official records.

Municipal Admin Technician Workload

The Technician Workload view summarizes active assignments by Technician and links to the relevant profile. It lets municipal personnel inspect distribution of work before making a coordination decision. Workload visibility is an administrative function; it does not grant the Administrator permission to perform the underlying insemination, pregnancy confirmation, or treatment.

Request reassignment is supported through the administrative backend operation and its permitted request context. It changes responsibility rather than marking the service completed. The manuscript should distinguish an available workload view from a particular reassignment control and show the actual control used in the tested build. A route existing in the backend alone does not prove that every web screen exposes it.

To verify reassignment, retain the request identifier, previous Technician, replacement Technician, administrative response, and updated work views. Both Technicians' affected lists should be checked. Cancellation or completion of the same request can create a different operational state, so those scenarios should not be conflated with a successful reassignment test.

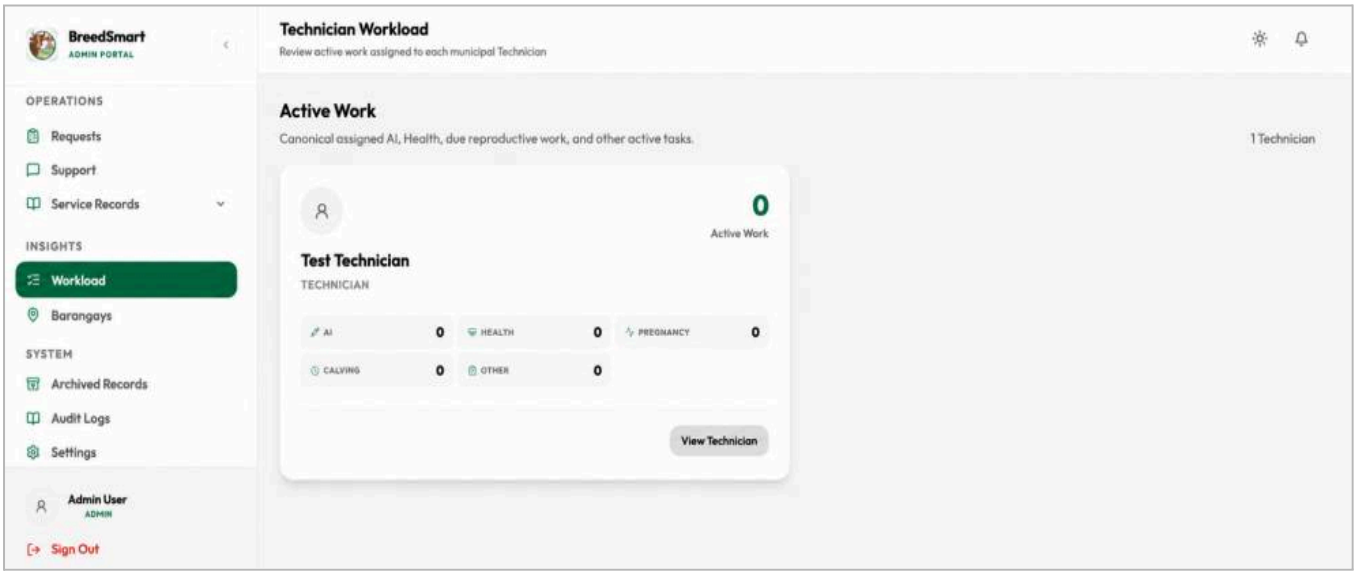


Figure. 54. Municipal Admin Technician Workload.

Municipal Admin Barangay Management

The Barangay Management interface allows the Municipal Administrator to monitor registered farmers, livestock, and pending service requests across different barangays. It provides summary statistics and a searchable table showing the number of farmers, animals, pending health requests, and artificial insemination requests per barangay. The administrator can also view individual barangay summaries for more detailed monitoring.

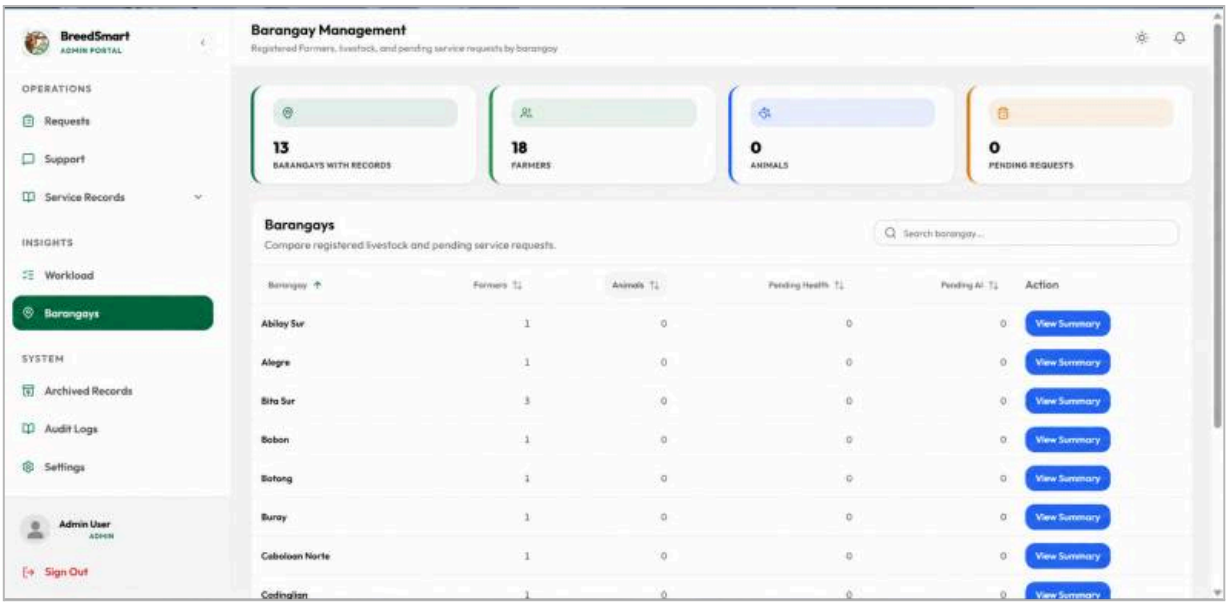


Figure 55. Municipal Admin Barangay Management Interface.

Municipal Admin Audit History

The Audit Logs screen presents administrative activity with category filtering, activity search, and pagination. An expanded entry can expose the relevant stored changes and actor context. This helps reconstruct who performed an administrative action and which entity was affected.

An audit entry is a record of a logged application operation. It is not a guarantee that every possible database change is captured, nor proof that the system has passed a security audit. The implementation includes sanitization of audit information, and the manuscript should describe the displayed administrative history without claiming complete compliance or tamper-proof storage.

A useful demonstration pairs an account-state change or authorized correction with the resulting audit entry. Verification should compare the target entity and actor with the operation performed. Sensitive fields should not be exposed in public manuscript captures. Missing audit evidence should remain explicitly unreported rather than be filled with a sample event presented as actual testing.

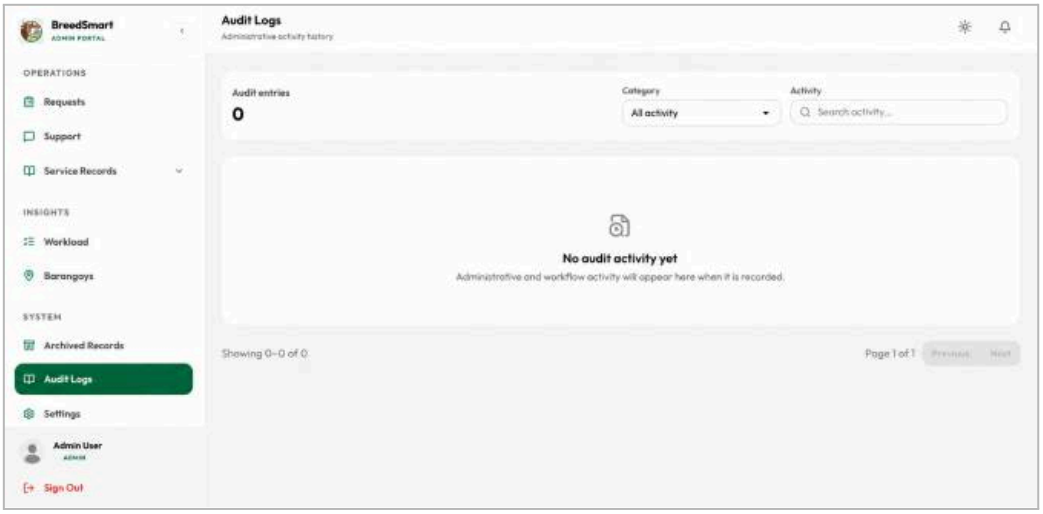


Figure 56. Municipal Admin Audit History.

Municipal Admin Archived Records and Restoration

The Archived Records view separates archived users and animals from normal active lists. Search helps locate an item, and an explicit confirmation precedes restoration. The intent is to restore a soft-deleted record while retaining its existing identity and available history.

Archival differs from permanent deletion. Restoring the same record should not be described as registering a new animal or creating a new user. The underlying authorization and restoration rules still apply. Related clinical records and ownership links should be reviewed to ensure that the restored item is represented consistently.

The evidence package should include the original identifier, archive state, restoration response, and returned directory entry. A single confirmation screenshot establishes only the requested action. The final record and linked history are necessary to support a finding that restoration completed correctly. A failed or unauthorized restoration needs its own recorded outcome.

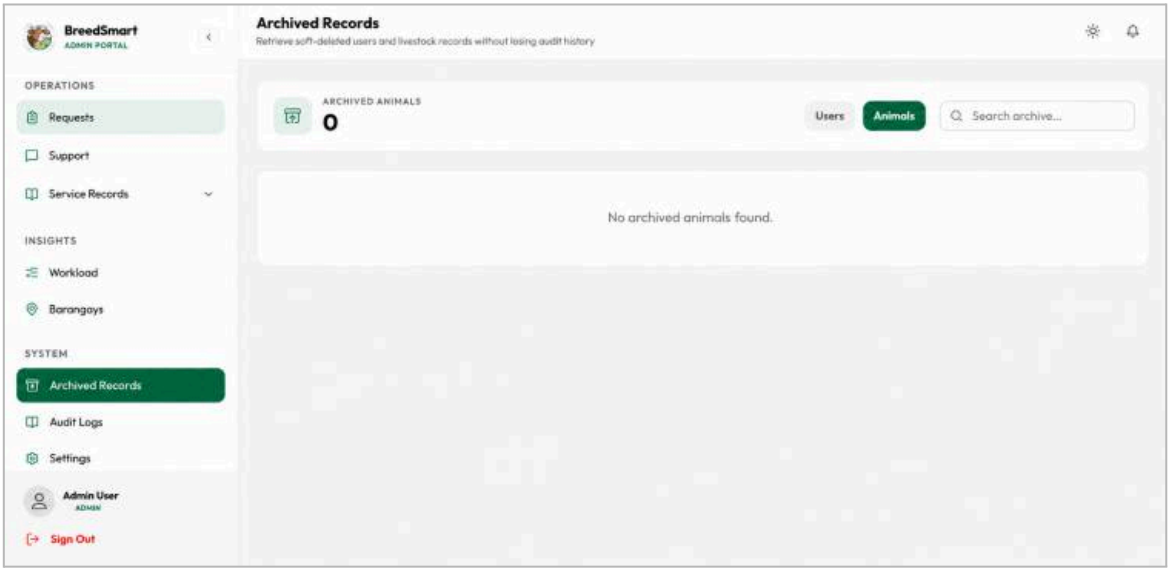


Figure 57. Municipal Admin Archived Records and Restoration.

Table 13**.** Municipal oversight responsibilities and clinical boundaries

Administrative function	Municipal purpose	Boundary
Account management	Create and maintain Farmer and Technician accounts.	Does not grant Farmer accounts clinical authority.
Livestock registry	Review identities, owners, and recorded history.	A registry row is not a clinical diagnosis.
Workload monitoring	Inspect active assignments and coordinate staffing.	Viewing work does not complete it.
Request reassignment	Transfer responsibility through the authorized operation.	Does not create a completed service record.
Oversight registers	Review pregnancy and calving entries.	Initial clinical recording follows Technician workflows.
Authorized correction	Correct official data through the supported path.	Keep the correction distinct from an ordinary field service.
Audit / Archive / Export	Review activity, restore soft-deleted entities, and extract supported data.	Requires specific evidence for each operation; no automatic external integration claimed.

5) Additional Farmer Record and Synchronization Views

Farmer Official Record Detail

The Farmer record view returns relevant official information for the Farmer's authorized animal. It lets the owner review what was recorded after a service rather than rely only on a request-status label. The detail depends on the event category: insemination information differs from a diagnosis, medical service, or delivery record.

Farmer access remains ownership-bound. Viewing an official detail does not authorize the Farmer to overwrite clinical fields or confirm a pregnancy independently. The record should be linked to the same animal and event shown in the request and Technician captures. This provides a readable end-to-end example without inventing an additional service.

Verification should compare the Farmer detail with the accepted Technician record and test another Farmer account against the same resource. A screenshot proves the visible content for one account; an ownership test requires a separate request and recorded response. Any missing record or attachment should be reported as observed.

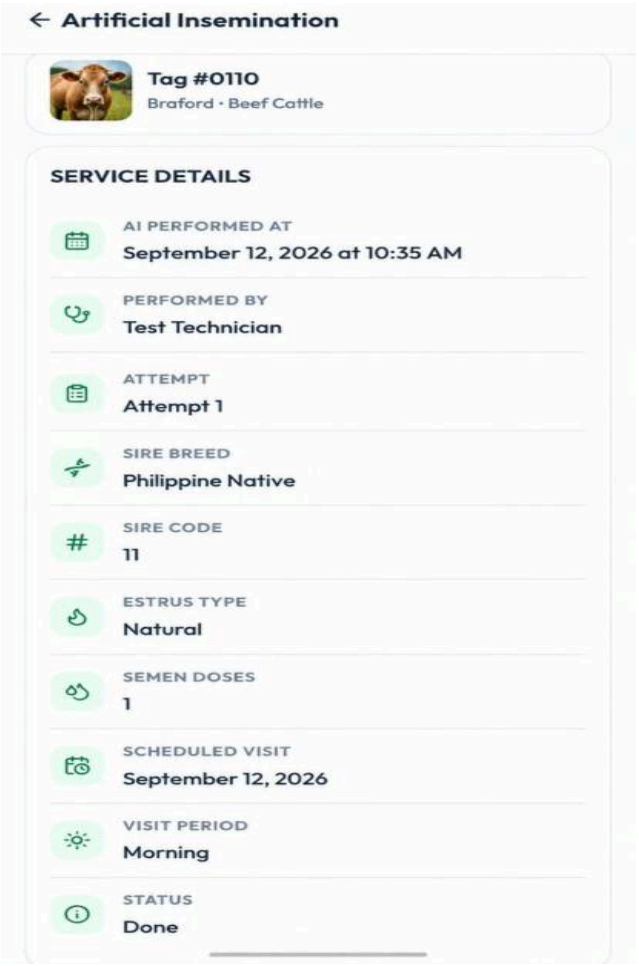


Figure. 58. Farmer Official Record Detail. Screenshot to be supplied.

Screenshot requirement: Capture an official service detail visible to its Farmer owner, showing the animal, service date, Technician, and category-specific information.

Mobile Synchronization Status and Recovery

Synchronization status distinguishes information retained locally from information accepted by the central backend. A pending item is waiting for processing or retry; a failed item requires attention according to the displayed error; a successful history entry represents a processed operation. These states should remain visible in the manuscript instead of describing all locally saved work as immediately synchronized.

Retry uses the originating account and the operation's identifiers. Dependent items wait for required earlier changes, including resolution of temporary animal references. If the earlier operation fails, the dependent action cannot safely proceed as though the referenced server record already exists. Account switching must not cause one user's queue to be submitted under another user's authority.

The strongest evidence follows one supported operation from local entry to server acceptance and compares the resulting identifier and values. A retry after a lost response can test duplicate prevention, while an invalid input can test failure presentation. Offline synchronization remains a limited capability and should be evaluated operation by operation.

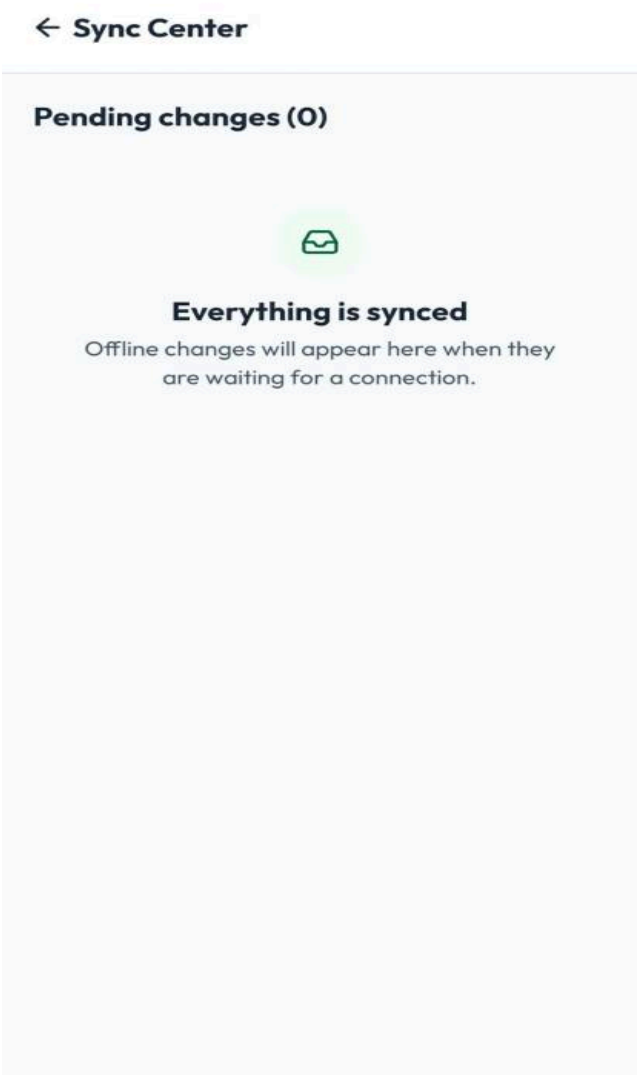


Figure. 59. Mobile Synchronization Status and Recovery. Screenshot to be supplied.

Screenshot requirement: Capture pending, failed, and successful synchronization/history states using supported operations. Provide the corresponding server records and any displayed failure messages.

B. Iterative Development and Release Progression

BreedSmart underwent successive development iterations that produced versioned Android application builds and documented changes to the system. The GitHub release history records four published releases available at the time of this discussion: v1.0.4, v1.0.5, v1.0.7-preview.1, and v1.0.8. Together, these releases document implementation checkpoints, workflow refinements, and preparation of application builds for continued testing [1]–[4].

Each release is associated with a Git tag and an Android application package (APK). The tags identify the corresponding repository revisions, while the APK assets provide installable mobile builds. This arrangement supports tracing development changes to particular versions. Publishing an APK establishes the availability of a build; its usability, workflow correctness, and readiness for municipal use require separate testing and evaluation evidence.

1) Summary of Published Releases

Table 14**.** Documented BreedSmart release progression

Version and date	Documented development changes	Evidence and qualification
v1.0.4 29 Aug 2026	Cumulative implementation of backend, mobile, and web features; refinement of service handling; addition of an Android release distribution workflow.	Release notes list merged contributions and an APK. This is the earliest published release in the retrieved history [1].
v1.0.5 12 Sep 2026	Technician insemination and health-service workflow refinements; linked Farmer and animal information corrections; dashboard and note handling changes; ear-tag generation.	Release notes describe the changes and provide an APK. They do not report complete workflow validation [2].
v1.0.7-preview .1 30 Sep 2026	Preparation for native update-management validation, compatible over-the-air updates, and Phase 3 stabilization.	Explicitly identified as an internal field-validation build and marked as a prerelease [3].
v1.0.8 3 Oct 2026	Changes to request skipping, calving outcome presentation, linked health dashboard entries, candidate data exposure, and web input error feedback.	APK publication is documented. Changes are supported by the tag comparison because the release description is empty [4], [5].

Dates are based on GitHub publication timestamps converted to Philippine Standard Time (UTC+8).

2) Implementation Baseline in v1.0.4

Release v1.0.4 provides a cumulative record of contributions incorporated into BreedSmart by that checkpoint. Its release notes list database configuration, authentication, backend routes and controllers, Farmer profile and records screens, animal profile management, and the development of web and mobile interfaces. The same history includes livestock registry functions, Farmer dashboard and Technician management work, and changes to Technician screens, pagination, and records [1].

The release also records work to simplify mobile workflows and refine service handling, followed by an Android release distribution workflow. These entries show that application implementation and build distribution were developed alongside workflow corrections. Since the notes summarize accumulated contributions, they should be read as a baseline of development up to v1.0.4 rather than as evidence that every listed feature was first implemented on its publication date.

3) Workflow Refinement in v1.0.5

Release v1.0.5 focuses on specific workflow and interface changes. Its notes identify improvements to the Technician artificial insemination service workflow and health-request and service-recording flow. They also document corrections to the Farmer and animal information linked to requests, changes to Technician dashboard presentation, and improvements to health-observation and Farmer-note handling. An ear-tag generator was added for animal registration, together with general interface and workflow corrections [2].

These changes addressed how users identify an animal, inspect the information associated with a request, and document a service. Their relevance to the study lies in refining the digital steps used for livestock-service coordination and record management. The release notes establish that these areas were changed, but they do not quantify gains in task completion, record accuracy, or user satisfaction.

4) Internal Validation Build in v1.0.7 Preview 1

The v1.0.7-preview.1 release was explicitly described as an internal field-validation build and was marked as a prerelease. Its notes identify native update-management validation, silent compatible over-the-air (OTA) updates, and recent Phase 3 stabilization fixes. OTA updates concern the delivery of compatible application updates, while the accompanying APK provides an installable preview build [3].

This release documents a testing checkpoint for recent stabilization and update-management work. Its stated purpose does not establish that a comprehensive field evaluation was completed. In particular, the preview designation and description should not be interpreted as evidence of complete user acceptance or municipal-wide deployment.

5) Subsequent Refinements in v1.0.8

Release v1.0.8 provides a later APK and is associated with the main branch. Although the release page contains no written description, the repository comparison between v1.0.7-preview.1 and v1.0.8 identifies changes incorporated into the later tagged revision [4], [5]. The comparison includes changes to backend, mobile, and web files, as well as test and release-check configuration.

The mobile changes restore the Technician request-skipping flow and redirect the user to the request list after a successful skip. Calving-record presentation was also expanded to distinguish live birth, stillbirth, and mixed delivery, with calf counts displayed when available. In the backend, linked health tasks are excluded from duplicate dashboard agenda entries when the associated health request is already listed. Candidate request responses were also narrowed by removing Farmer contact and detailed location fields from the affected responses [5].

The web changes include input error styling and invalid-state handling in the shared input component. The comparison also contains revisions to test fixtures and the configuration used for release checks. These changes provide concrete evidence of continued correction and integration work. Their presence in the tagged revision does not establish final test pass totals or complete end-to-end coverage; those results must be reported from verified test execution.

6) Discussion of Release Progression

The documented progression moves from a cumulative implementation baseline in v1.0.4 to targeted workflow refinements in v1.0.5, an internal validation build in v1.0.7-preview.1, and further corrections in v1.0.8. It supports the iterative development approach described in the methodology by showing that the application was revised and repackaged across successive checkpoints.

The release history establishes development activity and the availability of versioned mobile builds. The project remains technically functional and is still undergoing workflow checking and usability refinement. Comprehensive testing of the remaining workflows and broader user evaluation are needed before conclusions about operational effectiveness or municipal deployment readiness can be supported. Technical testing results and stakeholder evaluation should therefore be presented as separate evidence in Chapter IV.

Source References

[1] BreedSmart v1.0.4 release and cumulative changelog.
https://github.com/Error-404-not-found-org/Ilo-Agricultures-Office-System/releases/tag/v1.0.4

[2] BreedSmart v1.0.5 release notes.
https://github.com/Error-404-not-found-org/Ilo-Agricultures-Office-System/releases/tag/v1.0.5

[3] BreedSmart 1.0.7 Preview 1 release notes.
https://github.com/Error-404-not-found-org/Ilo-Agricultures-Office-System/releases/tag/v1.0.7-preview.1

[4] BreedSmart 1.0.8 release and APK asset.
https://github.com/Error-404-not-found-org/Ilo-Agricultures-Office-System/releases/tag/v1.0.8

[5] Repository comparison of v1.0.7-preview.1 and v1.0.8.
https://github.com/Error-404-not-found-org/Ilo-Agricultures-Office-System/compare/v1.0.7-preview.1...v1.0.8

GitHub sources accessed 4 October 2026

V. CONCLUSION AND RECOMMENDATIONS

A. Introduction

This chapter provides a comprehensive synthesis of the BreedSmart system's development outcomes. It evaluates the achievement of the research objectives, interprets the technical and expert evaluation findings, and offers strategic recommendations for future enhancements to maximize the system's utility, sustainability, and operational readiness for small and medium-sized agricultural communities and the Local Government Unit (LGU).

B. Summary of the Proposed Study Design and Implementation

This study aimed to design and develop BreedSmart, a centralized web and mobile-based livestock insemination and health management system tailored for the Municipality of Oton, Iloilo. The system was developed to transition existing fragmented, paper-based livestock tracking into an organized digital workflow that integrates role-based access for Farmers, Livestock Technicians, and Municipal Administrators.

BreedSmart facilitates the end-to-end management of animal profiles, artificial insemination (AI) service requests, health-service coordination, and reproductive monitoring. Through the mobile application, Farmers can register livestock, request veterinary services, and track reproductive milestones. Meanwhile, the web and mobile interfaces for Livestock Technicians enable the scheduling of farm visits, the execution of clinical documentation (such as pregnancy diagnoses and calving), and the tracking of assigned work. To accommodate rural field conditions, the system architecture incorporates a localized offline queueing function that allows supported operations to be saved without connectivity and synchronized once a network connection is restored.

C. Summary of Findings

The development of BreedSmart demonstrated that the decentralized livestock-service activities of the Municipality of Oton can be successfully unified within a centralized web and mobile-based architecture. The platform effectively established role-based access, securely connecting animal profiles, service requests, reproductive workflows, and official records through a common backend.

To rigorously validate the technical architecture, the system was evaluated by a panel of Information Technology experts using the ISO/IEC 25010 Software Quality model. The evaluation demonstrated highly robust technical performance across all core domains:

1. Functional Stability: Measuring completeness, correctness, and appropriateness, the system was rated "Very Satisfactory" with a mean of 3.93.
2. Compatibility: Assessing co-existence and interoperability, the system achieved a "Very Satisfactory" mean of 4.00.
3. Interaction Capability: Encompassing usability, learnability, and user error protection, the system scored highly with a mean of 4.12 ("Very Satisfactory").
4. Reliability: Evaluated through maturity, availability, and fault tolerance, the system earned a "Very Satisfactory" mean of 4.00.
5. Security: Protocols ensuring confidentiality, integrity, and accountability were rated at 3.90 ("Very Satisfactory").
6. Flexibility: Indicating high adaptability and replaceability, this domain secured the highest individual rating of 4.15 ("Very Satisfactory").

Cumulatively, the IT experts' assessment yielded a grand mean of 4.02, officially interpreting the system's technical implementation as "Very Satisfactory."

Table 15. Evaluation Results for the Proposed System

Criteria	Mean	Description
Functional Stability	3.93	Very Satisfactory
Compatibility	4	Very Satisfactory
Reliability	4	Very Satisfactory
Interaction Capability	4.12	Very Satisfactory
Security	3.9	Very Satisfactory
Flexibility	4.15	Very Satisfactory

Initial pilot testing with the principal livestock technician client confirmed that the core reproductive and scheduling workflows are functionally viable in real-world scenarios. The integration of offline synchronization proved technically successful, providing a crucial safeguard for field operations facing intermittent connectivity. However, the study also identified continuing limitations, specifically the need for expanded device-condition testing, a larger pilot population, and further refinement of the Administrator module's export functions to fully assess long-term operational endurance.

D. Conclusion

Based on the system development, integration, and evaluation phases, BreedSmart substantially achieved its primary objective of creating a comprehensive livestock insemination and health management platform for the Municipality of Oton. The system proves that decentralized livestock records, reproductive monitoring timelines, and clinical service coordination can be seamlessly consolidated into a single, role-based digital environment.

A critical conclusion of this study is the delineation of the system’s operational boundary: BreedSmart functions as an advanced organizational and communication tool, rather than an automated clinical diagnostic instrument. It successfully structures service data and generates milestone reminders, but it strictly preserves the authority of veterinary personnel over actual clinical diagnoses and treatment decisions. Furthermore, supported by the "Very Satisfactory" technical rating from IT experts, the architecture is technically sound, with its offline synchronization capabilities providing a vital foundation for rural deployment.

While the software architecture is fully realized and technically validated, the system remains in a state of ongoing usability refinement. The study establishes the immediate technological impact of digitizing these workflows, but acknowledges that proving long-term improvements in biological conception rates or overarching municipal efficiency will require longitudinal operational deployment and further research.

E. Recommendations

In light of the technical findings, expert evaluations, and stakeholder feedback, the following recommendations are proposed to enhance the system's operational readiness:

1. Conduct Comprehensive End-to-End Validation: Execute rigorous manual and automated workflow testing across all user roles (Farmer, Technician, Administrator) to resolve minor interface inconsistencies prior to a full municipal rollout.
2. Expand Pilot Evaluation: Engage a broader demographic of local Farmers, additional Technicians, and municipal staff to gather representative feedback on device compatibility and workflow clarity across varied technological

literacy levels.

3. Integrate Local Language Localization: Incorporate Hiligaynon language support within the Farmer mobile application to ensure system instructions, notifications, and prompts are easily understood by the local rural user base.
4. Broaden SMS Notification Capabilities: Develop advanced SMS alerts for service reminders, appointment schedules, and status updates to accommodate Farmers who may lack continuous internet access or consistent smartphone usage.
5. Enhance Offline Field Operations: Expand the scope of the mobile application's offline queuing capabilities, allowing Technicians to log a wider array of clinical data during network outages, fortified by robust conflict-handling protocols during server synchronization.
6. Refine Administrative Reporting Tools: Collaborate directly with the Municipal Agriculture Office to customize dashboard analytics, export formats, and data filters, ensuring the system generates reports that strictly align with LGU and provincial auditing requirements.
7. Perform Diverse Environment Testing: Evaluate the mobile application across a wider spectrum of Android devices, screen dimensions, and fluctuating network speeds to identify and mitigate edge-case synchronization errors.
8. Initiate Longitudinal Impact Research: Following full implementation, future researchers should conduct a long-term study comparing pre- and post-deployment metrics to scientifically measure the system's impact on livestock conception rates, health outcomes, and municipal service response times.

F. Impact of the Study

The primary contribution of this study is the technological demonstration that highly fragmented, paper-based livestock workflows can be digitized and centralized into a secure, role-based ecosystem. BreedSmart provides a highly evaluated, practical model for connecting animal records, AI services, and administrative oversight.

For Farmers: The platform establishes a direct, mobile-accessible channel for requesting veterinary services, monitoring reproductive milestones, and accessing official animal histories without navigating bureaucratic delays.

For Livestock Technicians: The system creates a structured environment that streamlines work scheduling, standardizes clinical documentation, and organizes field visits, significantly reducing the administrative burden of cross-referencing disparate paper logbooks.

For the Municipal Agriculture Office: BreedSmart provides a foundational administrative dashboard that enables real-time oversight of municipal livestock assets, technician workloads, and critical disease or reproductive trends.

For the Academic Community: The project serves as a validated architectural blueprint for future IT researchers developing offline-capable, role-based agricultural information systems tailored to the infrastructural realities of the Philippines.

G. Summary

This chapter summarized the findings, conclusions, recommendations, and overall impact of the BreedSmart study. The project resulted in a technically functional web and mobile-based livestock management system that successfully integrates role-based records, service requests, reproductive workflows, and administrative oversight.

Validated by Information Technology experts with a "Very Satisfactory" rating (4.02), the system's architecture—particularly its offline synchronization and robust security capabilities—proves highly capable of handling municipal data demands. While the current implementation effectively standardizes agricultural service processes and delineates clear user responsibilities, further environmental testing and broader user participation are recommended to ensure seamless, long-term adoption. Ultimately, BreedSmart provides a powerful, scalable foundation for modernizing livestock information coordination within the Municipality of Oton.

REFERENCES

[1] Inventory of Livestock and Poultry Farms, Municipal Agricultural Office, 2021.

[2] N. M. Trendov, S. Varas, and M. Zeng, Digital Technologies in Agriculture and Rural Areas – Status Report. Rome, Italy: Food and Agriculture Organization of the United Nations, 2019.

[3] World Bank, Future of food: Harnessing digital technologies to improve food system outcomes, World Bank Group, 2019.

[4] F. D. Davis, "Perceived usefulness, perceived ease of use, and user acceptance of information technology," MIS Quarterly, 13(3), 319–340, 1989.

[5] P. J. Hansen, "Reproductive efficiency in cattle: Challenges and solutions," Journal of Animal Science, 98(4), 1–10, 2020.

[6] Food and Agriculture Organization of the United Nations, Digital technologies in livestock production systems, FAO, 2023.

[7] S. Wolfert and L. Ge and C. Verdouw and M. J. Bogaardt, "Big data in smart farming," Agricultural Systems, 153, 69–80, 2017.

[8] F.S.Lima, F.T. Silvestre, F. Peñagaricano, and W.W. Thatcher, “Early genomic prediction of daughter pregnancy rate is associated with improved reproductive performance in Holstein dairy cows,” American Dairy Science Association, 2020.

[9] Mimansha Raj and M. Prahadeeswaran, “Revolutionizing agriculture: a review of smart farming technologies for a sustainable future,” 2025.

[10] T. J. Beyene, F. Asfaw, Y. Getachew, T. B. Tufa, I. Collins, A. F. Beyi, and C. W. Revie, "A Smartphone-Based Application Improves the Accuracy, Completeness, and Timeliness of Cattle Disease Reporting and Surveillance in Ethiopia," Frontiers in Veterinary Science, vol. 5, art. 2, 2018, doi: 10.3389/fvets.2018.00002.

[11] R. M. Briones and I. M. R. Galang and J. S. Latigar, "Transforming Philippine Agri-Food Systems with Digital Technology: Extent, Prospects, and Inclusiveness," Philippine Institute for Development Studies (PIDS) Discussion Paper Series, No. 2023-29, 2023.

[12] Gwaka L, May, J and Tucker, W, "The impacts of digital infrastructure transformation on livestock system sustainability in rural communities," 2020.

[13] A. K. Tegegnie, T. D. Dagne, and T. Alemu, "SMS Based Agricultural Information System for Rural Farmers," in Advances of Science and Technology (ICAST 2018), ser. Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, vol. 268. Cham, Switzerland: Springer, 2019, pp. 366–379.

[14] Department of Agriculture, Digital Agriculture Roadmap 2023–2028, Republic of the Philippines, 2023.

[15] W. H. DeLone and E. R. McLean, "The DeLone and McLean model of information systems success: A ten-year update," Journal of Management Information Systems, 19(4), 9–30, 2003.

[16] Department of Agriculture, Digital agriculture roadmap 2022–2028, Republic of the Philippines, 2022.

[17] International Livestock Research Institute, Livestock data for decisions: Enhancing livestock productivity through digital systems, ILRI, 2020.

[18] O. Mapiye, G. Makombe, A. Molotsi, K. Dzama, and C. Mapiye, "Information and communication technologies (ICTs): The potential for enhancing the dissemination of agricultural information and services to smallholder farmers in sub-Saharan Africa," Information Development, vol. 39, no. 3, pp. 638–658, 2021. doi: 10.1177/02666669211064847.

[19] K. C. Laudon and J. P. Laudon, Management information systems: Managing the digital firm (17th ed.), Pearson Education, 2021.

[20] Philippine Statistics Authority, Livestock and poultry production statistics report, PSA, 2024.

[21] R. S. Pressman and B. R. Maxim, Software engineering: A practitioner’s approach (9th ed.), McGraw-Hill Education, 2020.

[22] E. M. Rogers, Diffusion of innovations (5th ed.), Free Press, 2003.

[23] I. Sommerville, Software engineering (10th ed.), Pearson Education, 2016.

[24] IISTR Journals, "The Use of IoT on Smart Agriculture in the Philippines: A Narrative Review," International Journal of Innovative Science

and Technology Research, 2023.

[25] World Economic Forum, Innovation with a purpose: The role of technology in sustainable agriculture, World Economic Forum, 2022.

[26] World Organisation for Animal Health, "Animal health surveillance," Terrestrial Animal Health Code, ch. 1.4, 2021.

[27] R. Singh, W. Slotznick, and D. Stein, "Digital tools for rural agriculture extension: Impacts of mobile-based advisories on agricultural practices in Southern India," Journal of the Agricultural and Applied Economics Association, vol. 2, pp. 4–19, 2023 .

[28] Food and Agriculture Organization of the United Nations, Development of Integrated Multipurpose Animal Recording Systems, FAO Animal Production and Health Guidelines No. 19. Rome, Italy: FAO, 2016.

[29] World Organisation for Animal Health, "General principles on identification and traceability of live animals," Terrestrial Animal Health Code, ch. 4.2, 2021.

[30] O. L. Bondoc, "Genetics and economics of artificial insemination of cattle and carabaos in the Philippines," Philippine Agricultural Scientist, vol. 82, no. 2, pp. 213–229, 1999.

[31] M. K. Njenga et al., "High real-time reporting of domestic and wild animal diseases following rollout of mobile phone reporting system in Kenya," PLoS ONE, vol. 16, no. 9, e0244119, 2021, doi: 10.1371/journal.pone.0244119.

[32] E. P. Atabay, E. C. Atabay, C. F. Dela Cruz, E. R. S. Maylem, J. P. R. Apolinario, and R. D. Tadeo, "Improved artificial insemination efficiency through an ovarian structure-based controlled breeding program in post-partum buffaloes," Philippine Journal of Veterinary and Animal Sciences, vol. 50, no. 2, pp. 70–79, 2024.

[33] R. D. Tadeo, E. P. Atabay, E. C. At abay, C. F. Dela Cruz, and J. P. R. Apolinario, "Accuracy of pregnancy-associated glycoprotein-enzyme-linked immunosorbent assay (PAG-ELISA) in early pregnancy detection in water buffaloes," Philippine Journal of Veterinary and Animal Sciences, vol. 50, no. 1, pp. 13–20, 2024.

[34] J. A. Parish, J. E. Larson, and R. C. Vann, "The Estrous Cycle of Cattle," Mississippi State University Extension Service, Publication 2616, 2023. [Online]. Available: https://www.accessibility.extension.msstate.edu/publications/the-estrous-cycle-cattle

[35] J. Statham, "Pregnancy Determination in Cattle," Merck Veterinary Manual, updated Sept. 2024. [Online]. Available: https://www.merckvetmanual.com/management-and-nutrition/management-of-reproduction-cattle/pregnancy-determination-in-cattle

[36] Y. Resti, G. G. Reynoso, L. Probst, S. Indriasari, G. P. Mindara, A. Hakim, and M. Wurzinger, "A review of on-farm recording tools for smallholder dairy farming in developing countries," *Tropical Animal Health and Production*, vol. 56, no. 5, art. no. 168, May 2024

[37] T. Muromba, D. Keeni, and M. Fuyuki, "Application of the Technology Acceptance Model to mobile agricultural services in sub-Saharan Africa: A synthesis of empirical studies," *Journal of Agricultural Informatics*, vol. 12, no. 2, pp. 45–58, Nov. 2023.

[38] Y. S. Gaundare, S. A. Joshi, N. K. Punde, K. G. Bhave, T. V. Shirsath, V. V. Potdar, and V., "An efficient online offline data collection software solution for creating reference population to established genomic selection under small holders' dairy system in India," *International Journal of Computer Science Engineering and Information Technology*, vol. 13, no. 4, pp. 1–13, Aug. 2023.

[39] M. Sivakumar, R. K. Kumar, S. Natarajan, P. Murugesan, and V. Sundaram, "Systematic evaluation and comparative analysis of smartphone applications for agricultural stakeholders in India," *Journal of Agricultural and Extension Education*, vol. 30, no. 3, pp. 285–302,. 2022.

APPENDICES

This appendix is reserved for the data-gathering instruments used to support the requirements-gathering activities described in Chapter III. It should contain the actual copies of the researcher-developed questionnaire, interview or consultation guide, and observation checklist used during the study.

A.1 Livestock Technician Questionnaire

FOR AI (ARTIFICIAL INSEMINATIONS) TECHNICIANS

1. How many AI procedures do you handle in an average week?

- ☐ 1-5
- ☐ 6-10
- ☐ 11-20
- ☐ 21+

2. How do you receive most of your service requests?

- ☐ Direct calls/texts
- ☐ Via the Municipal Office
- ☐ Walk-ins

3. What is your primary tool for recording data in the field?

- ☐ Field Logbook
- ☐ Personal Phone (Notes/Texts
- ☐ Bond papers

4. How long does it take you to find a specific farmer's record in your files?

- ☐ A few seconds
- ☐ Several minutes
- ☐ I have to go home/to office to check

5. How do you schedule your follow-up Pregnancy Diagnoses (PD)?

- ☐ Manual reminders
- ☐ Checking my logbook daily
- ☐ Waiting for farmer to call

6. When compiling monthly reports, what is the most time-consuming part?

- ☐ Tallying numbers
- ☐ Verifying dates
- ☐ Locating missing data

7. How would you describe your current paperwork burden?

- ☐ Manageable

- ☐ Stressful
- ☐ Overwhelming

8. Do you often encounter cows with no previous breeding history available?

- ☐ Often
- ☐ Sometimes
- ☐ Rarely

9. Have you ever lost data due to a damaged logbook or lost phone?

- ☐ Yes
- ☐ No

10. How do you verify if a reported calf is actually from your AI?

- ☐ Cross-referencing dates
- ☐ Visual check of breed
- ☐ Farmer's word

11. What is your average conception (success) rate per month? _____

12. Is there a stable 4G/5G signal in the barangays you visit?

- ☐ Always
- ☐ Mostly
- ☐ Rarely/Dead spots

13. What data do you wish was "automatic" for you?

- ☐ Monthly Reports
- ☐ Success percentage
- ☐ PD reminders
- ☐ Others (Please specify): _____

14. How do you feel about shifting from paper to a mobile app?

- ☐ Very excited
- ☐ Neutral/Open
- ☐ Hesitant

15. What feature would make your field work 10x easier? _____

A.2 Interview and Consultation Guide — Insert the finalized questions used during interviews or consultations with livestock personnel and intended users. TO BE ADDED

FOR FARMERS (LIVESTOCK OWNERS)

1.How long have you been in the livestock business?

- ☐ 1-5 years

- ☐ 6-10 years
- ☐ 11+ years

2. How many cows are you currently managing?

- ☐ 1-2 heads
- ☐ 3-5 heads
- ☐ 6+ heads

3. What breed do you mostly raise?_____

4. When you notice a cow is ready for breeding (in heat), how do you usually tell?

- ☐ Behavioral changes
- ☐ Physical signs
- ☐ Using a breeding calendar

5. How do you usually get in touch with an AI technician?

- ☐ Private Call/Text
- ☐ Messenger/Social Media
- ☐ Personal visit to the office

6. Which type of technician do you usually contact for the service?

- ☐ Government / Municipal Technician (LGU)
- ☐ Private / Independent Technician
- ☐ Whoever is available first

7. What is the main reason you choose that specific technician?

- ☐ Cost/Affordability
- ☐ Faster response time
- ☐ Higher success rate
- ☐ Trusted friend

8. After you make the request, how long do you usually wait for them to arrive?

- ☐ Same day
- ☐ 1-2 days
- ☐ 3+ days

9. Do you mark down the dates when your cows are bred in a notebook or calendar?

- ☐ Always
- ☐ Sometimes
- ☐ Never (I rely on memory)

10. After the procedure, do you receive an official government form (AI Form) or a private receipt?

- ☐ Official Government Form
- ☐ Private/Handwritten Note

☐ No record given

11. How do you personally keep track of when the cow is expected to give birth?

☐ Manual counting/Mental note
☐ Marked on a calendar
☐ Technician reminds me

12. Who usually checks if the cow is pregnant (Pregnancy Diagnosis) after a few months?

☐ The same technician
☐ A different Vet
☐ No one (I just wait for the belly to grow)

13. If a cow does NOT get pregnant after the first try, do you report it to the Agriculture Office?

☐ Yes, always
☐ Only sometimes
☐ No, I just call the tech again

14. If you could see your cow's history and the technician's status on your phone, would that be helpful?

☐ Very helpful
☐ Somewhat helpful
☐ Not a priority

15. If we developed a simple app to help you request services and track your cows, would you try it?

☐ Yes, definitely
☐ Maybe (if it's easy)
☐ No

FOR MUNICIPAL AGRICULTURE OFFICE

1. How many active AI technicians are currently reporting to this office? _____

2. What is the current format of your "Master List" for AI?

☐ Hardbound Logbooks
☐ Folders of paper reports
☐ Excel/Google Sheets

3. How often are reports submitted to your office?

☐ Weekly
☐ Monthly
☐ Quarterly

4. How long does it take to consolidate all technician reports into one?

☐ 1-2 hours
☐ Half a day

☐ 2+ days

5. How accurate do you feel the data is when it reaches the office?

☐ 90-100%
☐ 70-89%
☐ Below 70%

6. Is there a current issue with delayed report submissions?

☐ Frequently
☐ Occasionally
☐ Rarely

7. Can you easily identify which Barangay has the highest AI success rate?

☐ Yes, instantly
☐ Yes, but requires manual calculation
☐ No

8. What is the most requested data from the Provincial Office ? _____

9. How do you track the "Total Calf Births" from AI?

☐ Direct reports
☐ Estimated
☐ We don't track this yet

10.Have you ever had records destroyed by pests, moisture, or fire?

☐ Yes
☐ No

11. How much time is spent responding to "Status Inquiries" from farmers?

☐ Significant time
☐ Some time
☐ Very little

12. Do you have a dedicated staff member for data encoding?

☐ Yes
☐ No (Technicians do it)

13. How do you feel about a centralized dashboard for the whole municipality?

☐ Highly needed
☐ Good to have
☐ Not a priority

14. What is your main concern with a digital system?

☐ Data security

- ☐ Technical skills of users
- ☐ System maintenance
- ☐ Other (Please specify): _____

15. If we developed a simple app to help you request services and track your cows, would you try it?

- ☐ Yes, definitely
- ☐ Maybe (if it's easy)
- ☐ No

\

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

ISO 25010 Software Evaluation Form

Assessment and Damage Reporting

Name of Juror: _____

Position/ Designation: _____

II. Instructions. Please rate by putting a checkmark on how you agree or disagree with the following statements.

1 - Poor

2 - Fair

3 - Satisfactory

4 - Very Satisfactory

5 - Excellent

	1	2	3	4	5
A. Functional Suitability					
1. Functional completeness. The system covers all the specified tasks and user objectives					
2. Functional correctness. The system provides the correct results with the needed degree of precision.					
3. Functional appropriateness. The system facilitates the accomplishment of specified tasks and objectives.					

B. Performance Efficiency					

1. Time behavior. The system response and processing times perform its functions and meet requirements.					
2. Resource utilization. The system amounts and types of resources perform its functions and meet requirements.					
3. Capacity. The system has the Maximum limits/capacity that meet requirements.					
C. Compatibility					
1. Co-existence. The system can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.					
2. Interoperabili ty. Two or more system components can Exchange information that has been exchanged.					
D. Usability					
1. Appropriateness recognizability. The users can recognize whether the system is appropriate to their needs.					
2. Learnability. The system can be used by specified goals of learning with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.					
3. Operability. The system has attributes that makes it easy to operate and to control.					

4. User error protection. The system protects users against making errors.					
5. User interface aesthetics. The system has a user interface that enables pleasing and satisfying interaction for the user.					
6. Accessibility. The system can be used by people with widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.					
E. Reliability					
1. Maturity. The system meets needs for reliability under normal operation.					
2. Availability. The system is operational and accessible when required for use.					
3. Fault tolerance. The system operates as intended despite the presence of hardware and software faults.					
4. Recoverability. The system can recover the data directly affected and re- establish the desired state of the system in the event of an interruption or a failure.					
F. Security					
1. Confidentiality. The system ensures that data are accessible only to those authorized to have access.					
2. Integrity. The system prevents unauthorized access to, or modification of computer programs or data.					
3. Non-repudiation. Actions or events can be proven to have taken place so that the events or actions cannot be repudiated later in the system.					

4. Accountability. The actions of an entity can be traced uniquely to the entity in the system.					
5. Authenticity. The identity of a subject or resource can be proved to be the one claimed in the system.					
G. Maintainability					
1. Modularity. The system is composed of discrete components such that a change to the component has minimal impact on other components.					
2. Reusability. A system asset can be used in more than one system, or in building other assets.					
3. Analyzability. The system is effective and efficient with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, or to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified.					
4. Modifiability. The system can be effectively and efficiently modified without introducing defects or degrading existing product quality.					
5. Testability. The system is effective and efficient with which test criteria can be established for a system, product or component and tests can be performed to determine whether those criteria have been met.					
H. Portability					

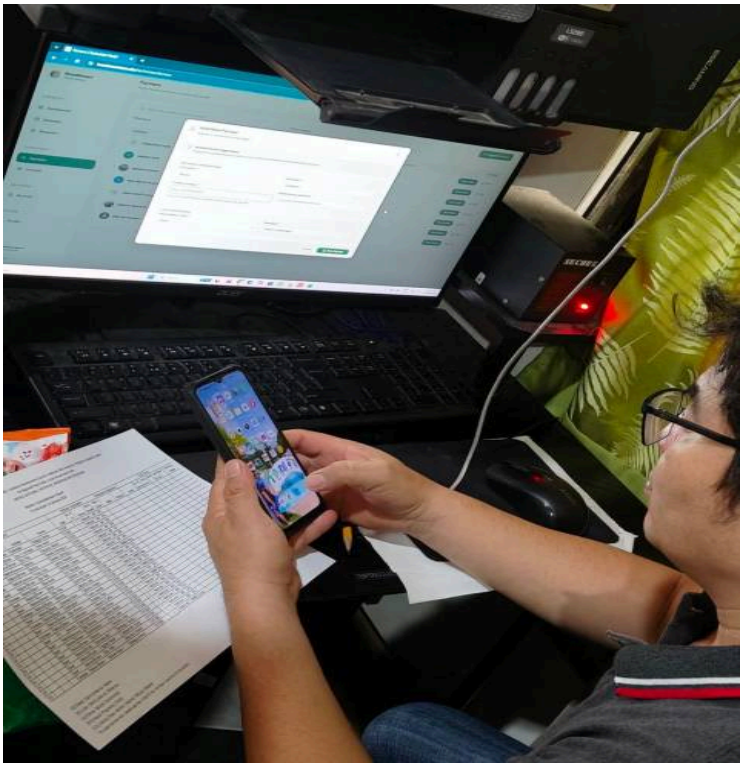
1. Adaptability. The system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.					
2. Installability. The system is effective and efficient with which a system can be successfully installed and/or uninstalled in a specified environment.					
3. Replaceability. The system can replace another specified software for the purpose in the same environment.					

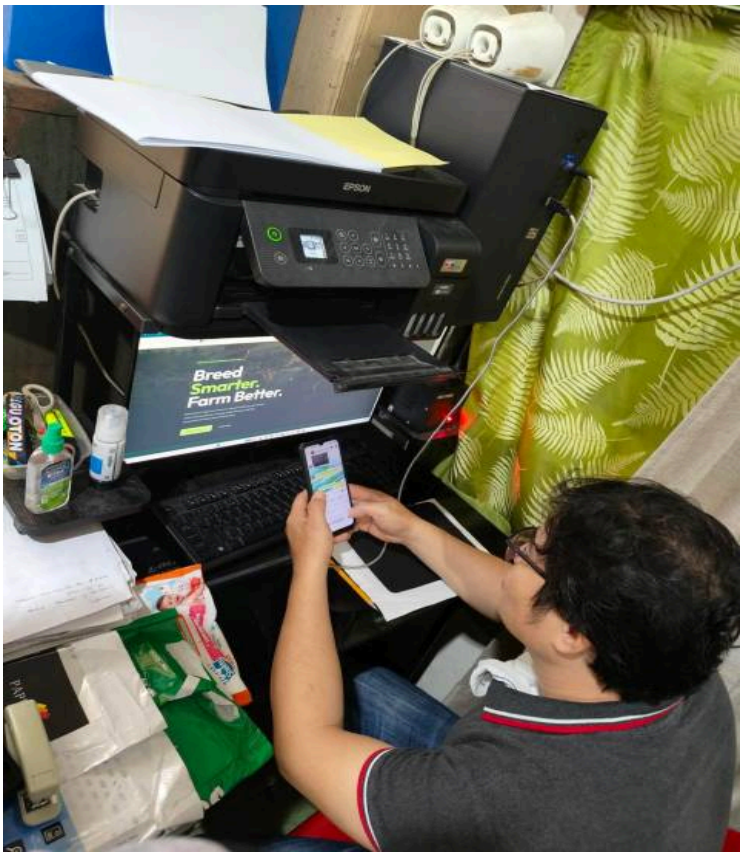
APPENDIX B: Field Visit and Research Documentation

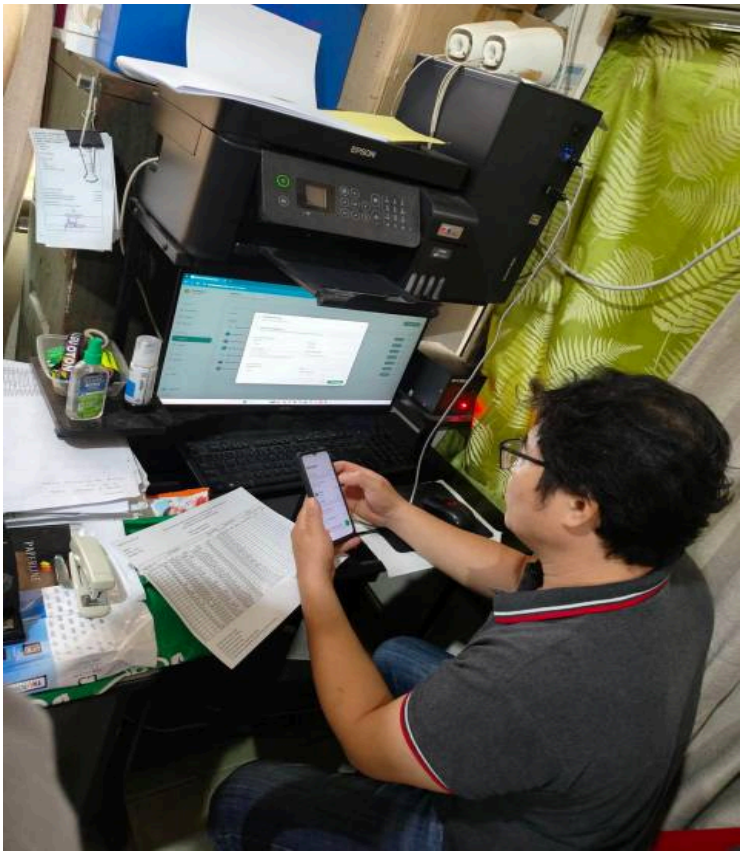
This appendix is reserved for photographic documentation of the research team's field activities in the Municipality of Oton. Photographs should be dated where possible and should use short captions that describe only the activity visible in the image.

September 12, 2026






Breed
Smarter.
Farm Better.





B.1 Initial Coordination and Consultation — photographs from the initial visit or consultation with the Municipal Agriculture Office and livestock personnel. TO BE ADDED

UNIVERSITY OF ILOILO
PHINMA EDUCATION

PHINMA University of Iloilo
College of Information Technology Education

CITE

John Lloyd P. Cabanig
PHINMA University of Iloilo
College of Information Technology Education
Rizal Street, Iloilo City
jopo.cabanig.ui@phinmaed.com
09922501413

OFFICE OF THE MAYOR
RECEIVED
02/24/26
DATE
BY
OTON, ILOILO

February 21, 2026

Ms. ROSA P. CADANO
Municipal Agriculturist
Municipality of Oton
Rizal St, Oton, 5020 Iloilo, Philippines

Thru: Hon. Sofronio L. Fusin Jr.
Municipal of Mayor

forward to MAEO for NOTE: Pls ensure that all data gathered will remain strictly confidential & will be used for this research only. TY.

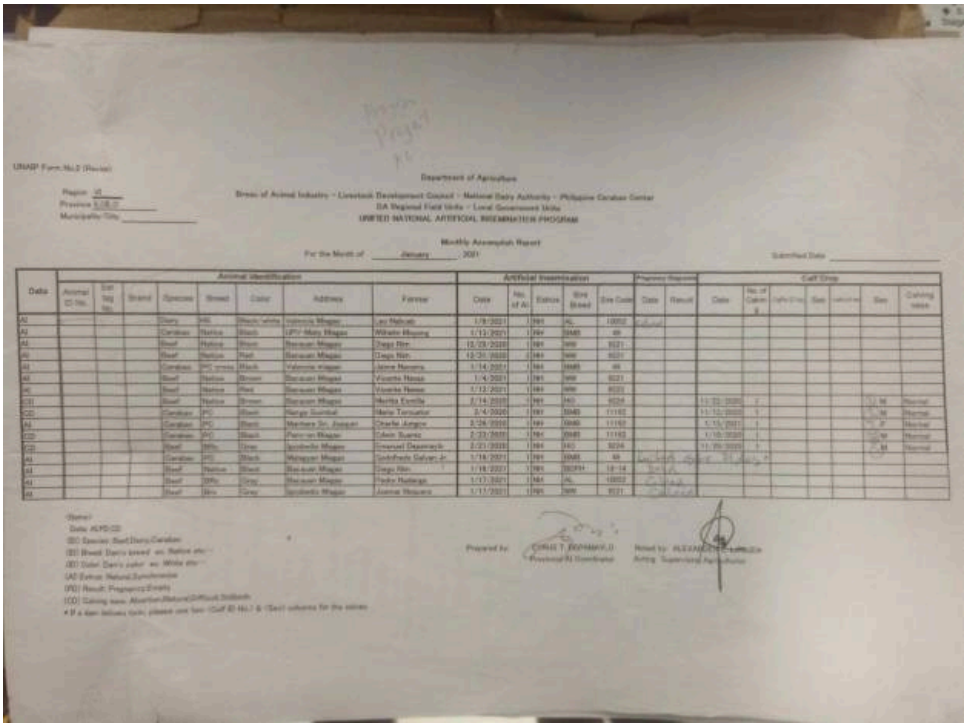
Subject: *Request for Consultation and Information Gathering Regarding Artificial Insemination Workflow for "SmartBreed: A Livestock Artificial Insemination Management Application"*

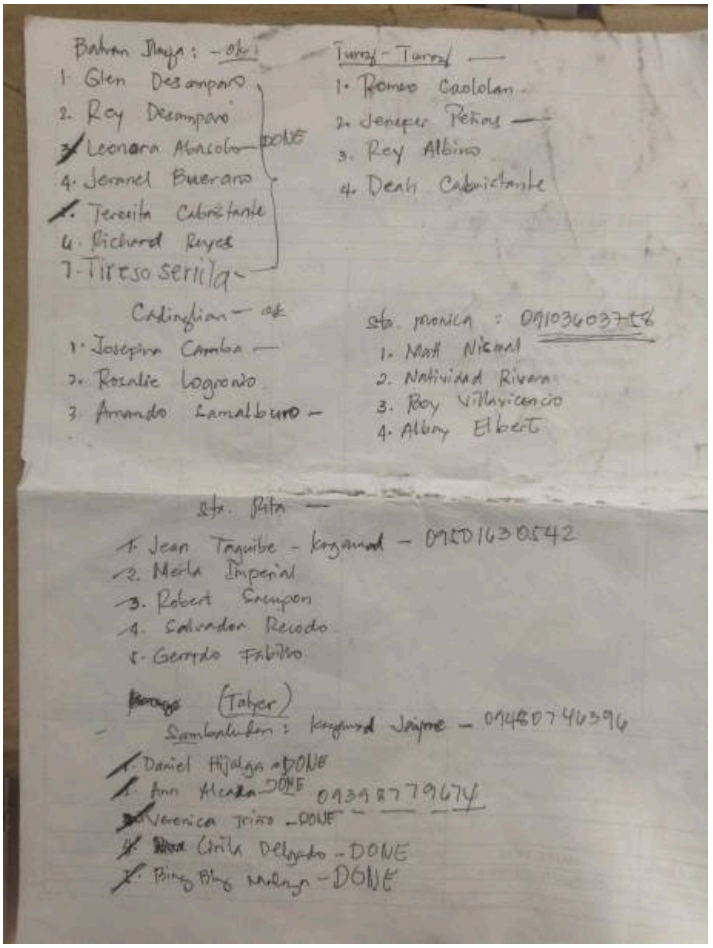
Dear Sir/Madam,

We are Bachelor of Science in Information Technology students from **PHINMA University of Iloilo**, currently under the **College of Information Technology Education**. As part of our capstone project requirements, we have developed an application titled **"SmartBreed: A Livestock Artificial Insemination Management Application."**

SmartBreed is a proposed application designed to assist in managing artificial insemination records, technician activities, and livestock breeding information in a more organized and

B.2 Requirements Gathering and Records Observation — Insert photographs documenting interviews, consultation, observation of existing livestock records, or other requirements-gathering activities.e. TO BE ADDED





B.3 System Demonstration, Pilot Use, and Evaluation Activities — Insert photographs taken

during actual BreedSmart demonstrations, pilot-use sessions, or user-evaluation activities, October 11, 2026




















Kids Town


DOWNLOAD



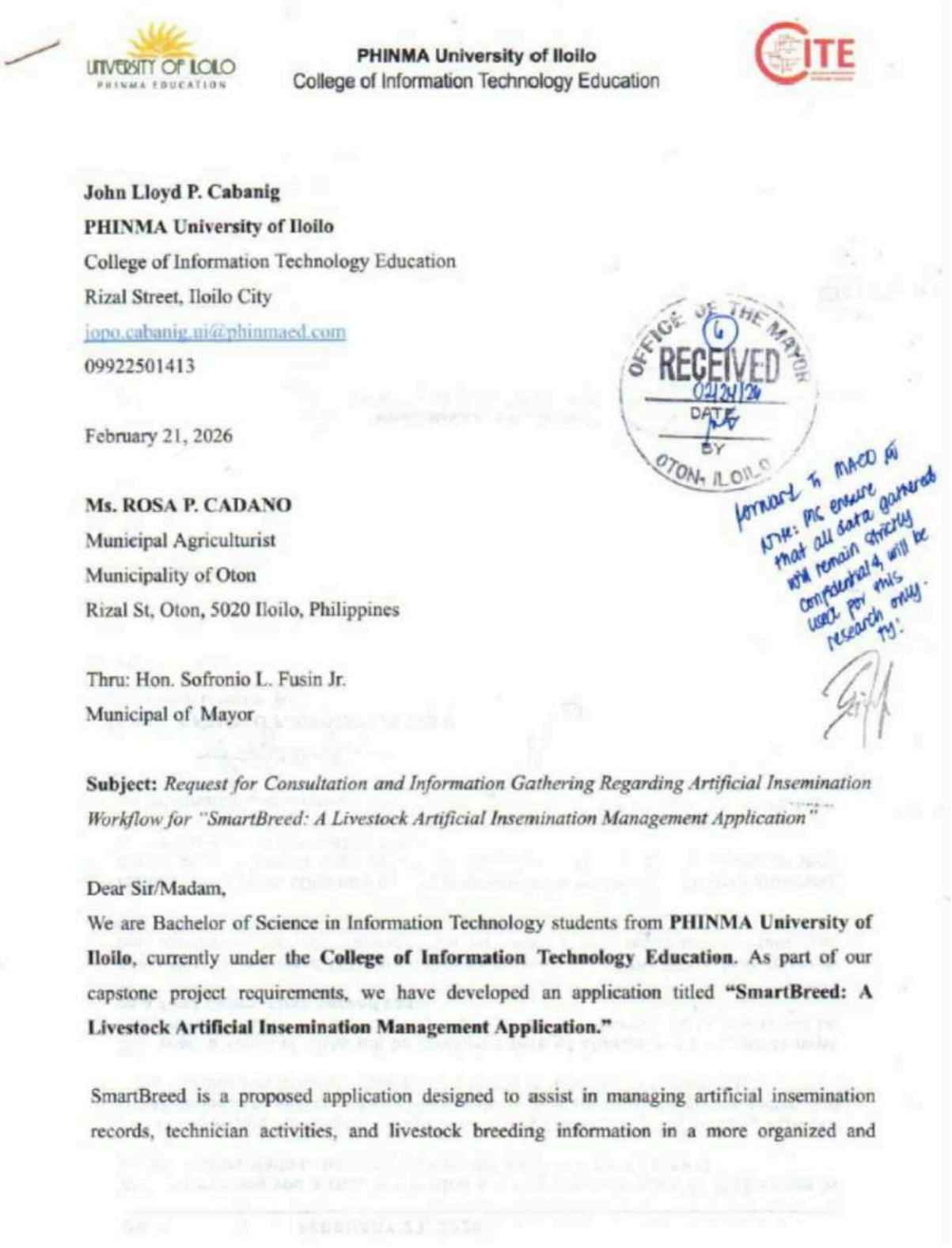

BreedSmart
LIVESTOCK MANAGEMENT
Smart livestock management
for farmers and technicians.
Track. Manage. Grow.
Smarter decisions for healthier herds.
Simple. Reliable. Built for farmers.
All your herd data in one place.
SCAN TO
DOWNLOAD
BE OUR TESTER
ANDROID APK
https://www.breedsmartoton.site/download-app
Municipality of Oton


DOWNLOAD
BreedSmart App
Smart livestock management for farmers and technicians.
Scan the QR code to download the Android APK.
BreedSmart
LIVESTOCK MANAGEMENT
Scan to Download
Android APK
Version 1.0.8
Municipality of Oton
BreedSmart Livestock Management

APPENDIX C: Letters, Permissions, and Research Transmittals TO BE ADDED

This appendix is reserved for copies of official letters, permission requests, endorsements, transmittals, and other correspondence used to coordinate the study with PHINMA University of Iloilo and the Municipal Agriculture Office of Oton. Arrange the documents chronologically where practical. Each item should retain its original date, addressee, signatures, receiving marks, and other official details.

Insert the signed or received copies of the applicable research letters and permissions in this section.



PHINMA University of Iloilo
College of Information Technology Education

John Lloyd P. Cabanig
PHINMA University of Iloilo
College of Information Technology Education
Rizal Street, Iloilo City
jopo.cabanig.ui@phinmaed.com
09922501413

OFFICE OF THE MAYOR
RECEIVED
02/24/26
DATE
BY
OTON, ILOILO

February 21, 2026

Ms. ROSA P. CADANO
Municipal Agriculturist
Municipality of Oton
Rizal St, Oton, 5020 Iloilo, Philippines

Thru: Hon. Sofronio L. Fusin Jr.
Municipal of Mayor

Subject: *Request for Consultation and Information Gathering Regarding Artificial Insemination Workflow for "SmartBreed: A Livestock Artificial Insemination Management Application"*

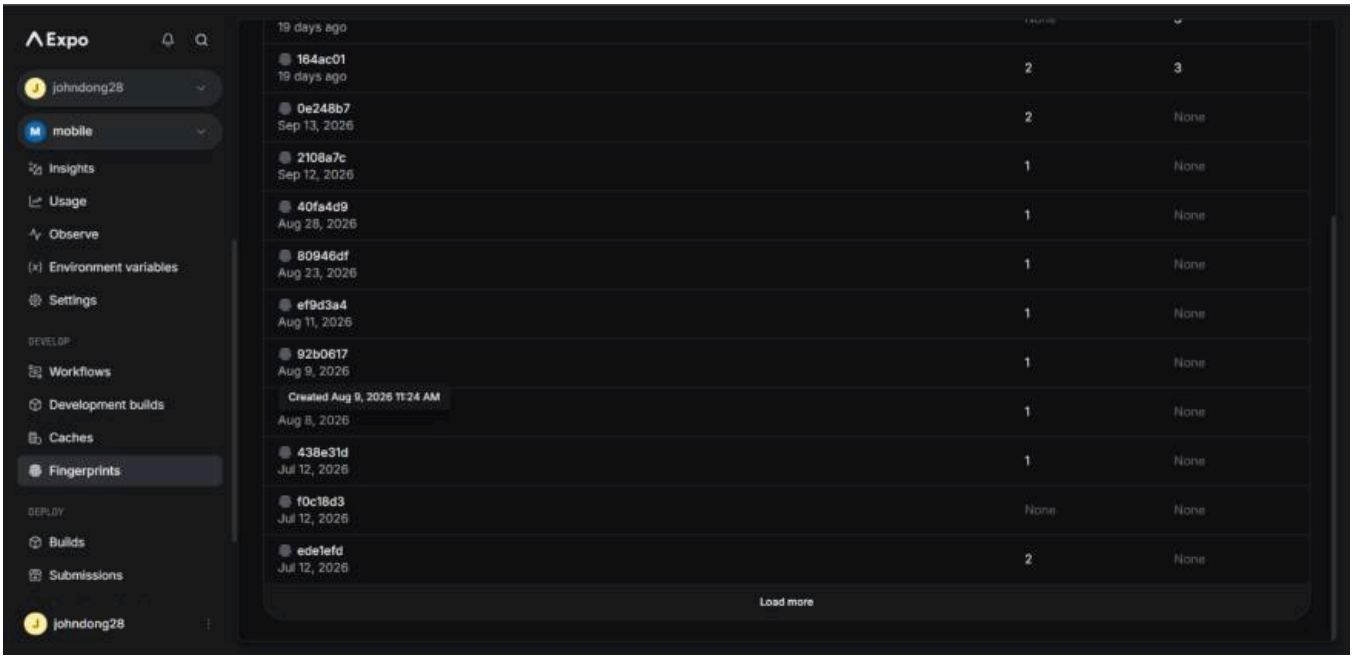
Dear Sir/Madam,

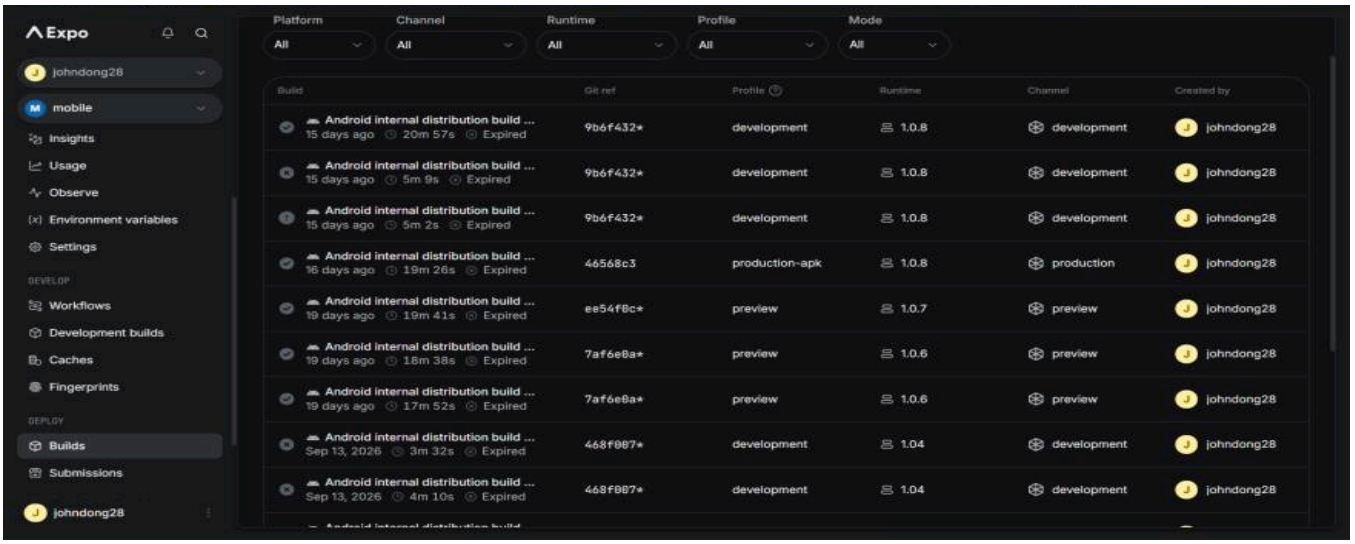
We are Bachelor of Science in Information Technology students from **PHINMA University of Iloilo**, currently under the **College of Information Technology Education**. As part of our capstone project requirements, we have developed an application titled **"SmartBreed: A Livestock Artificial Insemination Management Application."**

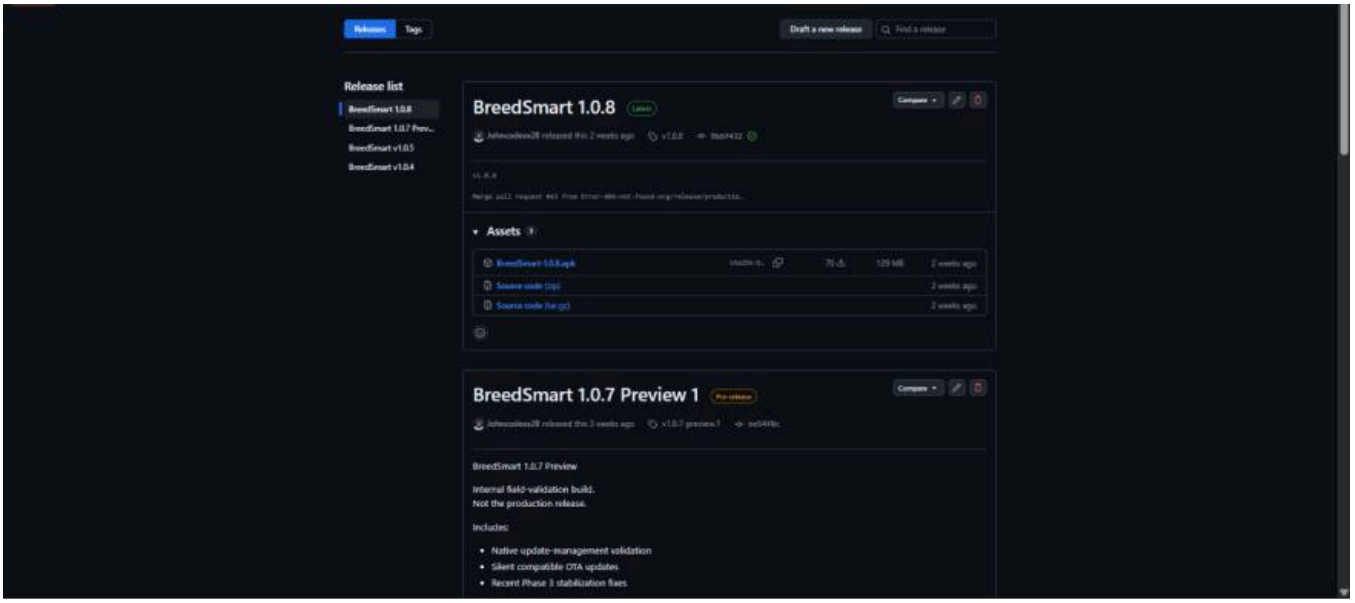
SmartBreed is a proposed application designed to assist in managing artificial insemination records, technician activities, and livestock breeding information in a more organized and

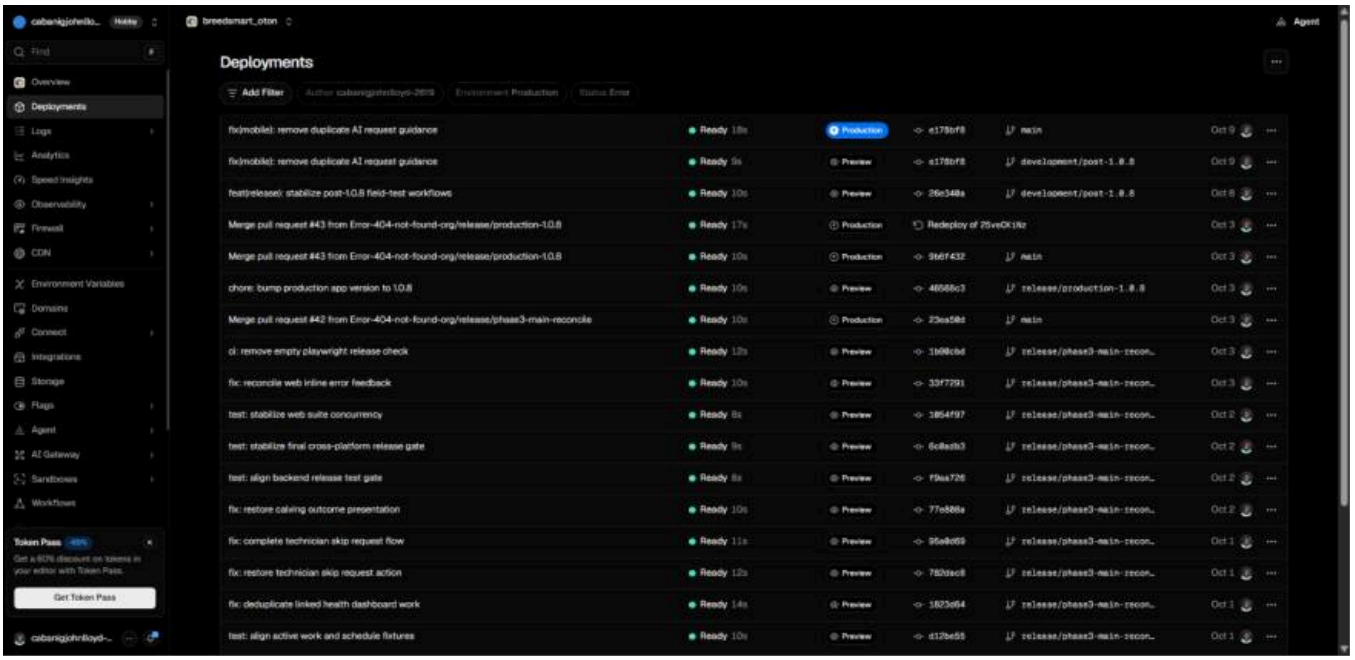
APPENDIX E: Verified Software and Deployment Configuration

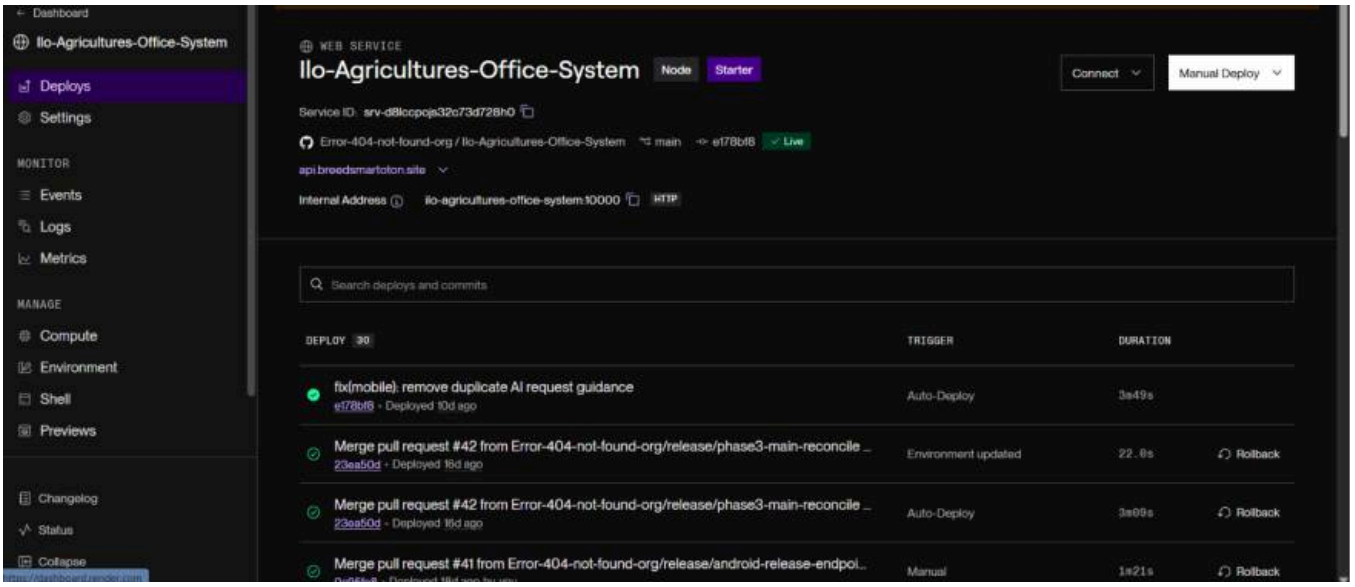


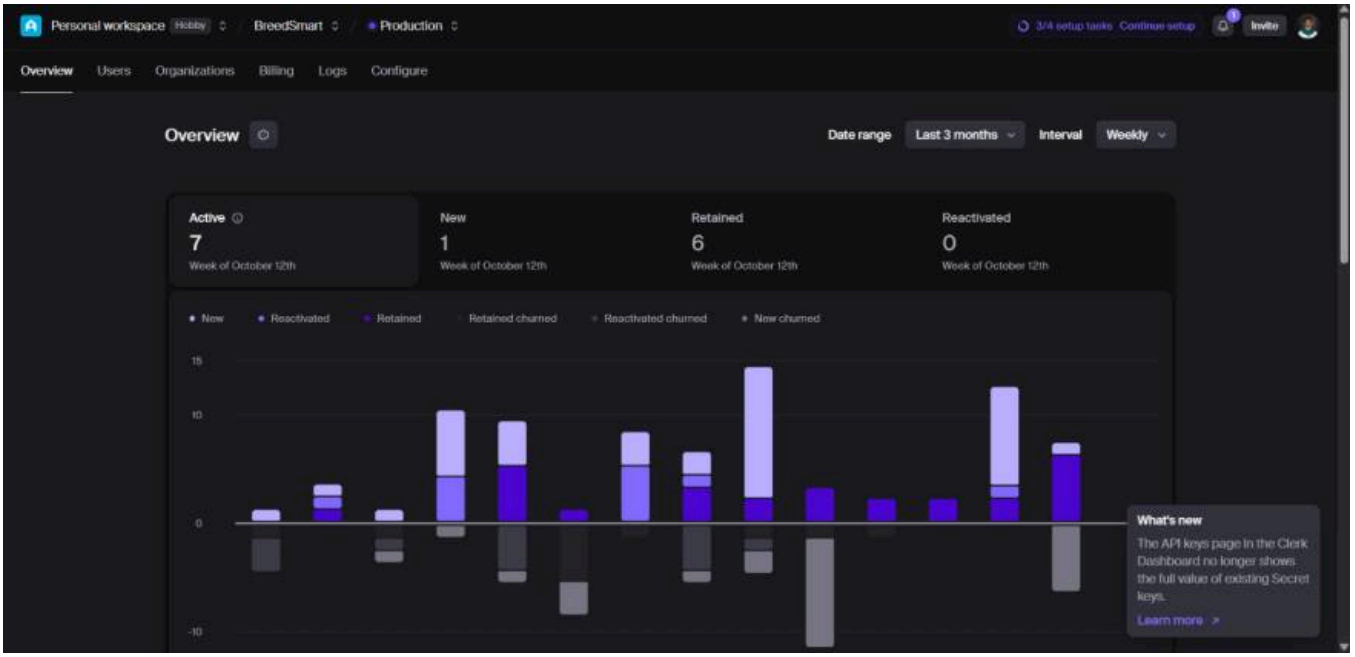


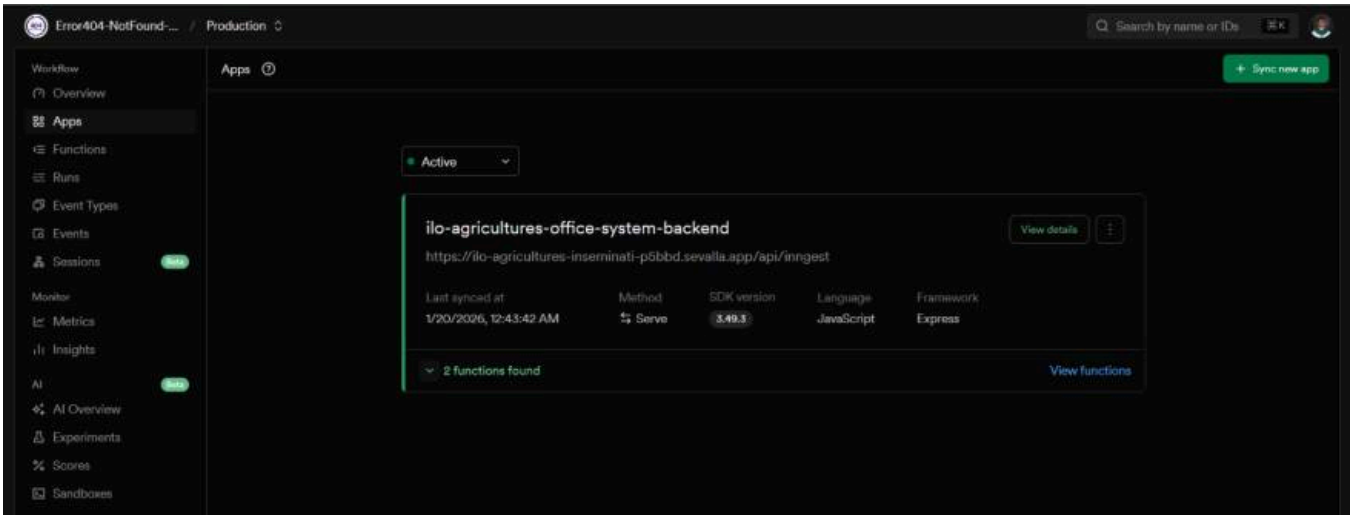

Expo
johndong28
mobile
Insights
Usage
Observe
Environment variables
Settings
DEVELOP
Workflows
Development builds
Caches
Fingerprints
DEPLOY
Builds
Submissions
johndong28
Platform
Channel
Runtime
Profile
Mode
All
Build
Git ref
Profile
Runtime
Channel
Created by
Android internal distribution build ... 15 days ago 20m 57s Expired 9b6f432* development 1.0.8 development johndong28
Android internal distribution build ... 15 days ago 5m 9s Expired 9b6f432* development 1.0.8 development johndong28
Android internal distribution build ... 15 days ago 5m 2s Expired 9b6f432* development 1.0.8 development johndong28
Android internal distribution build ... 16 days ago 19m 26s Expired 46568c3 production-apk 1.0.8 production johndong28
Android internal distribution build ... 19 days ago 19m 41s Expired ee54f8c* preview 1.0.7 preview johndong28
Android internal distribution build ... 19 days ago 18m 38s Expired 7af6e8a* preview 1.0.6 preview johndong28
Android internal distribution build ... 19 days ago 17m 52s Expired 7af6e8a* preview 1.0.6 preview johndong28
Android internal distribution build ... Sep 13, 2026 3m 32s Expired 468f087* development 1.04 development johndong28
Android internal distribution build ... Sep 13, 2026 4m 10s Expired 468f087* development 1.04 development johndong28

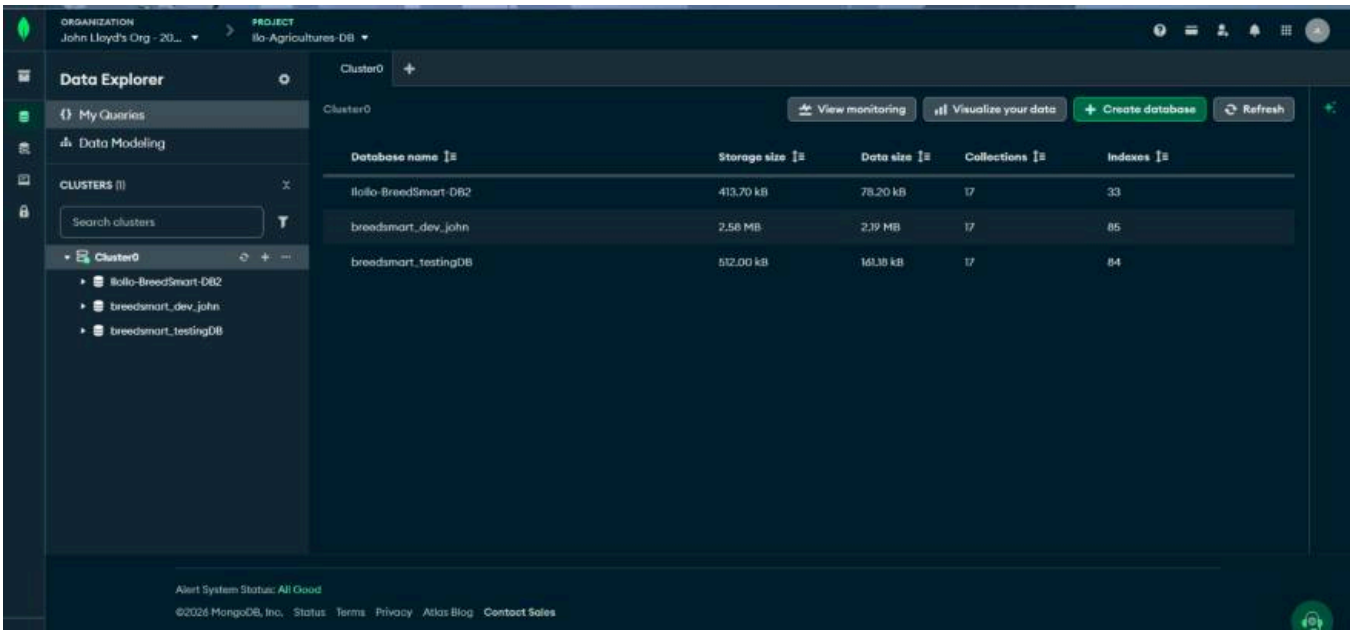

Releases
Tags
Draft a new release
Release list
BreedSmart 1.0.8
BreedSmart 1.0.7 Prev...
BreedSmart v1.0.5
BreedSmart v1.0.4
BreedSmart 1.0.8
Assets
BreedSmart 1.0.7 Preview 1
BreedSmart 1.0.7 Preview
Internal field-validation build.
Not the production release.
Includes:
Native update-management validation
Silent compatible OTA updates
Recent Phase 3 stabilization fixes


breedsmart_oton
Agent
Deployments
Overview
Deployments
Logs
Analytics
Speed Insights
Observability
Firewall
CDN
Environment Variables
Domains
Connect
Integrations
Storage
Flags
Agent
AI Gateway
Sandboxes
Workflows
Token Pass
Get Token Pass
Add Filter
fix(mobile): remove duplicate AI request guidance Ready 18s Production main Oct 9
fix(mobile): remove duplicate AI request guidance Ready 9s Preview development/post-1.0.8 Oct 9
feat(release): stabilize post-1.0.8 field-test workflows Ready 10s Preview development/post-1.0.8 Oct 8
Merge pull request #43 from Error-404-not-found-org/release/production-1.0.8 Ready 17s Production Redeploy of 25veOk1Nz Oct 3
Merge pull request #43 from Error-404-not-found-org/release/production-1.0.8 Ready 10s Production 9b6f432 main Oct 3
chore: bump production app version to 1.0.8 Ready 10s Preview release/production-1.0.8 Oct 3
Merge pull request #42 from Error-404-not-found-org/release/phase3-main-reconcile Ready 10s Production main Oct 3
ci: remove empty playwright release check Ready 12s Preview release/phase3-main-recon... Oct 3
fix: reconcile web inline error feedback Ready 10s Preview release/phase3-main-recon... Oct 3
test: stabilize web suite concurrency Ready 8s Preview release/phase3-main-recon... Oct 2
test: stabilize final cross-platform release gate Ready 9s Preview release/phase3-main-recon... Oct 2
test: align backend release test gate Ready 8s Preview release/phase3-main-recon... Oct 2
fix: restore calving outcome presentation Ready 10s Preview release/phase3-main-recon... Oct 2
fix: complete technician skip request flow Ready 11s Preview release/phase3-main-recon... Oct 1
fix: restore technician skip request action Ready 12s Preview release/phase3-main-recon... Oct 1
fix: deduplicate linked health dashboard work Ready 14s Preview release/phase3-main-recon... Oct 1
test: align active work and schedule fixtures Ready 10s Preview release/phase3-main-recon... Oct 1


Dashboard
Ilo-Agricultures-Office-System
Deploys
Settings
MONITOR
Events
Logs
Metrics
MANAGE
Compute
Environment
Shell
Previews
Changelog
Status
Collapse
WEB SERVICE
Ilo-Agricultures-Office-System
Node
Starter
Connect
Manual Deploy
Service ID: srv-d8lccpojs32c73d728h0
Error-404-not-found-org / Ilo-Agricultures-Office-System
main
e178bf8
Live
api.breedsmartofon.site
Internal Address
ilo-agricultures-office-system:10000
HTTP
Search deploys and commits
DEPLOY 30
TRIGGER
DURATION
fix(mobile): remove duplicate AI request guidance
e178bf8 · Deployed 10d ago
Auto-Deploy
3m49s
Merge pull request #42 from Error-404-not-found-org/release/phase3-main-reconcile ...
23ea50d · Deployed 18d ago
Environment updated
22.0s
Rollback
Merge pull request #42 from Error-404-not-found-org/release/phase3-main-reconcile ...
23ea50d · Deployed 18d ago
Auto-Deploy
3m09s
Rollback
Merge pull request #41 from Error-404-not-found-org/release/android-release-endpoi...
Manual
1m21s
Rollback


Personal workspace
Hobby
BreedSmart
Production
3/4 setup tasks
Continue setup
Invite
Overview
Users
Organizations
Billing
Logs
Configure
Overview
Date range
Last 3 months
Interval
Weekly
Active
7
Week of October 12th
New
1
Week of October 12th
Retained
6
Week of October 12th
Reactivated
0
Week of October 12th
New
Reactivated
Retained
Retained churned
Reactivated churned
New churned
15
10
0
-10
What's new
The API keys page in the Clerk Dashboard no longer shows the full value of existing Secret keys.
Learn more


Error404-NotFound-...
Production
Search by name or ID
⌘K
Workflow
Overview
Apps
Functions
Runs
Event Types
Events
Sessions
Monitor
Metrics
Insights
AI
AI Overview
Experiments
Scores
Sandboxes
Apps
Sync new app
Active
ilo-agricultures-office-system-backend
View details
https://ilo-agricultures-inseminati-p5bbd.sevalla.app/api/inngest
Last synced at
1/20/2026, 12:43:42 AM
Method
Serve
SDK version
3.49.3
Language
JavaScript
Framework
Express
2 functions found
View functions


ORGANIZATION
John Lloyd's Org - 20...
PROJECT
Ilo-Agricultures-DB
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
Ilo-Agricultures-DB2
breedsmart_dev_john
breedsmart_testingDB
Cluster0
Cluster0
View monitoring
Visualize your data
Create database
Refresh
Database name
Storage size
Data size
Collections
Indexes
Ilo-Agricultures-DB2
413.70 kB
78.20 kB
17
33
breedsmart_dev_john
2.58 MB
2.19 MB
17
85
breedsmart_testingDB
512.00 kB
161.18 kB
17
84
Alert System Status: All Good
©2026 MongoDB, Inc. Status Terms Privacy Atlas Blog Contact Sales

APPENDIX H: Curriculum Vitae



John Lloyd P. Cabanig

Jaro, Iloilo City, Western Visayas, Iloilo |
cabanigjohnlloyd@gmail.com |
09922501413

Educational Attainment

- Bachelor of Science in Information in Technology

PHINMA University of Iloilo, Western, Visayas, 4th Year Level (2026) |

Major in System Development and Business Analytics

Technical & Professional Competencies

- **Modeling & Prototyping:** LucidChart, Draw.io , Canva , Wireframing
- **AI Tools:** Claude, ChatGPT, Gemini, OpenCode
- **Data Entry & Documentation:** Google Sheets (Pivot table, formula functions, dashboard), Google Powerpoint, Google Docs, Microsoft Word, Microsoft Excel, Microsoft Powerpoint.
- **Database & Core Programming:** MongoDB, Node JS, Express, Python, Php, HTML/CSS.

Academic Projects & Case Studies

- **Parking System :** A system designed to help manage parking spaces, vehicle records, and basic parking transactions more efficiently at the university.
- **BreedSmart:** an animal welfare system designed to support stray animal assistance and make adoption access easier
- **Personal Portfolio Website –** A personal website created to showcase my skills, projects, experience, and background in information technology.



Nelmar B. Buenafe

Sarah, Iloilo, Western Visayas, Iloilo |

neba.buenafe.ui@phinmaed.com |

09817673396

Educational Attainment

- Bachelor of Science in Information in Technology

PHINMA University of Iloilo, Western, Visayas, 4th Year Level (2026) |

Major in System Development and Business Analytics

Technical & Professional Competencies

- **Business Analysis & Modeling:** Flow Chart Diagram, Process Flowchart, Detailed Flowchart.
- **Modeling & Prototyping:** Figma,Draw.io , Canva , Wireframing
- **Data Entry & Documentation:** Google Sheets (Pivot table, formula functions, dashboard), Google Powerpoint, Google Docs, Microsoft Word, Microsoft Excel, Microsoft Powerpoint.
- **Database & Core Programming:** Supabase, MySQL, Node JS, Express, React, Php, HTML/CSS.

Academic Projects & Case Studies

- **Phinma UI Network System:** a centralized network topology using CISCO Packet Tracer, Configure VLANs, IP addressing, DHCP, trunk connections, and basic network security.
- **BreedSmart:** an animal welfare system designed to support stray animal assistance and make adoption access easier.
- **Parking System :** A system designed to help manage parking spaces, vehicle records, and basic parking transactions more efficiently at the university.



John Arvy Z. Lopez

City Proper, Iloilo City, Western Visayas, Iloilo | joza.lopez.ui@phinmaed.com |

09703918920

Educational Attainment

- Bachelor of Science in Information in Technology

PHINMA University of Iloilo, Western, Visayas, 4th Year Level (2026) |

Major in System Development and Business Analytics

Technical & Professional Competencies

- **Business Analysis & Modeling:** Functional Requirements, Value Stream Mapping, SIPOC Analysis, Flow Chart Diagram, Process Flowchart, Detailed Flowchart.
- **Modeling & Prototyping:** LucidChart, Draw.io, Wireframing
- **Data Entry & Documentation:** Google Sheets (Pivot table, formula functions, dashboard), Google Powerpoint, Google Docs, Microsoft Word, Microsoft Excel, Microsoft Powerpoint.
- **Database & Core Programming:** MongoDB, Node JS, Express, Python, Php, HTML/CSS., React JS, React Native, Tailwind.

Academic Projects & Case Studies

- **Parking System :** A system designed to help manage parking spaces, vehicle records, and basic parking transactions more efficiently at the university.
- **BreedSmart:** an animal welfare system designed to support stray animal assistance and make adoption access easier.
- **Personal Portfolio Website –** A personal website created to showcase my skills, projects, experience, and background in information technology.



Gian Rovik B. Somes
Jalandoni Street, Jaro, Iloilo City, Western Visayas, Iloilo | gibu.somes.ui@phinmaed.com | 00969477526

Educational Attainment

- Bachelor of Science in Information in Technology

PHINMA University of Iloilo, Western, Visayas, 4th Year Level (2026) |
Major in System Development and Business Analytics

Technical & Professional Competencies

- **Business Analysis & Modeling:** Functional Requirements, Value Stream Mapping, SIPOC Analysis, Flow Chart Diagram, Process Flowchart, Detailed Flowchart.
- **Modeling & Prototyping:** LucidChart, Draw.io , Canva , Wireframing
- **Data Entry & Documentation:** Google Sheets (Pivot table, formula functions, dashboard, Data Mining, Data Cleaning), Google Powerpoint, Google Docs, Microsoft Word, Microsoft Excel, Microsoft Powerpoint.
- **Database & Core Programming:** MongoDB, MySQL, Node JS, Express, Python, Php, HTML/CSS.

Academic Projects & Case Studies

- **Breedsmart System:** a mobile/web-based system for managing cattle breeding and health record management.
- **Carezone Adoption Center System:** an animal welfare system designed to support stray animal assistance and make adoption access easier
- **Company Job Offering System: –** a capstone proposal system focused on helping job seekers discover available employment opportunities.



Justin A. Balmores

Pototan, Western Visayas, Iloilo |

usap.balmores.ui@phinmaed.com |

00969477526

Educational Attainment

- Bachelor of Science in Information in Technology

PHINMA University of Iloilo, Western, Visayas, 4th Year Level (2026) |

Major in System Development and Business Analytics

Technical & Professional Competencies

- **Business Analysis & Modeling:** Context Diagram, Flow Chart Diagram, Process Flowchart
- **Modeling & Prototyping:** Draw.io , Canva
- **Data Entry & Documentation:** Google Sheets (Pivot table, formula functions, dashboard), Google Powerpoint, Google Docs, Microsoft Word..
- **Database & Core Programming:** Mysql, Node JS, Express, Python, Php, HTML/CSS.